

Stochastic and Thermal Exactness

Diophantine certificates for exact events
in uniformly convergent systems

Positive stochastic erasure, exact value iteration of stochastic games, and thermal Hamiltonians with undecidable confinement boundaries

AI-assisted research report prepared for Vladimir Reshetnikov from three manuscripts
(batch 78 of ProveIt's incoming reports, manuscripts 02, 10 and 03), October 2, 2026

Abstract

Three independent manuscripts share one pattern: a numerical approximation converges uniformly, at a rate known in advance, while an *exact* event of the same system is undecidable.

Part I (source 02) compiles integer matrix mortality into strictly positive rational column-stochastic matrices with no common invariant complex line, each entry within any prescribed band around $1/n$ and every product contracting total variation by η^T . A product has rank one exactly when the source projection is mortal, so exact erasure is undecidable for at most eight positive 7×7 matrices and many-one complete for a nine-state family with a variable alphabet. Explicit horizon- T natural quartics have one witness per erasing labelled word.

Part II (source 10) builds a fixed turn-based stochastic game with discount $1/2$, reward $1/4$, dyadic transition rows of full support and the known value vector $\frac{1}{2}\mathbf{1}$. Value iteration converges at rate 2^{-n} , yet exact attainment of $\frac{1}{2}\mathbf{1}$ from a dyadic terminal payoff is computably enumerable complete. Each horizon has an orthant-nonnegative quadratic certificate with exactly one complete natural witness.

Part III (source 03) turns a natural arithmetic circuit into a diagonal Hamiltonian $H = (1 + \sum_i n_i)Q$ of degree at most five whose zero states biject with the witnesses and whose positive spectrum is effectively confined. Compact resolvent and finite thermal trace are equivalent to finite witness multiplicity and are undecidable on a padded universal family; a uniformly computable critical fugacity t_* satisfies $t_* < 1$ exactly on halting inputs.

The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq, and priority is not certified for any Part.

Status. Proofs and executable finite checks; not kernel-checked. Section 1 states the common spine, the notation, the relation to neighbouring reports and to the repository's reviews of these manuscripts, and what is not claimed; Appendix F gives the provenance.

Contents

1	Scope, common spine and conventions	6
1.1	The three Parts	6
1.2	The common spine: uniform approximation, undecidable exactness	7
1.3	Notation across the Parts	7
1.4	Relation to neighbouring reports and to the repository’s reviews	9
1.5	Status and what is not claimed	10
I	Exact erasure without a common invariant line (source 02)	11
2	Research question, contributions, and provenance	12
2.1	A concrete question that the construction answers	12
2.2	Main results and their logical dependencies	13
2.3	Relationship to the inspected ProveIt material	13
3	Exact forgetting and the zero-mass representation	14
4	An explicit positive affine lift	15
5	Two guards eliminate every common invariant line	17
5.1	Small numerical consequences	18
6	A direct correspondence-problem compiler	19
6.1	A four-dimensional integer front end	19
7	Excluding all small common invariant subspaces	21
8	Decidable boundaries and a sharp commuting horizon	22
8.1	An exact dimension correspondence	22
8.2	Commuting information dynamics	22
9	Random schedules, exact forgetting, and a hidden-state interpretation	23
9.1	A labelled hidden-state model with trivial output statistics	24
9.2	Finite-horizon probabilities	25
10	Explicit natural-domain Diophantine certificates	25
10.1	A full-prefix certificate with no minors	25
10.2	A smaller certificate in information coordinates	26
10.3	When the numerical matrices become polynomial parameters	27
10.4	Exact counts of solutions and limits of uniqueness	28

11 Fixed unbounded Diophantine representations and impossibility of bounds	29
11.1 What DPRM supplies, and what it does not	29
11.2 No uniform computable exactness horizon	29
12 Exact erasure is not robust under perturbation	30
13 Worked examples and executable verification	31
13.1 A completely printed three-state example	31
13.2 A seven-state, eight-letter numerical packet	32
13.3 End-to-end PCP example	32
13.4 Scope and results of the executable checks	33
14 Integration and formalization plan for ProveIt	34
15 Further research questions and concrete next targets	34
15.1 Reduce the tensor overhead without losing the line guarantee	34
15.2 Irreducibility beyond a missing invariant line	35
15.3 Low-dimensional classifications via the exact correspondence	35
15.4 Regularly constrained and non-independent scheduling	35
15.5 Sharper canonical certificates and fully charged arithmetic	35
15.6 Fixed-alphabet universality with an ordinary input interface	35
15.7 Structural non-erasure certificates	36
15.8 Precision-sensitive versus robust observation predicates	36
15.9 A formal proof of the guard compiler and the projection bijection	36
16 Conclusion	36
II Exact convergence as computation (source 10)	36
17 Research question, contribution, and source boundary	37
17.1 The central theorem	38
17.2 Why the source comparison matters	38
17.3 What is and is not computed	39
18 Conventions and the three observation problems	39
19 A homogeneous counter-machine compiler	40
19.1 Source instruction set	40
19.2 Configuration coordinates and exact zero tests	40
19.3 Two halt conventions	41

20 From signed circuits to genuine stochastic games	42
20.1 Layering and normalization	42
20.2 Paired channels and a baseline state	42
20.3 The invariant subspace	43
20.4 Initialization is part of the compiler	43
21 Uniform mixing preserves exact computation	44
21.1 A common-reset lemma	44
21.2 Contraction estimates	45
22 Proof of fixed-game universality and its variants	45
22.1 Uniformly stopping reachability games	46
22.2 Zero running rewards and scalar observations	46
22.3 One-sided errors are easy	46
23 Two decidable boundaries and one Skolem boundary	47
23.1 No averaging: finite selector dynamics	47
23.2 No choices: whole-vector zeros are easy	47
23.3 Scalar equality: a dyadic, full-support Skolem specialization	48
24 Canonical quadratic Diophantine certificates	49
24.1 Integer scaling of zero-reward runs	49
24.2 A min/max gate with unique gaps	49
24.3 The complete polynomial	50
24.4 The fixed-point theorem’s nonzero reward	51
24.5 Size and precision	51
25 Worked examples and complete emitted objects	52
25.1 A two-state polynomial that can be checked by hand	52
25.2 A compiled countdown program	52
26 Reproducibility, checks, and verification limits	53
27 From bounded quadratics to a fixed Diophantine relation	55
27.1 A quartic existence corollary, with the correct caveat	55
27.2 No computable global search cutoff	55
28 Precision, robustness, and why approximation is not a decision	56
29 Research agenda and concrete next targets	57
29.1 Instantiate and minimize a fixed universal game	57

29.2 Remove one player's genuine choices	57
29.3 Prescribed randomness and sparse supports	57
29.4 One-dimensional observations versus whole-vector attainment	58
29.5 Witness-economical native certificates	58
29.6 Integrate with the repository's fixed-arity MRDP route	58
29.7 Formalize the proof-critical interfaces	58
30 Conclusion	59
III Thermal arithmetic (source 03)	59
31 Research question, scope, and relationship to ProveIt	60
31.1 Why another computational substrate?	60
31.2 Source snapshot and limits of the audit	60
31.3 Established foundations and the proposed contribution	61
31.4 Main conclusions at a glance	61
32 Arithmetic conventions and the witness-faithful compiler	62
32.1 Natural domains and representations	62
32.2 Natural arithmetic circuits	62
32.3 A small example with a complete resource count	64
33 The operator and its effectively confined positive spectrum	64
33.1 Domains are part of the construction	64
33.2 Energy layers and an exact finite search	64
33.3 Separating the zero projection from a computable compact part	65
34 Multiplicity, compactness, and thermal finiteness	66
34.1 A spectral equivalent of finite-fold representability	67
34.2 An unconditional undecidable compactness family	68
35 Regulated partition functions and their analytic boundary	69
35.1 Two thermal parameters	69
35.2 A sharp analytic dichotomy	70
36 A computable critical fugacity with an undecidable endpoint	71
36.1 An instance-independent thermal separation	71
36.2 No computable uniform distance from the boundary	72
36.3 Witness heights determine the crossover scale	72

37 What remains undecidable despite the positive-energy gap	73
37.1 Bottom-of-spectrum separation	73
37.2 Bounded operator perturbations versus coefficient errors	74
38 Worked examples and exact computational checks	74
38.1 A single witness that moves toward the boundary	74
38.2 Infinite zero multiplicity with a computable excited trace	75
38.3 An arithmetic infinite fiber: Pell witnesses	75
38.4 Why the radial factor must contain every mode	76
38.5 Compiler and verification package	76
38.6 Certified truncation without enormous denominators	77
39 Further research questions	78
39.1 Can radial isolation be achieved at degree four?	78
39.2 Can four-mode positive terms be reduced to three-mode terms?	78
39.3 Instantiate a numerical universal arithmetic family	78
39.4 Optimize compiled witness height, not only gate count	79
39.5 Find unconditional all-fiber compact subclasses	79
39.6 Characterize effective compactness on restricted families	79
39.7 Classify witness-series singularities	79
39.8 What survives non-diagonal bounded couplings?	79
39.9 Can a physically motivated cutoff retain a useful conditional theorem?	80
39.10 Formalize the arithmetic and effective-analysis interface	80
40 Dependency and claim ledger	80
41 Conclusion	81
A A fully instantiated horizon-two polynomial schema	81
B Reproduction and source ledger	82
C Source audit and provenance	82
D A compact implementation contract	83
E Reproduction and file inventory	84
F Provenance of this report	84
F.1 Sources	85
F.2 Where the merge had to choose	85

1 Scope, common spine and conventions

[write] This section was written when the three manuscripts were merged. Parts I, II and III print the three sources; text there without a [write] marker is the source’s own.

1.1 The three Parts

This report is built from three AI-assisted research manuscripts delivered with batch 78 of ProveIt’s incoming reports (manuscripts 02, 10 and 03; Appendix F gives the full provenance). They are called *source 02*, *source 10* and *source 03* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. None cites another, they were written from three different snapshots of the repository on the same day, and they share no lemma.

- **Part I (source 02), *Exact erasure without a common invariant line*.** A finite alphabet of rational column-stochastic matrices is driven by an external schedule; a word *erases* when its product has rank one, so that the output distribution no longer depends on the input distribution. Two invertible tensor guards lift integer mortality into strictly positive stochastic matrices with no common invariant complex line and the exact rank identity $\text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}$ (Theorem 5.1). Rank-one existence is undecidable for at most eight positive 7×7 matrices (Corollary 5.3) and many-one complete for nine states with a variable alphabet (Corollary 6.3). Under independent full-support switching, existence of an erasing word, almost-sure finite erasure and finite expected erasure time coincide (Theorem 9.1). Horizon- T natural quartics have exactly one witness per erasing labelled word (Theorems 10.1 and 10.2).
- **Part II (source 10), *Exact convergence as computation*.** A counter program is compiled through homogeneous dyadic min/max circuits into a genuine two-player turn-based stochastic game with discount $1/2$, running reward $1/4$, at most two actions per state and every transition probability at least $1/(2N)$. The Bellman operator is a $1/2$ -contraction with the known fixed point $\frac{1}{2}\mathbf{1}$, and exact finite attainment of that fixed point from a dyadic terminal payoff is computably enumerable complete for a suitable fixed game (Theorem 17.1). Point-mass selectors and chance-only games give decidable boundaries, and chance-only scalar equality is Skolem-equivalent (Section 23). Each horizon has an orthant-nonnegative quadratic with exactly one complete natural witness (Theorem 24.2).
- **Part III (source 03), *Thermal arithmetic*.** A natural arithmetic circuit becomes a diagonal Hamiltonian $H = (1 + \sum_i n_i)Q$ on occupation modes, of joint degree at most five, whose zero-energy basis states are in bijection with the witnesses while every positive energy is at least $1 + \sum_i n_i$ (Theorem 32.1). Compact resolvent and finite thermal trace hold exactly when the witnesses are finitely many (Theorem 34.1), and both are undecidable on a padded universal family (Theorem 34.6). A uniformly computable critical fugacity satisfies $t_*(c) < 1$ exactly on halting inputs (Theorem 36.1).

Each Part keeps its own definitions, letters and numbering of statements. Parts I and II are both about stochastic operators with full support (stochastic games extend stochastic matrices by minima and maxima); Part III shares no lemma with either, and it is placed here because of the spine of Section 1.2, not because of a shared construction.

1.2 The common spine: uniform approximation, undecidable exactness

In each Part an approximation statement holds with a computable rate that does not depend on the instance, and an exact statement about the same object is undecidable. Table 1 lines them up.

Part	Uniform approximation	Exact predicate and its status	Bounded or static certificate
I (02)	$\text{TV}(S_w p, S_w q) \leq \eta^T \text{TV}(p, q)$ for every word of length T (Proposition 4.2)	some product has rank one: undecidable on the seven-state class, c.e.-complete on the nine-state class (Corollaries 5.3, 6.3)	horizon- T quartic, $T((n-1)^2 + k)$ natural witnesses, one per erasing labelled word (Theorem 10.2)
II (10)	$\ \Phi^n(v) - \frac{1}{2}\mathbf{1}\ _\infty \leq$ $2^{-n} \ v - \frac{1}{2}\mathbf{1}\ _\infty$, and $4^{-n}/2$ on encoded inputs (Section 21)	$\exists n \Phi^n(v) = \frac{1}{2}\mathbf{1}$: c.e.-complete for a fixed game (Theorem 17.1)	horizon- T quadratic, $(N+2b)T$ natural witnesses, one complete witness (Theorem 24.2)
III (03)	excited trace, regulated partition function and t_* uniformly computable (Theorems 33.3, 35.1, 36.1)	$t_*(c) < 1$: c.e.-complete; compact resolvent, finite thermal trace: co-c.e.-complete on a padded family (Theorems 36.1, 34.6)	static polynomial of degree at most five with a fixed number of modes, one zero state per witness (Theorem 32.1)

Table 1: [write] The common spine of the three Parts.

Three further features recur. First, every Part proves that no total computable function of the input bounds the time or the witness height of a positive instance (Theorem 11.2, Corollary 27.1, Theorem 37.1); this is the bounded-search argument of `cdc:bd:prop:nobound` in [22] and `pqc:qm:cor:nobound` in [21], applied to new predicates. Second, a fixed positive tolerance restores decidability or a finite cutoff in Parts I and II ($\eta^T \leq \varepsilon$; Proposition 28.1), while in Part III no computable positive separation from the boundary exists (Corollary 36.3). Third, each Part also states an MRDP consequence of fixed arity and degree at most four obtained by circuit quadratization (Proposition 11.1; Section 27; Corollary 32.3 with Theorem 32.1); none of them controls the multiplicity of the original MRDP witnesses.

The slogan itself is not new in the collection: Parts V and VII of [21] already prove contraction-versus-exactness results, and sources 02 and 10 say so themselves (Sections 2 and 17). What each Part adds is stated in Section 1.5.

1.3 Notation across the Parts

No symbol was renamed and no normalization was changed. Each Part keeps its source's letters, and a letter means what its own Part says it means. Table 2 lists the letters that the Parts use differently.

Table 2: [write] Letters used differently in the three Parts.

Letter	Part I (source 02)	Part II (source 10)	Part III (source 03)
n, N	n : state dimension; N : nilpotent shift, also N_T (erasing words of length T) and the input bit length in $b(N)$	N : number of game states; n : iteration count	N_i : number operators; N : index of the example H_N ; $n \in \mathbb{N}^m$: occupation tuple
K	common lift scale, $K \geq 2$	layer normalizer $K = K_L$, a power of two	K : halting set; K : compact remainder operator
D	common column sum $D = nK$	common probability denominator	$D = (B + 1)^m$: tuples in a truncation box
M	largest entry of $ \Delta_i $	pipeline period $M = L + 1$	number of zero states $\#Z$
T	word length or horizon; $T_i = p_i S_i$ labelled transitions	horizon	T_q : trace-class remainder
H	H_n : zero-mass hyperplane; H_i : integer restriction matrices	first halting time	the Hamiltonian
Q	PCP connector $Q = c\ell$; Q_T : full quartic	Q_T : complete quadratic	compiled quartic penalty
S, s	S_i : stochastic matrices; s_t : selector residuals	S : game operator; s : clock register	$s(n) = \sum_i n_i$: total occupation
E	coordinate map E	E_T : endpoint sum	energy level
L	guard L ; length of an erasing word	final circuit layer	minimum zero-state height
R	guard R	power of two in the Skolem lift	excited sum $R(q, t)$
P	cyclic permutation (Section 7)	integer transition numerators $P_{ia,j}$; chance-only matrix	source polynomial; P_0 : zero projection
η	contraction band	reset weight	requested width in the proof of Theorem 36.1
m, k	m source matrices, k letters	m control states, k counters	m modes, k source witnesses
q, t	q : probability of a block; t : time	t : time step	q : Boltzmann parameter; t : fugacity

Watch for these readings:

- **Erasure.** In Part I a word *erases* when its stochastic product has rank one: every input distribution is sent to one common output distribution, which need not be uniform (a *uniform* erasing word gives J_n). In Part II the *erasing map* F_{era} and the *erasing halt convention* send a halted counter configuration, and then the whole simulated pipeline vector, to the zero vector; the exact event observed there is attainment of the fixed point $\frac{1}{2}\mathbf{1}$, not rank one of a product. The two notions are unrelated constructions that happen to share a word.

- **Uniform.** Part I: a uniform erasing word (target J_n), uniform switching (equal letter probabilities), and the uniform-input quartic (matrix entries as parameters). Part II: uniformly mixing (every row of full support) and the common uniform reset. Part III: uniformly computable (one algorithm for every instance).
- **Contraction and confinement.** Part I contracts total variation by η per letter; Part II contracts the supremum norm by $\delta = 1/2$ and the span by $\delta(1 - \eta)$. Part III has no dynamics at all: its *confinement* is the spectral inequality $H > 0 \Rightarrow H \geq 1 + s$, and its diagonal evolution only changes phases (Section 37).
- **Certificate.** The certificates of Parts I and II are horizon-indexed: their arity grows with T . The Hamiltonian of Part III has a fixed number of modes for a fixed circuit. None of the three is a fixed-arity single-fold representation of a computably enumerable complete set.
- **Fugacity, thermal.** In Part III these are mathematical parameters of a positive trace; no physical reservoir, laboratory implementation or physical decision procedure is assumed (source 03’s own caveat).

1.4 Relation to neighbouring reports and to the repository’s reviews

This report sits in the collection’s `hilbert-tenth-problem` category beside four earlier reports and the signal-machine report written in the same batch. Its closest neighbour is the report on probabilistic, quantum and continuous computation [21], cited here by label names (it is a separate document):

- *Part II of [21] (quantum mortality).* Theorem `ppc:qm:thm:47` imports the undecidability of mortality for six integer 3×3 matrices and packs it into seven rational 4×4 Kraus operators, so that a zero-probability outcome word exists exactly when the source is mortal. Corollary 5.3 runs the same pipeline with a different exact predicate (rank one of a positive stochastic product) and Theorem 11.2 is the argument of `ppc:qm:cor:nobound`. Proposition `ppc:qm:prop:nonnegative` decides mortality for nonnegative matrices; Part I shows that this positivity boundary does not transfer to rank one (note after Corollary 5.3), which bears on Question `ppc:q:signed`.
- *Part V of [21].* Source 13’s witness-faithful quartic energy (`ppc:ql:thm:compiler`) is a bounded Boolean energy landscape over $\mathbb{R}$ and expressly treats no Gibbs distribution; Part III is the unbounded natural counterpart with a thermal trace. Source 17’s reversible chains have a uniform spectral gap and an undecidable exact stationary equality, the same slogan as Part II for a different object.
- *Part VII of [21].* Source 12 proves r.e.-complete exact extinction for a 2-adic 1/2-contraction (`ppc:ad:cor:complete`); source 14 a real piecewise-linear 1/2-contraction whose point-to-point reachability is decidable (`ppc:sc:thm:main` (iii)); source 15 the unsquared complementarity device and the degree-two unique-witness trace polynomial (`ppc:pm:lem:clamp`, `ppc:pm:thm:trace`). Part II is complementary to all three: a real, order-preserving 1/2-contraction whose exact fixed-point attainment is c.e.-complete because its clock is erased at halting (note after Section 17’s comparison), and its quadratic specializes source 15’s device to min/max gates. Part II also answers Part VII’s unlabelled question “Structural subclasses” negatively in part (note after the table of Section 23).
- *Parts VIII and IX of [21].* Corollary `ppc:tu:cor:fixedquartic` and its remark (a fixed

MRDP quartic whose witness multiplicity is uncontrolled) are the observation that Part III's spectral classification makes exact. Theorem `ppc:iq:thm:geometric` (geometric stopping clocks with undecidable records) is related only to Part I's zero-one law.

The report on canonical Diophantine certificates [22] supplies four results that the Parts reprove, or use without citation, and that are printed here as pointers or marked second routes, never as new: the unique-extension quadratization `cdc:wf:lem:quadratization` (and `cdc:cs:lem:quartic`), which is the core of Theorem 32.1 and of the fixed-arity quartic corollaries of Parts I and II; the absence of a computable witness-height bound `cdc:bd:prop:nobound`; the finite-fold substrate equivalence `cdc:cs:thm:finitfold`, the template of Theorem 34.4; and the positivity-restricted degree classification `cdc:of:thm:classification`, which shows that radial isolation below degree four is impossible for a universal family (note in Section 39). The reports *liveness-beyond-halting*, *group-theoretic-substrates* and *signal-machine-collision-certificates* have no overlap with this one.

The Hilbert's-tenth-problem research programme (read-only for this report) had already reviewed all three manuscripts before they were placed here; this report does not present its placement as a first review:

- [23] reviewed sources 02 and 03 (commit 44b28c395). It accepts source 02 within its stated domains, replays its suite and all eight certificates and adds independent checks; it reports no defect. It accepts source 03's mathematics for natural arithmetic DAGs and finds that the delivered low-level `Compiler/Atom` interface accepts invalid descriptors; it supplies the tested patch `thermal_exact_dag_guards.patch`. The shipped compiler here is the unpatched original (note in Section 38.5).
- [24] reviewed source 10 (commit 126028588). It accepts the mathematics and finds two implementation defects: the independent checker does not bind its scaled initial vector to the declared input, and an explicitly supplied integer scale in the optional counter loader is divided in floating point; it supplies the patches `exact_convergence_input_binding.patch` and `exact_convergence_integer_scale.patch`. The shipped programs here are the unpatched originals (notes in Section 26 and Appendix D).
- The note `matrix_pcp_trace.md` [25] uses the same classical radix-three encoding of Post correspondence tiles as Lemma 6.1, with a leading sentinel instead of Part I's fourth coordinate and idempotent connector.

The formal project. The report sits in the collection, not in `Computability/HilbertTenthProblem`, and placement beside a Lean/Rocq development confers no formal status. The three sources import MRDP only as a classical theorem; source 10 names the project's interface, the theorem `Diophantine.mrdp` with its variants `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` in `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean` (unchanged from source 10's pin to this write). The project has formalized none of this report's statements: it has no stochastic-matrix, stochastic-game, Bellman-operator or occupation-number Hamiltonian module.

1.5 Status and what is not claimed

The report claims conventional mathematical proofs, by its sources, of the results summarized in Section 1.1. What each source presents as its own contribution is narrower: source 02 the two-guard no-common-line compiler, its higher-dimensional subspace exclusion, the exact-forgetting interpretation and the counted certificate interface; source 10 the stochastic-game realization under

the simultaneous restrictions of Theorem 17.1 with its complete arithmetic interface; source 03 the radial factor $1 + \sum_i n_i$ with its effective spectral decomposition, the multiplicity classification and the critical-fugacity construction.

Every limitation stated by a source is kept in its Part. Collected here, the report does **not** claim:

- historical priority for any Part (all three sources say their searches were targeted), or any refereeing, proof-assistant verification, or rebuild of the repository’s Lean or Rocq developments; the finite checks (50,582, 26,077 and 7,936 assertions) illustrate and do not prove;
- an improvement of the repository’s 75-operation complete-certificate or 87-operation universal-polynomial records; no Part instantiates a numerical universal polynomial, alphabet or game;
- a finite-fold or single-fold Diophantine representation of computably enumerable sets; the bounded certificates of Parts I and II have horizon-dependent arity, and the fixed-arity MRDP quartics leave the original witness multiplicity uncontrolled;
- *Part I*: irreducibility (a common invariant hyperplane remains); one fixed universal stochastic alphabet with an ordinary input loader (the seven-state result is a family of alphabets parameterized by the input); many-one completeness of the eight-letter bound (only of the nine-state variable-alphabet family); anything about undecidable output entropy of hidden Markov models; optimality of the dimension, alphabet or certificate counts. Its headline answer to a MathOverflow question rests on an external discussion that is not archived in the repository;
- *Part II*: a numerical universal game (the shipped 128-state games are one-counter examples); hardness of ordinary value iteration from zero or of solving finite discounted games; a resolution of the Skolem problem or of the one-player Bellman reachability frontier; robustness to perturbed tables or floating point; novelty of the chance-only Skolem specialization (Vahanwala);
- *Part III*: a resolution of the finite-fold problem; physical hypercomputation or autonomous step-by-step dynamics; a numerical value of the mode count of the universal family; that t_* is a noncomputable real (it is computable; its exact endpoint is undecidable); geometric locality from four-mode support; robustness to coefficient errors.

Part I

Exact erasure without a common invariant line (source 02)

[write] Part I prints batch-78 manuscript 02, *Exact Erasure without a Common Invariant Line: Positive stochastic computation, tensor guards, and Diophantine certificates*, “Research report prepared for Vladimir Reshetnikov; Mathematical development and implementation: ChatGPT”, October 2, 2026. Its labels carry the prefix **ste:ee:**. Its abstract and status statement follow; its two appendices are Appendices A and B.

Abstract (source 02). We give an explicit reduction from integer matrix mortality to exact finite-time forgetting by strictly positive rational stochastic matrices. Two tensor-factor guards

remove every common invariant complex line without creating spurious zero products. From m matrices of dimension d , the construction produces $m+2$ matrices of dimension $2d+1$, with every entry arbitrarily close to the uniform stochastic matrix and a prescribed uniform total-variation contraction factor. A product has rank one precisely when its source-letter projection is mortal. Consequently, rank-one product existence is undecidable already for at most eight positive rational 7×7 column-stochastic matrices having no common invariant complex line. Clearing one common denominator gives the same conclusion for strictly positive integer matrices. This answers the specific no-common-invariant-line restriction raised in a June 2026 MathOverflow discussion; it does not establish irreducibility.

A separate, fully printed four-dimensional correspondence-problem compiler gives a many-one complete nine-state stochastic family with a variable-size alphabet. A higher tensor construction excludes all invariant subspaces of dimensions $1, \dots, r-1$. For independent full-support switching, existence of an erasing word is equivalent both to almost-sure finite exact forgetting and to finite expected forgetting time, despite deterministic exponential approximate forgetting on every instance. Explicit mass-compressed horizon- T natural-domain quartics have $T((n-1)^2 + k)$ witnesses and $T((n-1)^2 + 1) + (n-1)^2$ squared residuals, with exactly one witness per accepted labelled word. We also prove a sharp $n-1$ erasure horizon when the induced matrices on the zero-mass hyperplane commute, and exhibit arbitrarily small non-erasing perturbations preserving all common invariant subspaces. The package includes exact-arithmetic code, independent certificate replay, and 50,582 passing finite checks.

Status (source 02). Full mathematical proofs are supplied below. The new constructions are not proof-assistant verified or independently refereed. Established mortality, PCP, and DPRM results are cited as dependencies. The literature search is targeted, not a certification of global priority. No improvement to ProveIt’s fixed universal polynomial operation count is claimed.

2 Research question, contributions, and provenance

A stochastic matrix usually suggests information loss rather than universal computation. If every entry is positive and close to $1/n$, every step loses a large fraction of the distinction between two initial distributions. It is therefore tempting to expect that determining whether all distinction is ever lost exactly should be straightforward. The distinction between *small* and *zero*, however, changes the problem completely.

This report studies a finite alphabet $S_1, \dots, S_k$ of rational stochastic matrices. A word is an externally selected schedule. The question is whether some finite schedule makes the output distribution independent of the input distribution. Equivalently, does a matrix product have rank one? The control alphabet and rational coefficients are finite data; the word length is unbounded.

2.1 A concrete question that the construction answers

In a MathOverflow question posted on June 2, 2026, Victor Miller asked about the rank-one analogue of matrix mortality. Benjamin Steinberg observed that $A \mapsto A \oplus [1]$ transfers mortality to rank-one product existence. The subsequent discussion isolated the restriction that the generators have no common invariant line, and suggested an affine modification without a complete argument for that restriction [4]. We prove the missing restricted statement, with positivity and strong stochastic contraction added. ([write] The MathOverflow question and its June 3 discussion are external sources, not archived in the repository; that the construction answers their restriction is this source’s claim, not checked at the write.)

The answer has a necessary qualification. Every column-stochastic matrix preserves the hyperplane $\mathbf{1}^\top x = 0$. Our construction has no common invariant *line*, even over $\mathbb{C}$, but it is not an irreducible matrix family. It answers the stated line restriction, not a stronger irreducibility requirement that may arise in a particular application. It also does not settle any general question about computing hidden-Markov entropy.

2.2 Main results and their logical dependencies

The principal construction, Theorem 5.1, is elementary once a matrix mortality instance is supplied. Its central identity is

$$\text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}, \quad (1)$$

where π deletes the two guard letters. The guards are invertible on their private tensor factor, so inserting them anywhere cannot fabricate a mortal source computation. One guard removes possible shared eigenvectors in a transverse direction; an affine displacement in the other removes a common stationary line.

For the small numerical undecidability bounds, we import only the established mortality bounds of Cassaigne, Halava, Harju, and Nicolas [1]. Their conventions allow oracle Turing reductions, so we do not silently promote every numerical bound in their paper to a many-one completeness statement. Instead, Section 6 prints a separate PCP reduction. That route gives many-one completeness in dimension nine, with alphabet size part of the input. The classical halting-to-PCP dependency is standard and has also been treated by formal reduction proofs [2].

The Diophantine contribution is explicit at a fixed horizon. Its witness bijection is proved directly, rather than inferred from DPRM. The latter theorem is used only for the separate fixed-arity, unbounded representation in Section 11. Its use does not preserve a unique witness per unbounded computation.

2.3 Relationship to the inspected ProveIt material

The repository was inspected at the pinned commit

928ea97017a25ebe56d240c84f27d2275d818c75.

The relevant sources are its Hilbert-tenth-problem README, the October 2 substrate review, and the README of the combined probabilistic/quantum/continuous report [5, 6, 7]. These already contain extensive work on exact observations, contractions, matrix substrates, and bounded certificates. We do not present the generic claim “computably enumerable implies Diophantine” or “contraction can coexist with undecidable exactness” as new.

The proposed contribution is more specific: the two-guard no-invariant-line compiler, its higher-dimensional invariant-subspace exclusion, its exact stochastic forgetting interpretation, and the ensuing explicitly counted certificate interface. The targeted inspection did not amount to rereading every manuscript in the repository, and the public-source search is not an exhaustive priority audit.

The October 2 review emphasizes the difference between an external-horizon polynomial and one fixed polynomial accepting arbitrarily long computations [6]. We retain that distinction throughout. The review’s stated 87-operation fixed universal polynomial is not improved by any count in this report. State dimension, residual count, arithmetic-circuit size, and number of unbounded Diophantine witnesses are different resources.

[write] After the pin, the research programme reviewed this manuscript, byte for byte the same package [23]: it accepts the results within their stated domains and quantifiers, replays the suite and all eight certificates, adds independent tensor-rank, natural selector/prefix and signed-selector checks, and reports no defect. The collection’s report [21] has a parallel pipeline in its Part II (Theorem `qqc:qm:thm:47`: six integer 3×3 mortality matrices compiled into seven rational 4×4 Kraus operators, zero-probability word $\Leftrightarrow$ mortal); Corollary 5.3 below is its positive stochastic counterpart with a different exact predicate. Neither that report nor the research notes contain the no-common-line compiler, the idempotent PCP connector, the stochastic zero-one law or the compressed quartic. Section 1.4 gives the comparisons.

3 Exact forgetting and the zero-mass representation

All stochastic matrices in this article are *column-stochastic*. Probability distributions are columns, so $p \mapsto Sp$. We set $\mathbb{N} = \{0, 1, 2, \dots\}$. Matrices called positive have every entry strictly greater than zero. Integer matrix mortality allows negative entries in the source matrices.

For a word $w = i_1 \cdots i_T$ in chronological order, define

$$S_w = S_{i_T} \cdots S_{i_1}, \quad S_\epsilon = I.$$

The same convention applies to every matrix alphabet. A nonempty word is required for mortality and erasure.

Definition 3.1. An *erasing word* for $S_1, \dots, S_k$ is a nonempty word w for which $S_w p$ is the same distribution for every initial distribution p . A *uniform erasing word* satisfies $S_w = J_n$, where

$$u_n = \frac{1}{n} \mathbf{1}_n, \quad J_n = u_n \mathbf{1}_n^\top.$$

Lemma 3.2 (Equivalent erasure tests). *For a nonnegative column-stochastic $n \times n$ matrix S , the following are equivalent: S erases all initial distributions; all its columns are equal; $\text{rank } S = 1$; and S vanishes on*

$$H_n = \{x \in \mathbb{R}^n : \mathbf{1}_n^\top x = 0\}.$$

If S is also row-stochastic, these conditions are equivalent to $S = J_n$.

Proof. The images of the point-mass distributions are precisely the columns of S . Thus erasure is equivalent to equal columns. Equal columns have rank one, not zero, since their sums are one. Conversely, rank one makes the columns scalar multiples of a fixed nonzero vector; their equal nonzero column sums force all these multiples to be equal. The vectors $e_j - e_1$, $j > 1$, span H_n , giving the hyperplane condition. Finally, a matrix with equal columns has the form $p \mathbf{1}^\top$; its row sums are np_a , which are all one precisely when $p = u_n$. $\square$

Lemma 3.3 (Erasure persists). *If a word w is erasing, every word containing w as a contiguous block is erasing. Appending further letters after erasure can change the common output distribution, but cannot restore dependence on the initial distribution.*

Proof. Multiplication on either side cannot increase matrix rank. Every product remains column-stochastic and so has rank at least one. A product containing the rank-one factor S_w therefore has rank exactly one. $\square$

To separate total mass from information, let $h = n - 1$ and introduce

$$E = \begin{pmatrix} I_h \\ -\mathbf{1}_h^\top \end{pmatrix}, \quad F = [I_h \ 0] - \frac{1}{n} \mathbf{1}_h \mathbf{1}_n^\top, \quad F_0 = nF. \quad (2)$$

These are explicit rational coordinate maps, not existentially chosen bases.

Lemma 3.4 (Coordinates). *One has*

$$FE = I_h, \quad Fu_n = 0, \quad \mathbf{1}_n^\top E = 0, \quad EF = I_n - J_n.$$

Every $x \in \mathbb{Q}^n$ has the unique decomposition

$$x = Ey + su_n, \quad y = Fx, \quad s = \mathbf{1}_n^\top x.$$

For any rational column-stochastic S , its matrix in these coordinates is

$$\begin{pmatrix} C & b \\ 0 & 1 \end{pmatrix}, \quad C = FSE, \quad b = FSu_n, \quad (3)$$

and $\text{rank } S = 1 + \text{rank } C$.

Proof. The displayed identities follow by multiplying the matrices in (2). They prove existence and uniqueness of the decomposition. Column-stochasticity preserves s , while applying F to $S(Ey + su_n)$ gives $Cy + sb$. To compute the rank, subtract from each of the first h output rows its last-column entry times the last row. This leaves the block diagonal matrix $\text{diag}(C, 1)$ and hence $\text{rank } 1 + \text{rank } C$. $\square$

The same algebra holds over $\mathbb{C}$. We will use the complexified hyperplane and coordinates when excluding invariant lines. The distinction between a common invariant subspace and a common eigenvector is important: a one-dimensional invariant subspace is generated by a common eigenvector, but its eigenvalue may depend on the generator.

4 An explicit positive affine lift

Let (C_i, b_i) , $1 \leq i \leq k$, be finitely many integer affine data, with $C_i \in \mathbb{Z}^{h \times h}$ and $b_i \in \mathbb{Z}^h$. Fix a rational number $0 < \eta < 1$, chosen independently of the input coefficients. Define

$$\Delta_i = EC_i F_0 + nEb_i \mathbf{1}_n^\top, \quad (4)$$

$$M = \max_{i,a,b} |(\Delta_i)_{ab}|, \quad K = \max\{2, 1 + \lceil M/\eta \rceil\}, \quad D = nK, \quad (5)$$

$$B_i = K \mathbf{1}_n \mathbf{1}_n^\top + \Delta_i, \quad S_i = B_i/D. \quad (6)$$

Here B_i and D are integers. The same K and D are used for every generator.

Theorem 4.1 (Positive affine lift). *The matrices in (6) are positive rational column-stochastic and satisfy the strict entrywise bounds*

$$\frac{1 - \eta}{n} < (S_i)_{ab} < \frac{1 + \eta}{n}. \quad (7)$$

In the coordinates of Lemma 3.4,

$$S_i \sim \begin{pmatrix} K^{-1}C_i & K^{-1}b_i \\ 0 & 1 \end{pmatrix}. \quad (8)$$

For every word w of length T ,

$$\text{rank } S_w = 1 + \text{rank } C_w. \quad (9)$$

All entries of B_i are strictly positive integers and all its column sums equal D .

Proof. Since $K > M/\eta$, each entry of $\Delta_i/(nK)$ has absolute value less than η/n . Adding J_n proves (7) and positivity. The column sums of Δ_i vanish because $\mathbf{1}^\top E = 0$, so the column sums of S_i are one and those of B_i are D . Using $F_0 = nF$, rewrite the lift as

$$S_i = J_n + K^{-1}EC_iF + K^{-1}Eb_i\mathbf{1}_n^\top.$$

On $Ey + su_n$ this gives

$$S_i(Ey + su_n) = E(K^{-1}C_iy + K^{-1}b_is) + su_n,$$

which proves (8). Multiplying these upper block-triangular matrices gives a top-left block $K^{-T}C_w$. Its translation block need not vanish, but Lemma 3.4 shows that it does not affect the rank. Since $K \neq 0$, (9) follows. $\square$

Proposition 4.2 (Uniform approximate forgetting). *For all probability distributions p, q and every word w of length T ,*

$$\text{TV}(S_w p, S_w q) \leq \eta^T \text{TV}(p, q), \quad \text{TV}(p, q) = \frac{1}{2} \|p - q\|_1. \quad (10)$$

Proof. The lower bound in (7) allows the decomposition

$$S_i = (1 - \eta)J_n + \eta Q_i,$$

where Q_i is nonnegative and column-stochastic. Such a matrix is nonexpansive in ℓ^1 : for any real vector z ,

$$\|Q_i z\|_1 \leq \sum_a \sum_b (Q_i)_{ab} |z_b| = \|z\|_1.$$

For $z = p - q$ one has $J_n z = 0$, giving a factor η in one step. Iteration proves the result. $\square$

In particular, for a fixed accuracy $\varepsilon > 0$, a horizon with $\eta^T \leq \varepsilon$ works for every input instance and every schedule. This is a bound on the difference between two trajectories driven by the same schedule. When translations are present, it is *not* a claim that arbitrary switching converges to one common stationary distribution.

Corollary 4.3 (Doubly stochastic special case). *For $A_1, \dots, A_m \in \mathbb{Z}^{d \times d}$, apply the lift with $h = d$, $C_i = A_i$, and $b_i = 0$. Then S_i is doubly stochastic, and*

$$S_w = J_{d+1} + K^{-T}EA_wF. \quad (11)$$

Thus $A_w = 0$ if and only if S_w has rank one, if and only if $S_w = J_{d+1}$.

Proof. The zero translations imply $S_i u_n = u_n$, which is row-stochasticity. The block product has zero translation and top-left block $K^{-T}A_w$. Conjugating it back gives (11); the equivalences follow from $FE = I_d$. $\square$

The baseline lift alone is a direct transport of mortality, not the main no-line result: its generators share the stationary line $\mathbb{C}u_n$.

5 Two guards eliminate every common invariant line

The obstacle in the baseline is structural. Merely adding an affine term need not remove eigenvectors already shared by the original matrices inside H_n . We therefore reserve a tensor factor for two invertible guards.

Set

$$R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (12)$$

Given $A_1, \dots, A_m \in \mathbb{Z}^{d \times d}$, take $h = 2d$, $n = 2d + 1$, and use the following $m + 2$ affine data:

$$(C_i, b_i) = (A_i \otimes I_2, 0), \quad 1 \leq i \leq m, \quad (13)$$

$$(C_R, b_R) = (I_d \otimes R, 0), \quad (14)$$

$$(C_L, b_L) = (I_d \otimes L, e_1). \quad (15)$$

The vector e_1 in (15) is the first standard basis vector of $\mathbb{Z}^{2d}$. Apply the common-scale positive affine lift to all these pairs.

For a word w in the augmented alphabet, let $\pi(w)$ be its source-letter subword and $\gamma(w)$ its guard-letter subword. Both retain chronological order. An empty source subword has product I_d .

Theorem 5.1 (Line-free mortality compiler). *The construction (13)–(15) is effective and has the following properties.*

- (i) All $m + 2$ generators are positive rational column-stochastic matrices of size $2d + 1$, satisfying (7) and (10) for the prescribed η .
- (ii) The generators have no common invariant one-dimensional subspace of $\mathbb{C}^{2d+1}$.
- (iii) Every word w satisfies

$$C_w = A_{\pi(w)} \otimes G_{\gamma(w)}, \quad \text{rank } S_w = 1 + 2 \text{rank } A_{\pi(w)}, \quad (16)$$

where $G_{\gamma(w)}$ is a product of R, L and is invertible.

- (iv) The source family is mortal if and only if the target family has an erasing word, if and only if it has a uniform erasing word.

The same rank-one existence equivalence and no-common-line property hold for the positive integer numerators $B_1, \dots, B_{m+2}$.

Proof. The first assertion is Theorem 4.1 and Proposition 4.2.

For the tensor identity, each source linear part acts only on the first tensor factor and each guard only on the second. Thus source factors commute with guard factors. Moving source factors past guards, without changing the order within either subword, gives the first identity in (16). Both R and L have determinant one. Therefore $G_{\gamma(w)}$ is invertible, including $G_\epsilon = I_2$. Multiplying on the second factor by its inverse shows

$$\text{rank}(A_{\pi(w)} \otimes G_{\gamma(w)}) = \text{rank}(A_{\pi(w)} \otimes I_2) = 2 \text{rank } A_{\pi(w)}.$$

The affine rank formula now proves (16).

We next rule out a common invariant complex line ℓ . There are two cases. If $\ell \subseteq H_n^{\mathbb{C}}$, use the E coordinates. A nonzero vector spanning it would be an eigenvector of both $I_d \otimes R$ and $I_d \otimes L$; scaling both by K^{-1} does not affect their eigenvectors. The only eigenvalue of each unipotent guard is one. Their eigenspaces are respectively

$$\mathbb{C}^d \otimes \text{span}(e_1) \quad \text{and} \quad \mathbb{C}^d \otimes \text{span}(e_2),$$

whose intersection is zero. This is impossible.

If $\ell \not\subseteq H_n^{\mathbb{C}}$, choose $v \in \ell$ with $\mathbf{1}^T v = 1$. Every column-stochastic generator acts on v by its line eigenvalue, and preservation of total mass forces that eigenvalue to be one. Write $v = Ey + u_n$. The unshifted R guard gives

$$y = K^{-1}(I_d \otimes R)y.$$

The matrix on the right has only eigenvalue $K^{-1} \neq 1$, since $K \geq 2$. Consequently $y = 0$ and $v = u_n$. But the shifted L guard sends

$$u_n \mapsto u_n + K^{-1}Ee_1 \neq u_n,$$

contradicting invariance. This proves (ii) over $\mathbb{C}$, not just over $\mathbb{Q}$.

If $A_v = 0$ for a nonempty source word v , use the same word without guards. Its affine translations are zero, so its lifted product is exactly J_n . Conversely, if S_w has rank one, (16) gives $A_{\pi(w)} = 0$. The projection cannot be empty, since $A_e = I_d \neq 0$. This establishes (iv).

Finally, each $B_i = DS_i$ with a common nonzero scalar D . Scaling a generator by a nonzero scalar preserves all its invariant subspaces, and $B_w = D^{|w|}S_w$ preserves every product rank. The integer conclusion follows. $\square$

Remark 5.2 (Why the guard arguments must both be present). The two unipotent guards without an affine displacement would still fix u_n . An affine displacement without a protected tensor factor could leave an invariant line inside the information hyperplane. Our two cases address these independent obstructions. Invertibility of the guards is equally essential: a singular guard could destroy source information and introduce false erasing words.

5.1 Small numerical consequences

Cassaigne et al. prove undecidability of mortality for the four parameter pairs $(d, m) = (3, 6), (5, 4), (9, 3), (15, 2)$ [1]. Applying the two explicit lifts gives Table 3. The entries are sufficient bounds transported from that paper, not a claim to list an optimal or exhaustive current frontier. ([write] Part II of [21] imports the same six-matrix 3×3 bound from Neary [20, Corollary 12], through binary tag systems and the Post correspondence problem for five pairs of words. Both attributions are kept; this report does not decide which publication first proved the $(3, 6)$ bound.)

Source mortality (d, m)	Doubly stochastic (n, k)	No common line (n, k)
(3, 6)	(4, 6)	(7, 8)
(5, 4)	(6, 4)	(11, 6)
(9, 3)	(10, 3)	(19, 5)
(15, 2)	(16, 2)	(31, 4)

Table 3: Undecidable exact-erasure families. Generator counts are upper bounds. All target entries may be confined to any prescribed band $((1 - \eta)/n, (1 + \eta)/n)$ with rational $0 < \eta < 1$.

Corollary 5.3. *For any fixed rational $0 < \eta < 1$, it is undecidable whether a finite product of at most eight positive rational 7×7 column-stochastic matrices has rank one, even under both promises that the entrywise band (7) holds and that the generators have no common invariant complex line. The corresponding rank-one problem for positive integer matrices with one common positive column sum is also undecidable.*

Proof. Apply Theorem 5.1 to the (3,6) mortality family. An algorithm on the promised target class would decide every source instance, since the compiler always meets the promises and preserves the answer. $\square$

This is a family of finite alphabets parameterized by the source input. It is not a claim that the displayed three-state example below, or one fixed numerical eight-generator alphabet, simulates every program. Rational precision is unbounded across input instances; fixing all numerator and denominator bounds would leave only finitely many instances.

[write] *Relation to the nonnegative-mortality boundary.* Part II of [21] proves that mortality is decidable for entrywise nonnegative matrices (Proposition `pqc:qm:prop:nonnegative`) and asks, in Question `pqc:q:signed`, for further decidable classes, among them families with a common invariant cone. The generators of Corollary 5.3 are strictly positive and share the invariant cone of nonnegative vectors, yet their rank-one problem is undecidable: the positivity boundary for the zero predicate does not transfer to rank one, because rank one in dimension n is mortality of the signed induced matrices in dimension $n - 1$ (Proposition 8.1).

6 A direct correspondence-problem compiler

To separate completeness from the numerical mortality bounds, we now give a standalone many-one reduction. The starting problem is PCP: for a finite list of pairs (u_i, v_i) , does some nonempty index word satisfy

$$u_{i_1} \cdots u_{i_T} = v_{i_1} \cdots v_{i_T}?$$

We use its classical computably enumerable completeness. The halting-to-PCP reduction is an external dependency, not re-proved here; see the formal treatment of PCP-related reductions by Forster, Heiter, and Smolka [2]. Restriction to a two-letter alphabet does not lose generality: encode a finite alphabet injectively by equal-length binary codewords, which preserves equality of all concatenations.

6.1 A four-dimensional integer front end

Use letters a, b as digits 1, 2 in base three, read most significant digit first. Write $\text{val}(s)$ for the resulting nonnegative integer, including $\text{val}(\epsilon) = 0$. This encoding is injective, and

$$\text{val}(st) = 3^{|t|} \text{val}(s) + \text{val}(t). \quad (17)$$

Define

$$M_i = \begin{pmatrix} 3^{|u_i|} & 0 & \text{val}(u_i) & 0 \\ 0 & 3^{|v_i|} & \text{val}(v_i) & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \ell = \begin{pmatrix} 1 & -1 & 0 & 1 \end{pmatrix}, \quad Q = c\ell. \quad (18)$$

The letter q denotes the extra generator Q . It is deliberately idempotent: $\ell c = 1$ and $Q^2 = Q \neq 0$.

Lemma 6.1 (PCP block identity). *For a nonempty source word $v = i_1 \cdots i_T$, let $U_v = u_{i_1} \cdots u_{i_T}$ and $V_v = v_{i_1} \cdots v_{i_T}$. Then*

$$M_v = \begin{pmatrix} 3^{|U_v|} & 0 & \text{val}(U_v) & 0 \\ 0 & 3^{|V_v|} & \text{val}(V_v) & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \ell M_v c = \text{val}(U_v) - \text{val}(V_v).$$

The corresponding scalar for the empty word is $\ell I_4 c = 1$, not zero.

Proof. The product convention is chronological. Multiplying a later tile on the left updates the first translation by $3^{|u_i|} \text{val}(U) + \text{val}(u_i) = \text{val}(Uu_i)$, and similarly for the second. Induction proves the matrix formula. The scalar follows by direct multiplication. For the empty word use $M_\epsilon = I_4$; the block formula with a zero final diagonal entry is asserted only for a nonempty word. $\square$

Theorem 6.2 (Full-word PCP reduction). *The integer alphabet $M_1, \dots, M_m, Q$ is mortal if and only if the PCP instance has a solution. More precisely, write any word containing $s \geq 1$ connector letters as*

$$w = v_0 q v_1 q \cdots q v_s,$$

where each v_j is a possibly empty source word. Its product is zero if and only if at least one internal block v_j , $1 \leq j < s$, is a nonempty PCP solution. Words without a connector never have zero product.

Proof. By $Q = c\ell$ and the chronological convention,

$$M_w = (M_{v_s} c) \left(\prod_{j=1}^{s-1} \ell M_{v_j} c \right) (\ell M_{v_0}). \quad (19)$$

Both endpoint vectors are nonzero. For a nonempty v_s , the third coordinate of $M_{v_s} c$ is one; for an empty block it is the nonzero vector c . For nonempty v_0 , the first coordinate of ℓM_{v_0} is $3^{|U_{v_0}|} > 0$; for an empty block it is $\ell \neq 0$. Thus the outer product in (19) is nonzero, and vanishing occurs exactly when an internal scalar is zero. Lemma 6.1 identifies those zero scalars with nonempty PCP matches; an empty internal block contributes one. Without Q , the third diagonal entry of every nonempty product is one, excluding mortality.

Conversely, a PCP match v gives the zero word qvq . The entire map from tiles to matrices is an explicit computable map, so this is a many-one reduction. $\square$

The fourth coordinate has a specific purpose. A more naive connector based on a scalar comparison that vanishes for the empty word would have $Q^2 = 0$, making every instance mortal. Here the extra coordinate is killed by the first genuine tile but makes the empty-block scalar one. This is a boundary condition, not a cosmetic padding coordinate.

[write] The three-dimensional radix-three tiles are the classical encoding that the research note [25] also uses, there with a leading sentinel digit so that the empty word has a nonzero code. The fourth coordinate and the idempotent connector $Q = c\ell$, $\ell c = 1$, are this source's own.

Corollary 6.3 (A many-one complete stochastic substrate). *For any fixed rational $0 < \eta < 1$, exact erasure is computably enumerable complete under many-one reductions for positive rational 9×9 column-stochastic families with a variable-size alphabet, even with no common invariant complex line and with the entrywise band (7).*

Proof. A PCP instance with m tiles maps by Theorem 6.2 to $m + 1$ integer 4×4 matrices, then by Theorem 5.1 to $m + 3$ stochastic 9×9 matrices. Both maps are many-one and preserve the exact answer. Erasure is computably enumerable: enumerate all finite words and multiply their rational matrices exactly. This proves completeness on the promised class. Equivalently, one may regard the source tile list as the input parameter of the compiled family, so promise validation is not part of the reduction. $\square$

This gives a precise sense in which the substrate supports universal finite computation. A halting instance can be transformed into a finite rational gate alphabet such that a finite exact-forgetting schedule exists exactly when the computation halts. It does not supply an autonomous universal stochastic machine with one fixed gate alphabet and a separate ordinary input loader.

7 Excluding all small common invariant subspaces

The tensor guard method is not limited to lines. Let $r \geq 2$, and let P be the cyclic permutation matrix on $\mathbb{C}^r$, while

$$Z_r = \text{diag}(1, 2, \dots, r).$$

For an integer mortality instance of dimension d , use $h = rd$, $n = rd + 1$, and

$$(A_i \otimes I_r, 0), \quad (I_d \otimes P, 0), \quad (I_d \otimes Z_r, e_1). \quad (20)$$

Apply the same common-scale lift. The name Z_r here denotes a diagonal matrix, not an integer witness domain.

Lemma 7.1 (The guard algebra). *The complex unital algebra generated by P and Z_r is $\text{Mat}_r(\mathbb{C})$. Every subspace of $\mathbb{C}^d \otimes \mathbb{C}^r$ invariant under $I_d \otimes P$ and $I_d \otimes Z_r$ has the form $W \otimes \mathbb{C}^r$ for some $W \leq \mathbb{C}^d$.*

Proof. Since the diagonal entries of Z_r are distinct, Lagrange interpolation produces the coordinate projections:

$$E_{jj} = \prod_{a \neq j} \frac{Z_r - aI}{j - a}.$$

For each i, j choose a power P^t taking e_j to e_i . Then $E_{ii}P^tE_{jj} = E_{ij}$, so the generated algebra contains every matrix unit.

A subspace invariant under the two generators is invariant under their products and complex linear combinations, hence under all $I_d \otimes E_{ij}$. Define $W = \{v : v \otimes e_1 \text{ belongs to the subspace}\}$. Applying $I_d \otimes E_{1j}$ to $\sum_j v_j \otimes e_j$ shows that every coefficient v_j belongs to W . Conversely, applying $I_d \otimes E_{j1}$ shows that $W \otimes \mathbb{C}^r$ is contained in the subspace. This proves equality. $\square$

Theorem 7.2 (Invariant-subspace and product-rank gaps). *For the lift of (20), every common invariant complex subspace has dimension congruent to 0 or 1 modulo r . There is no common invariant line and, consequently, no nonzero common invariant subspace of any dimension $1, \dots, r-1$. Every product satisfies*

$$\text{rank } S_w = 1 + r \text{ rank } A_{\pi(w)}. \quad (21)$$

In particular, a product is either rank one or has rank at least $r+1$. Mortality, erasure, and existence of a uniform erasing word remain equivalent.

Proof. Both P and Z_r are invertible. The source/guard tensor factorization therefore proves (21) exactly as before. Zero translations on source-only words give the uniform-target equivalence.

Let U be a common invariant subspace. Its intersection with H_n^C is invariant under the linear guard restrictions and has dimension divisible by r , by Lemma 7.1. If $U \subseteq H_n^C$, this is its full dimension. Otherwise, restriction of the nonzero mass functional to U has a one-dimensional image, so $\dim U = \dim(U \cap H_n^C) + 1$. This proves the congruence statement.

A common invariant line cannot be inside H_n^C because its dimension would be a positive multiple of r . Outside H_n^C normalize a spanning vector to mass one. The unshifted P guard forces its transverse part y to satisfy $y = K^{-1}(I_d \otimes P)y$. Every eigenvalue of P has modulus one, whereas $K > 1$, so $y = 0$. The translated Z_r guard does not fix u_n , again a contradiction. Excluding dimension one and using the congruence statement excludes all positive dimensions smaller than r . $\square$

The rank gap makes the exactness phenomenon sharper: products cannot drift through ranks $2, \dots, r$ on their way to erasure. Their entries can nevertheless approach a rank-one matrix arbitrarily closely without ever becoming rank one. The common hyperplane of dimension rd remains, so the theorem must not be reported as an irreducible-semigroup result.

For example, applying this construction to the $d = 3$, $m \leq 6$ mortality bound gives undecidable erasure for at most eight positive stochastic matrices in dimension $3r + 1$, with no invariant subspace of dimensions $1, \dots, r - 1$. Applying it to the explicit PCP front end gives many-one completeness in dimension $4r + 1$ with variable alphabet size.

8 Decidable boundaries and a sharp commuting horizon

8.1 An exact dimension correspondence

Proposition 8.1. *For $n \geq 2$, rank-one product existence for rational column-stochastic $n \times n$ matrices reduces effectively, without changing the number of generators, to mortality for integer $(n - 1) \times (n - 1)$ matrices. Conversely, integer mortality in dimension $n - 1$ reduces to erasure for positive rational doubly stochastic matrices of dimension n , in any prescribed band (7).*

Proof. For the first direction, compute $C_i = FS_iE$ using the explicit rational coordinates of Lemma 3.4. The restriction of S_w to H_n has coordinate matrix C_w , so erasure is equivalent to $C_w = 0$. Clear the rational denominators in each C_i , or use one common positive denominator. Every product is merely multiplied by a nonzero scalar, preserving zero products. The converse is Corollary 4.3. $\square$

This correspondence is useful independently of particular known dimension bounds. For example, the three-state positive doubly stochastic erasure problem has exactly the corresponding two-dimensional mortality difficulty. We do not infer a current open/closed classification of two-dimensional mortality from a dated source; the reduction equivalence itself is the assertion here.

8.2 Commuting information dynamics

The affine stochastic generators need not commute in the next theorem. It is enough that their restrictions to the zero-mass hyperplane commute.

Theorem 8.2 (Sharp commuting horizon). *Let $S_1, \dots, S_k$ be rational column-stochastic $n \times n$ matrices, $n \geq 2$. Suppose that the induced $(n - 1)$ -dimensional matrices $C_i = FS_iE$ commute pairwise. If an erasing word exists, one of length at most $n - 1$ exists. A finite exact rank-reduction algorithm finds such a word or certifies that no erasing word exists. The bound $n - 1$ is sharp, even for a single positive doubly stochastic generator.*

Proof. Set $h = n - 1$ and start with $P = I_h$, $V = \text{im } P = \mathbb{Q}^h$. Since P is a product of commuting C_i , every C_i leaves V invariant. If some generator satisfies $\text{rank}(C_i P) < \text{rank } P$, replace P by $C_i P$, append that letter, and continue. Each such step strictly decreases the nonnegative integer $\dim V$, so at most h steps are possible.

If $P = 0$, the chosen word is erasing. Otherwise suppose that no generator lowers the rank. Every $C_i : V \rightarrow V$ then has full rank on V and is invertible there. All products remain invertible on this nonzero space, so none can be the zero matrix on $\mathbb{Q}^h$. Thus failure to find a further rank-lowering generator is a correct no-erasure certificate. Conversely, if any zero product exists it annihilates V , which is impossible while all $C_i|_V$ are invertible. Hence the algorithm cannot stop at a nonzero V in a mortal instance.

For sharpness, take the $h \times h$ nilpotent shift N with ones immediately above the diagonal. It has $N^h = 0$ and $N^t \neq 0$ for $0 \leq t < h$. The single-generator positive doubly stochastic lift of N therefore first erases at time $h = n - 1$. $\square$

The algorithm uses rational matrix multiplication and Gaussian rank tests; it has no unbounded word search. The proof does not assume that a rank reduction already makes a matrix zero, and it allows arbitrary choices among available rank-lowering generators. The common-invariant-image property provided by commutation is what makes these greedy choices safe.

For one generator, erasure is exactly nilpotency of C , and $C^{n-1} = 0$ is a complete test. For $n = 2$, every induced C_i is a scalar, so erasure occurs if and only if at least one generator already has rank one. These are genuine finite boundaries, not numerical heuristics for the undecidable cases.

9 Random schedules, exact forgetting, and a hidden-state interpretation

Now choose each letter independently with probabilities $p_1, \dots, p_k > 0$, $\sum_i p_i = 1$. Define the exact-forgetting time

$$\tau = \inf\{T \geq 1 : \text{rank}(S_{I_T} \cdots S_{I_1}) = 1\},$$

with $\inf \emptyset = \infty$. The event concerns exact equality, not a numerical distance threshold.

Theorem 9.1 (A zero-one law for finite exact forgetting). *For a finite column-stochastic alphabet under independent full-support switching, the following conditions are equivalent:*

- (i) *an erasing word exists;*
- (ii) $\mathbb{P}(\tau < \infty) > 0$;
- (iii) $\mathbb{P}(\tau < \infty) = 1$;
- (iv) $\mathbb{E}\tau < \infty$.

If v is an erasing word of length L and $q = \prod_{j=1}^L p_{v_j} > 0$, then

$$\mathbb{P}(\tau > t) \leq (1 - q)^{\lfloor t/L \rfloor}, \quad \mathbb{E}\tau \leq L/q. \quad (22)$$

For uniform letter probabilities, $q = k^{-L}$.

Proof. If no erasing word exists, no finite prefix can have rank one, so $\tau = \infty$ for every schedule. Suppose instead that an erasing word v of length L exists. Partition the random schedule into successive disjoint blocks of length L . Each block equals v with probability q , independently of the others. The first matching block occurs after a geometric number G of blocks, with $\mathbb{E}G = 1/q$. By Lemma 3.3, regardless of the preceding prefix, that block makes the entire prefix product rank one. Hence $\tau \leq LG$, which proves (22) and finite expectation. Finite expectation implies almost-sure finiteness, and almost-sure finiteness implies positive probability. These implications establish all equivalences. $\square$

Corollary 9.2. *On the seven-state promised class of Corollary 5.3, both almost-sure finite exact forgetting and finiteness of its expected time are undecidable. They are many-one complete on the variable-alphabet nine-state family of Corollary 6.3. Nevertheless, the deterministic approximation bound (10) holds for every schedule on every instance.*

Proof. Theorem 9.1 identifies each of these random-schedule predicates with existence of a finite erasing word. The reductions and contraction estimates already proved therefore apply unchanged. $\square$

There is no contradiction with the higher logical complexity of many other almost-sure termination questions. Here a single finite erasing block is an absorbing certificate that recurs in independent full-support sampling. That special structure collapses the almost-sure predicate to an existential finite-word predicate. Removing independence, full support, or the absorbing rank-one property changes the argument.

[write] Part IX of [21] (Theorem `pqc:iq:thm:geometric`) also combines a geometric stopping clock with an undecidable exact event, for completed records of rational quantum instruments; the statements are different and neither implies the other.

9.1 A labelled hidden-state model with trivial output statistics

The stochastic interpretation can be made into a finite labelled transition model. For label i , use the nonnegative transition matrix

$$T_i = p_i S_i.$$

Since $\sum_i T_i$ is column-stochastic, these matrices specify joint probabilities of the next hidden state and the emitted label. Conditional on any current hidden state, label i has probability p_i , because $\mathbf{1}^\top T_i = p_i \mathbf{1}^\top$. Thus the emitted labels are independent with the prescribed common distribution.

After observing a word $w = i_1 \cdots i_T$, its probability is $\prod_t p_{i_t}$, independent of the initial hidden distribution. The normalized hidden-state update is exactly S_w . Consequently, exact independence of the posterior from the initial hidden distribution is precisely rank-one product existence.

The output entropy rate in this particular construction is simply $-\sum_i p_i \log p_i$. It is not an undecidable quantity. This observation is important: even when output statistics are completely elementary, deciding whether filtering ever forgets its initial distribution *exactly* can be undecidable.

The result addresses exact filtering, not the entropy question that motivated the MathOverflow discussion.

9.2 Finite-horizon probabilities

Under uniform switching, let N_T be the number of labelled words of length T with rank-one product. By persistence,

$$\mathbb{P}(\tau \leq T) = N_T/k^T. \quad (23)$$

This is an exact rational number computable by finite enumeration. The next section realizes N_T as the number of natural solutions of an explicit quartic. Duplicate numerical generators, when present, remain distinct labels; this matches the probability model and the certificate count.

10 Explicit natural-domain Diophantine certificates

Fix a numerical input alphabet $B_1, \dots, B_k \in \mathbb{Z}_{>0}^{n \times n}$, each with column sum $D > 0$, and set $S_i = B_i/D$. In this section $T \geq 1$ is an externally supplied horizon. A polynomial is produced for that horizon; its arity is not fixed as T varies.

10.1 A full-prefix certificate with no minors

Introduce natural selector variables $e_{t,i}$, $1 \leq t \leq T$, $1 \leq i \leq k$, and natural prefix matrices $X_t \in \mathbb{N}^{n \times n}$. The matrix $X_0 = I_n$ is constant. Use residuals

$$s_t = \sum_{i=1}^k e_{t,i} - 1, \quad (24)$$

$$r_{t,ab} = (X_t)_{ab} - \sum_{i=1}^k e_{t,i} \sum_{c=1}^n (B_i)_{ac} (X_{t-1})_{cb}, \quad (25)$$

$$q_{ab} = (X_T)_{ab} - (X_T)_{a1}, \quad 1 \leq a \leq n, \quad 2 \leq b \leq n. \quad (26)$$

Define the integer polynomial

$$\mathcal{Q}_T = \sum_{t=1}^T s_t^2 + \sum_{t=1}^T \sum_{a,b=1}^n r_{t,ab}^2 + \sum_{a=1}^n \sum_{b=2}^n q_{ab}^2. \quad (27)$$

Theorem 10.1 (Word-bijective full certificate). *The polynomial $\mathcal{Q}_T$ has total degree at most four in its natural witnesses. Its natural zero set is in bijection with the labelled erasing words of length T . It has exactly*

$$T(n^2 + k) \quad \text{witness variables,} \quad T(n^2 + 1) + n(n - 1) \quad \text{squared residuals.} \quad (28)$$

Proof. A sum of integer squares vanishes exactly when every residual vanishes. Natural selectors with sum one have exactly one entry equal to one and every other entry zero. No separate Boolean equation is needed. Let i_t be that unique selected letter. The prefix equations then force, successively,

$$X_t = B_{i_t} \cdots B_{i_1}.$$

All these entries are natural, and each column sum equals D^t . Thus the terminal equations say precisely that the stochastic matrix X_T/D^T has equal columns, which is equivalent to erasure by Lemma 3.2.

Conversely, each erasing word supplies those one-hot selectors and exact prefix products, and all residuals vanish. The selectors and the recurrence determine every witness uniquely. Different labelled words have different selector tuples, even when some numerical generators coincide. This proves the bijection. The prefix residuals have degree at most two because B_i are fixed numerical coefficients, and all other residuals are linear. The displayed counts simply count selectors, matrix entries, and equations. $\square$

The terminal test uses equal columns rather than 2×2 minors. It is sound because the common, nonzero column sum has already been established by the retained recurrence. If the target is specifically J_n , replace (26) by

$$(X_T)_{ab} - (X_T)_{11} = 0 \quad ((a, b) \neq (1, 1)).$$

This uses $n^2 - 1$ terminal residuals. It is unnecessary in the doubly stochastic case, but it distinguishes a general rank-one product from the uniform matrix in the line-free construction.

10.2 A smaller certificate in information coordinates

Mass conservation allows a substantial further reduction. Put $h = n - 1$ and compute the integer restriction matrices

$$(H_i)_{ac} = (B_i)_{ac} - (B_i)_{an}, \quad 1 \leq a, c \leq h. \quad (29)$$

These may have negative entries. They are the coordinates of the action of B_i on the zero-mass hyperplane: $B_i E = E H_i$. For a prefix product X_t ,

$$(H_{i_t} \cdots H_{i_1})_{ab} = (X_t)_{ab} - (X_t)_{an}. \quad (30)$$

Each entry on the right lies between $-D^t$ and D^t , since both nonnegative entries belong to columns of sum D^t .

Introduce natural matrices $Y_t \in \mathbb{N}^{h \times h}$ and use the fixed shifts

$$p_t = D^t \quad (0 \leq t \leq T), \quad Y_0 = I_h + \mathbf{1}_h \mathbf{1}_h^T.$$

The intended value is

$$(Y_t)_{ab} = (H_{i_t} \cdots H_{i_1})_{ab} + p_t.$$

Retain the selector residuals s_t and define

$$\rho_{t,ab} = (Y_t)_{ab} - p_t - \sum_{i=1}^k e_{t,i} \sum_{c=1}^h (H_i)_{ac} ((Y_{t-1})_{cb} - p_{t-1}), \quad (31)$$

$$\theta_{ab} = (Y_T)_{ab} - p_T. \quad (32)$$

Let

$$\mathcal{I}_T = \sum_t s_t^2 + \sum_{t,a,b} \rho_{t,ab}^2 + \sum_{a,b} \theta_{ab}^2. \quad (33)$$

Theorem 10.2 (Mass-compressed natural quartic). *The natural zeros of $\mathcal{I}_T$ are in bijection with the labelled erasing words of length T . Every residual has degree at most two, and the polynomial has degree at most four. Its exact counts are*

$$\boxed{T((n-1)^2 + k) \text{ witnesses,} \quad T((n-1)^2 + 1) + (n-1)^2 \text{ residuals.}} \quad (34)$$

There is a canonical bijection between the natural zero sets of $\mathcal{Q}_T$ and $\mathcal{I}_T$, preserving the selected labelled word.

Proof. The selectors again determine a unique word. Write $Z_t = Y_t - p_t \mathbf{1}_h \mathbf{1}_h^\top$. Equation (31) becomes $Z_t = H_{i_t} Z_{t-1}$, with $Z_0 = I_h$. Hence every Y_t is forced uniquely. The intended values are natural because (30) bounds each entry of Z_t below by $-D^t$. Thus no additional range witnesses or nonnegativity slack equations are needed.

The terminal residuals are zero exactly when $Z_T = 0$, which means that the full product vanishes on the zero-mass hyperplane. Lemma 3.2 identifies this with rank-one product existence. This proves both directions and the word-bijection. The counts and degree bounds follow directly from the formulas.

For the zero-set bijection, keep the selectors and map a full prefix certificate to the compressed one by

$$(Y_t)_{ab} = (X_t)_{ab} - (X_t)_{an} + D^t.$$

On valid zeros these values are natural. Conversely, the selectors in a compressed zero uniquely reconstruct the full prefix products. Both directions preserve the word and are inverse on the zero sets. $\square$

This is a bijection on the valid natural zero sets, not an identity between the two polynomials away from zero. The map from arbitrary full witness assignments need not preserve nonnegativity; only its zero-set behavior is asserted. The distinction matches the source-audit discipline emphasized in ProveIt’s October 2 review [6].

For the seven-state, eight-letter construction, the full certificate uses $57T$ witnesses and $50T + 42$ residuals. The compressed certificate uses

$$\boxed{44T \text{ natural witnesses and } 37T + 36 \text{ residuals.}} \quad (35)$$

The improvement is $13T$ witnesses and $13T + 6$ residuals. No assertion about an optimal arithmetic circuit or a smallest possible polynomial follows from these ledgers.

The constants $p_t = D^t$ are computed for the supplied horizon. Their bit lengths are $O(t \log D)$, and printing all of them has a horizon-dependent cost. They are not an uncharged fixed-arity exponentiation primitive. The next subsection makes the uniform-input version polynomial without treating those powers as free input constants.

10.3 When the numerical matrices become polynomial parameters

A degree statement must specify whether matrix entries are coefficients or variables. In (25), treating $(B_i)_{ac}$ as polynomial input parameters turns a term $e_{t,i}(B_i)_{ac}(X_{t-1})_{cb}$ into a cubic. It would therefore be incorrect to reuse the full-prefix quartic degree claim unchanged in that setting.

Here is a fully uniform compressed alternative for fixed n, k, T . Let the positive entries of B_i and the common column sum D be input parameters, subject to the stated stochastic-numerator

promise. Introduce natural variables $z_t, W_t \in \mathbb{N}^{h \times h}$, and $Y_t \in \mathbb{N}^{h \times h}$ at each step, in addition to the selectors. Set $z_0 = 1$ and the same constant Y_0 as before. Use the residuals

$$z_t - Dz_{t-1} = 0, \quad (36)$$

$$(W_t)_{ac} - \sum_i e_{t,i}((B_i)_{ac} + D - (B_i)_{an}) = 0, \quad (37)$$

$$(Y_t)_{ab} - z_t - \sum_c ((W_t)_{ac} - D)((Y_{t-1})_{cb} - z_{t-1}) = 0, \quad (38)$$

along with the selector equations and terminal equations $(Y_T)_{ab} - z_T = 0$. Square and sum all these residuals.

Proposition 10.3 (Uniform-input quartic). *On the positive, common-column-sum input domain, the resulting polynomial has total degree at most four, including its matrix and denominator input parameters. It has exactly*

$$T(k + 2(n-1)^2 + 1) \quad \text{natural witnesses} \quad (39)$$

and

$$T(2(n-1)^2 + 2) + (n-1)^2 \quad \text{squared residuals.} \quad (40)$$

Its natural zeros are again in bijection with the labelled erasing words of length T .

Proof. Every displayed residual is quadratic or linear in all inputs and witnesses. One-hot selectors force $z_t = D^t$ and

$$(W_t)_{ac} = (H_{i_t})_{ac} + D.$$

Since $0 \leq (B_i)_{ab} \leq D$, these selected values are natural. The Y recurrence then reduces exactly to (31), so the preceding proof supplies uniqueness and naturalness of the remaining witnesses. There are $k + 2h^2 + 1$ variables and $2h^2 + 2$ residuals per step, and h^2 terminal residuals, yielding the counts. $\square$

The input promises themselves are decidable finite equalities and inequalities. They are not counted as witnesses in (39). To make a polynomial whose accepted inputs also enforce these promises, include their finite arithmetic conditions in the input-domain representation; one should then charge those additional conditions separately. This proposition supplies the uniform schema and its proof. The shipped numerical certificate evaluators implement Theorems 10.1 and 10.2, not an expanded symbolic version of Proposition 10.3.

10.4 Exact counts of solutions and limits of uniqueness

The two numerical quartics have at most k^T natural zeros, exactly one per erasing labelled word. Combining the bijection with (23) gives

$$\mathbb{P}(\tau \leq T) = \frac{\#\{x \in \mathbb{N}^{T((n-1)^2+k)} : \mathcal{I}_T(x) = 0\}}{k^T}. \quad (41)$$

This identity is about a finite solution set for one externally fixed horizon. It is not a finite-fold Diophantine representation of arbitrary computably enumerable sets. A different polynomial is constructed for every T , and an unbounded union over those polynomials is not automatically one fixed-arity finite-fold polynomial.

11 Fixed unbounded Diophantine representations and impossibility of bounds

11.1 What DPRM supplies, and what it does not

Fix an effective encoding $e \in \mathbb{N}$ of finite rational stochastic alphabets. Validity of the finite rational data, positivity, and common-column-sum claims is decidable. The set of valid codes possessing an erasing word is computably enumerable by exact word enumeration. The Davis–Putnam–Robinson–Matiyasevich (DPRM) theorem identifies computably enumerable sets with Diophantine sets [3]. Hence there is an integer polynomial P and a fixed number of natural witnesses such that

$$e \text{ admits exact erasure} \iff \exists x_1, \dots, x_s \in \mathbb{N} : P(e, x_1, \dots, x_s) = 0. \quad (42)$$

The same conclusion holds for the compiled mortality or PCP input predicates.

Proposition 11.1 (Fixed-arity quartic consequence). *The unbounded erasure predicate has a fixed-arity integer-polynomial representation of total degree at most four over natural witnesses.*

Proof. Start from (42). Split $P = P_+ - P_-$, where P_+ and P_- have nonnegative integer coefficients. Evaluate these two polynomials by finite addition/multiplication circuits over natural numbers. Introduce one natural wire variable for each gate output and impose the appropriate equation $z - u - v = 0$, $z - uv = 0$, or $z - c = 0$. Identify the two final outputs by a linear equation. Squaring and summing all gate and final-output equations gives total degree at most four, including e as a parameter. For fixed original witnesses, all new circuit-wire values are unique and natural. Conversely, any zero of the sum of squares reproduces the two evaluations and satisfies $P = 0$. The circuit and its arity are fixed because P is fixed. $\square$

This is an existence-and-compilation theorem, not a printed small universal polynomial. It leaves the number and multiplicity of the original DPRM witnesses uncontrolled. The explicit horizon-dependent polynomials of the previous section are independent constructions with better traceability, not substitutes for this fixed-arity argument. None of these observations resolves a general finite-fold representation problem.

[write] The circuit quadratization in this proof is the unique-extension quadratization `cdc:wf:lem:quadratization` of [22], and the quartic lowering `pgc:lem:quarticlower` of [21]; it is printed as the source’s own route, without a novelty claim. Part II proves the same corollary for its own predicate (Section 27).

11.2 No uniform computable exactness horizon

Let the size of an input be the bit length of its finite binary encoding. The following statement applies to either the fixed seven-state undecidable family or the nine-state many-one complete family. ([write] “The fixed seven-state undecidable family” means the seven-state, at-most-eight-letter promised class of Corollary 5.3: a class of input-dependent alphabets of fixed dimension, not one fixed alphabet, as the paragraph after that corollary explains.)

Theorem 11.2 (No computable yes-instance horizon). *There is no total computable function $b : \mathbb{N} \rightarrow \mathbb{N}$ such that every erasing input of bit length N has an erasing word of length at most $b(N)$. The conclusion remains true at every fixed rational contraction factor $0 < \eta < 1$ used above.*

Proof. Such a bound would decide erasure: compute $b(N)$ and enumerate all words up to that length, multiplying exactly and checking equal columns. If a word erases, answer yes; otherwise the assumed bound permits a no answer. Applied to the compiler output, this would decide source mortality or PCP, contradicting the proved reduction and its external undecidability dependency. $\square$

By contrast, approximate forgetting has the uniform computable horizon $\eta^T \leq \varepsilon$, independently of the matrix coefficients. The undecidability concerns exact zero information, not the task of reaching any fixed positive tolerance.

[write] The proof of Theorem 11.2 is the bounded-search argument of `cdc:bd:prop:nobound` in [22] and `pqc:qm:cor:nobound` in [21], applied to exact erasure.

Corollary 11.3 (No computable uniform finite-expectation bound). *Under uniform independent switching, there is no total computable bound in the input bit length for $\mathbb{E}\tau$ on all inputs where that expectation is finite. Nor is there a computable function $b(N)$ guaranteeing $\mathbb{P}(\tau \leq b(N)) \geq 1/2$ on every such input.*

Proof. Let $L_{\min}$ be the shortest erasing-word length. Every finite value of τ is at least $L_{\min}$, so a finite upper bound for $\mathbb{E}\tau$ bounds $L_{\min}$. A bound with positive probability of erasure by a given horizon also implies existence of a word by that horizon. Either would contradict Theorem 11.2 after taking an integer ceiling when necessary. $\square$

For any fixed arity representation such as (42), an analogous argument excludes a computable bound on all necessary natural witness coordinates in terms of the input size. A complete computably enumerable family of *non-erasure* certificates is also impossible on the undecidable class: parallel enumeration of yes and no certificates would be a decision algorithm. These are uniform impossibility statements, not obstacles to finding useful certificates or tight bounds for particular instances.

12 Exact erasure is not robust under perturbation

The preceding hardness holds arbitrarily close to the uniform matrix. That is a statement about the location of the input family, not robustness of individual yes instances. Indeed, erasure is destroyed by arbitrarily small perturbations that preserve the invariant-subspace restrictions.

Theorem 12.1 (Subspace-preserving non-erasing perturbations). *Let $S_1, \dots, S_k$ be positive rational column-stochastic matrices of dimension $n \geq 2$. For arbitrarily small positive rational δ , all matrices*

$$S_i^{(\delta)} = (1 - \delta)S_i + \delta I_n \quad (43)$$

are invertible. They remain positive and column-stochastic, have exactly the same common invariant complex subspaces as the original family, and have no erasing word. If the original family contracts total variation by at most η , the perturbed family contracts it by at most $\eta_\delta = (1 - \delta)\eta + \delta < 1$ for $0 < \delta < 1$.

Proof. For each i , $f_i(\delta) = \det((1 - \delta)S_i + \delta I_n)$ is a polynomial and $f_i(1) = 1$. It therefore has only finitely many roots. Outside the finite union of these root sets, all perturbed generators are invertible. Every interval $(0, \varepsilon)$ contains a positive rational outside that finite set. A finite product of invertible matrices is invertible and, since $n \geq 2$, cannot have rank one.

Positivity and column-stochasticity follow immediately from the convex combination for $0 < \delta < 1$. For every subspace U , invariance under S_i implies invariance under $S_i^{(\delta)}$. Conversely,

$$S_i = \frac{S_i^{(\delta)} - \delta I_n}{1 - \delta}$$

shows the reverse implication whenever $\delta \neq 1$. Applying this to every generator proves equality of the common invariant-subspace families. Finally, for $z = p - q$,

$$\|S_i^{(\delta)} z\|_1 \leq (1 - \delta)\eta\|z\|_1 + \delta\|z\|_1,$$

which proves the contraction claim. $\square$

For a prescribed open entrywise band, first choose a compiled family strictly inside that band. Its positive distance from the finitely many band endpoints allows sufficiently small perturbations to remain inside it. Thus even within the line-free, strongly contracting promised class, non-erasing instances occur arbitrarily close to every constructed erasing instance.

This theorem also explains why floating-point rank tests cannot certify exact erasure. A determinant or singular value that is numerically tiny can be nonzero for a nearby rational input with a completely different exact answer. The shipped checks therefore use integer arithmetic and exact rational Gaussian elimination, with explicit rejection of floating-point and Boolean values in integer witness fields.

13 Worked examples and executable verification

13.1 A completely printed three-state example

Take the scalar source alphabet $A_0 = [0]$, tensor degree two, and $\eta = 1/2$. The compiler gives $K = 13$, $D = 39$, and

$$B_0 = \begin{pmatrix} 13 & 13 & 13 \\ 13 & 13 & 13 \\ 13 & 13 & 13 \end{pmatrix}, \quad B_R = \begin{pmatrix} 14 & 14 & 11 \\ 12 & 15 & 12 \\ 13 & 10 & 16 \end{pmatrix}, \quad B_L = \begin{pmatrix} 18 & 15 & 15 \\ 14 & 14 & 11 \\ 7 & 10 & 13 \end{pmatrix}. \quad (44)$$

Every column sums to 39, every entry divided by 39 lies between $1/6$ and $1/2$, and the family has no common invariant complex line by the guard proof. The source generator is already $S_0 = J_3$; this is a readable sanity-check instance, not a universal numerical machine.

The chronological word $(0, L)$ has numerator product

$$B_L B_0 = \begin{pmatrix} 624 & 624 & 624 \\ 507 & 507 & 507 \\ 390 & 390 & 390 \end{pmatrix}$$

and denominator $39^2 = 1521$. Thus its stochastic product is

$$\begin{pmatrix} 16/39 \\ 13/39 \\ 10/39 \end{pmatrix} \mathbf{1}_3^\top,$$

which is rank one but not J_3 . This explicitly demonstrates why the line-free construction requires a distinction between erasure and a specified uniform output, even though existence of the two kinds of word is equivalent.

The full horizon-two certificate has 24 natural witnesses and 26 residuals. Its compressed version has 14 witnesses and 14 residuals. For the compressed certificate, the restriction matrices are

$$H_0 = 0, \quad H_R = 3R, \quad H_L = 3L.$$

After the source letter, the information restriction is zero and remains zero. Hence the two Y matrices are respectively $39\mathbf{1}_2\mathbf{1}_2^\top$ and $152\mathbf{1}_2\mathbf{1}_2^\top$. These are the exact values in the delivered JSON certificate, not floating-point approximations.

Under uniform random switching, erasure occurs precisely when letter 0 first appears. Therefore

$$N_T = 3^T - 2^T, \quad \mathbb{P}(\tau \leq T) = 1 - (2/3)^T, \quad \mathbb{E}\tau = 3.$$

The enumerated values for $T = 1, \dots, 5$ are 1, 5, 19, 65, 211, matching the certificate counts. This instance also permits direct checks of the zero-one law and the difference between a uniform erasing product and an arbitrary rank-one product.

13.2 A seven-state, eight-letter numerical packet

The delivered seven-state instance has source dimension three and six labelled source generators:

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad I_3, \quad -I_3, \quad \text{diag}(2, 1, 1), \quad I_3 + E_{12}, \quad I_3 + E_{23}.$$

The two tensor guards give eight target letters. With $\eta = 1/10$, the compiler selects $K = 131$ and $D = 917$. All 392 integer entries of the eight numerator matrices are printed in `examples/seven_state_instance.json`. ([write] Shipped as `data/02-exact-erasure-seven_state_instance.json`; every `examples/*.json` file of source 02 is shipped as `data/02-exact-erasure-*.json`.)

The chronological word (N, N, N) erases because $N^3 = 0$, while neither N nor N^2 is zero. The numerator product is

$$B_N^3 = 110156459 \mathbf{1}_7 \mathbf{1}_7^\top, \quad D^3 = 771095213 = 7 \cdot 110156459.$$

The full certificate has 171 witnesses and 192 residuals. The compressed certificate has 132 witnesses and 147 residuals. Both independent evaluations return polynomial value exactly zero. A separate uniform-target full certificate checks that this source-only word produces precisely J_7 .

This finite example illustrates the format and counts of the undecidable class. It does not prove undecidability by enumeration; that conclusion rests on the all-input reduction and the cited mortality theorem.

13.3 End-to-end PCP example

Use the two tiles

$$(u_0, v_0) = (a, ab), \quad (u_1, v_1) = (ba, a).$$

The index word $(0, 1)$ is a match because both concatenations are aba . The four-dimensional front end has three generators, including connector $q = 2$. The chronological word $(q, 0, 1, q)$ is zero.

After the two-guard lift, this becomes an erasing word in a nine-state, five-letter positive stochastic instance.

Both that numerical instance and its certificates are included. The full certificate has 344 natural witnesses and 400 residuals; the compressed one has 276 natural witnesses and 324 residuals. Every residual evaluates to zero in both. The example exercises the complete chain

PCP match $\rightarrow$ integer zero product $\rightarrow$ positive stochastic erasure $\rightarrow$ natural quartic zero.

The general proof also handles arbitrary connector placement, adjacent connectors, and empty component strings, rather than checking only this successful shape.

13.4 Scope and results of the executable checks

The shipped suite is deterministic. It uses Python’s exact integers and `fractions.Fraction`; no external Python package is required. Rational matrix rank is computed by exact Gaussian elimination. The finite random source sample uses seed 20261002, recorded in the result file.

The completed run performed **50,582 passing assertions**. Its principal categories are shown in Table 4; the full category ledger is in `verification/check_results.json`. ([write] Shipped as `data/02-exact-erasure-check_results.json`; the suite was rerun at the write on a copy with the delivered layout and reproduced it apart from the recorded Python version.) Counts denote assertions, not distinct mathematical instances or independently proved theorems.

Check category	Assertions
Mortality versus erasure for enumerated words	10,188
Exact tensor product-rank formula	10,188
Full word coordinate/block identity	10,188
Independent full-prefix quartic replay	3,448
Independent compressed quartic replay	3,448
Full and compressed size ledgers, combined	6,896
Exact total-variation contraction inequalities	3,601
PCP arbitrary connector-word reduction	1,638
All remaining algebraic, domain, mutation, and example checks	987
Total	50,582

Table 4: Finite exact checks run for this report. The two numerical certificate formats are checked by a module that imports no generator code.

The suite includes scalar source alphabets with coefficients from -2 to 2 , eight seeded two-dimensional source examples, the seven-state packet, a higher tensor example, one-generator nilpotent examples attaining the commuting bound, and PCP examples with empty words and adjacent connectors. It also changes individual witness entries, changes selectors, and rejects values of type `float`, `bool`, and `str` in natural-number fields.

The independent checker evaluates the defining residuals directly, without calling the generator. It is independent at the implementation-module level, not an independent human review. The all-instance reduction, the nonexistence of common complex lines, the infinite-horizon probability theorem, and the DPRM dependency are established mathematically in the text, not by these finite experiments. The uniform-parameter schema has a hand proof but is not one of the implemented certificate formats.

14 Integration and formalization plan for ProveIt

The present package is self-contained and does not modify the repository. It is suited to a separate research-report directory, with a link from the substrate review rather than a change to the fixed universal-operation record. ([write] This report is that directory. A link from the substrate review is for the owner of the research programme to add; the review of this manuscript [23] already exists.)

A useful formalization order follows the mathematical dependency graph. First formalize the explicit maps E, F , the affine block decomposition, and the rank-one/equal-column equivalence over $\mathbb{Q}$. Next establish the tensor product identity and the two common-line obstructions over $\mathbb{C}$. The mortality undecidability theorem can initially be an explicitly declared external interface, rather than an unproved statement hidden inside an implementation. The PCP front end offers a separate path to a fully concrete reduction chain.

The certificate layer is especially suitable for an early machine-checked milestone. It needs only finite sums, natural one-hot selectors, a matrix-product recurrence, and the equivalence between a zero sum of squares and vanishing of each residual. The compressed form additionally needs the column-sum invariant, the difference formula (30), and the offset bounds. This proves an exact zero-set bijection, not merely soundness of a successful example.

For source integration, four interfaces should remain visibly separate:

1. the computable mortality/PCP-to-stochastic compiler, including its common scale and all numerical input coefficients;
2. the exact finite-word rank and invariant-subspace theorems;
3. the bounded polynomial family and its literal witness/residual counts;
4. the unbounded DPRM polynomial existence theorem and its unspecified original witness multiplicities.

No existing Lean or Rocq file in ProveIt was compiled during this task, and no new theorem here is presented as proof-assistant verified. The accompanying `PROOF_STATUS.md` states these boundaries explicitly. ([write] Shipped as `02-exact-erasure-PROOF_STATUS.md`, beside `02-exact-erasure-SOURCE_AUDIT.md`.)

15 Further research questions and concrete next targets

The following are research directions exposed by the present proofs. Except for the precisely cited MathOverflow restriction already answered, this list does not claim that every formulation is a previously catalogued open problem.

15.1 Reduce the tensor overhead without losing the line guarantee

The present no-line compiler uses dimension $2d + 1$ and two guard letters. Can one construct a uniformly valid compiler in dimension $d + 2$, or with one additional letter, while retaining positivity and a word-projection soundness theorem? The obstacle is not positivity, which the affine lift handles uniformly; it is removing all source invariant lines without allowing guards to create false zero products. A useful first goal is to characterize the affine guard systems whose linear products cannot lower the rank of a protected source word.

15.2 Irreducibility beyond a missing invariant line

Our stochastic models necessarily have a common invariant hyperplane. Thus no construction confined to normalized column-stochastic matrices can make the whole linear action irreducible. A meaningful stronger target is either irreducibility on the information hyperplane, or a positive integer matrix construction without a common invariant hyperplane after normalization is abandoned. The first target is not achieved here either: source invariant subspaces can survive as $W \otimes \mathbb{C}^r$. A reduction must explain how to remove those spaces while still decoding mortality.

15.3 Low-dimensional classifications via the exact correspondence

Proposition 8.1 translates state-dimension questions directly into mortality questions. It can be used to transport new decidability results, bounded-alphabet results, or structural restrictions in either direction. The three-state case is especially focused: its transverse dimension is two. Any claimed classification should consult current primary sources rather than inherit the status labels of a 2014 survey table.

15.4 Regularly constrained and non-independent scheduling

For a finite-state controller, only a regular language of schedules may be allowed. The unrestricted mortality compiler does not by itself preserve such constraints when guard letters are erased. One needs a product construction that tracks both source computation and controller state. On the probabilistic side, identify conditions under which a reachable erasing block recurs almost surely. Full-support independent switching is sufficient here, but zero probabilities, transient controller states, and forbidden blocks can invalidate the zero-one argument.

15.5 Sharper canonical certificates and fully charged arithmetic

The compressed quartic reduces natural witnesses by using the invariant mass rather than storing it. Can additional structure of the tensor restrictions reduce the h^2T information entries, while retaining a bijection for every labelled word? One possibility is to certify the source projection and guard product separately; another is to exploit the rank gap without recording a full matrix at every step. Every proposal should count input parsing, selectors, scale arithmetic, coefficient generation, and reconstruction obligations. An improvement to $44T$ witnesses is a bounded-compiler result, not automatically an improvement to the repository's 87-operation universal construction.

15.6 Fixed-alphabet universality with an ordinary input interface

The nine-state many-one complete family encodes the problem in the gate alphabet. A different objective is one fixed numerical stochastic alphabet with a separate ordinary integer or finite-word input representation. Such a construction needs a concrete universal correspondence or matrix system and a fully paid input loader. It would make the relationship to the fixed universal Diophantine polynomials in ProveIt substantially tighter.

15.7 Structural non-erasure certificates

The commuting case supplies a complete finite certificate: a nonzero invariant image on which all generators are invertible. Can useful larger decidable classes be characterized by analogous invariant-space certificates? Candidate restrictions include simultaneously triangularizable transverse actions, prescribed rank patterns, or semigroups with a bounded descending-image chain. A proposed algorithm must account for the possibility that noncommuting invertible-on-current-image maps carry that image elsewhere; the commuting proof cannot simply be reused without that invariant-image property.

15.8 Precision-sensitive versus robust observation predicates

The perturbation theorem shows that exact erasure is destroyed by generic small invertibilizing perturbations, even when the invariant-subspace restrictions are unchanged. A quantitative alternative is to study coefficient uncertainty sets together with a fixed tolerance on output dependence. Which robust predicates admit computable finite certificates, and how do their costs scale as the tolerance tends to zero? The universal bound $\eta^T \leq \varepsilon$ solves one natural approximate-forgetting question; more delicate targets could depend on a specified output distribution, a restricted schedule language, or nonuniform contraction rates.

15.9 A formal proof of the guard compiler and the projection bijection

A focused formal milestone is Theorem 5.1 together with Theorem 10.2, keeping mortality undecidability as a separately identified dependency. A second milestone is the complete PCP chain and its connector boundary condition. These targets are concrete enough to test the repository’s arithmetic and matrix APIs, and they would turn the present hand-checked compiler into a reusable trusted frontend.

16 Conclusion

Exact stochastic erasure supports computational hardness even when every gate is positive, every step uniformly contracts total variation, and no common invariant complex line exists. The mechanism is explicit: preserve the source computation on one tensor factor, use invertible guards on the other, and lift the resulting affine dynamics into the interior of the stochastic simplex.

The strongest precise line-restriction result in this report is the $(d, m) \mapsto (2d + 1, m + 2)$ compiler. It supplies the promised seven-state numerical undecidability bound, while the independent four-dimensional PCP front end supplies a nine-state many-one complete family. The probabilistic interpretation then identifies exact finite forgetting, almost-sure finite forgetting, and finite expected forgetting time without changing the underlying finite-word predicate.

The Diophantine layer is equally explicit at a bounded horizon. A zero-set bijection reduces the seven-state certificate from $57T$ to $44T$ natural witnesses by storing only shifted information coordinates. The fixed-arity unbounded representation remains a separate DPRM consequence. Together, these results give a concrete new computational frontend and a carefully delimited research contribution, rather than a claim that a generic invocation of DPRM or contraction has solved every arithmetic-compression problem.

Part II

Exact convergence as computation (source 10)

[write] Part II prints batch-78 manuscript 10, *Exact Convergence as Computation: Diophantine Certificates for Uniformly Mixing Stochastic Games*, subtitled *Fixed-game universality, canonical quadratic witnesses, and the distinction between scalar and vector reachability*, under the kicker “Diophantine computational substrates”, “Prepared for Vladimir Reshetnikov; Mathematical development and exact-arithmetic implementation: ChatGPT”, October 2, 2026. Its labels carry the prefix `ste:ec:.` Its research-status statement and abstract follow; its two appendices are Appendices C and D.

Research status (source 10). The central results below are supported by complete mathematical arguments and a delivered, executable compiler. The accompanying standard-library Python suite passes 26,077 exact-arithmetic checks. These checks are not a proof-assistant verification, and the manuscript is not peer reviewed. The literature and repository comparison is targeted rather than exhaustive; priority for the precise constrained simulation theorem is not established. No resolution of the general Skolem problem, no single-fold MRDP theorem, and no improved fixed universal-polynomial operation bound is claimed.

Abstract (source 10). We construct a finite, turn-based, two-player stochastic game whose exact value iteration is computationally universal despite unusually strong regularity. The discount is $1/2$, every running reward is $1/4$, every transition probability is dyadic, every available action assigns probability at least $1/(2N)$ to each of the N states, and the value vector is the known constant $\frac{1}{2}\mathbf{1}$. Nevertheless, for a suitable fixed game, deciding whether an input dyadic terminal-payoff vector ever reaches that value vector in finitely many Bellman iterations is computably enumerable complete.

The proof gives a compiler from counter programs to homogeneous linear/min/max circuits and then to genuine stochastic-game transition tables. Paired positive and negative channels remove signed coefficients; a correctly initialized cyclic evaluation pipeline removes nested gates; and a common uniform reset provides full support without destroying the simulated computation. An erasing halt convention turns computational halting into finite attainment of the fixed point. A zero-reward variant gives complete coordinate-equality and finite-consensus problems.

For each externally specified horizon T , the complete computation admits an orthant-nonnegative quadratic Diophantine certificate with exactly one natural witness tuple when the prescribed endpoint condition holds. With N game states and b binary-decision states, it uses $(N + 2b)T$ witnesses. Two boundary results clarify the source of hardness: deterministic min/max-selector games have a finite abstraction, whereas chance-only fixed-point attainment is a kernel-stabilization problem. Scalar equality for chance-only games, in contrast, is Skolem-equivalent; an explicit dyadic, full-support specialization is included and related to the existing ergodic Markov-chain literature. Complete example polynomials, exact tests, and an independent serialized-certificate checker accompany the article.

17 Research question, contribution, and source boundary

A computational substrate need not resemble a tape machine. Here the substrate is a familiar numerical recursion: take a vector, form several convex averages of its coordinates, choose a minimum or maximum at each coordinate, and repeat. All coefficients are rational. A discount makes the recursion converge rapidly. The question is whether such benign-looking iteration can still contain an unbounded exact computation.

The substantive question is not whether a computably enumerable relation has some Diophantine representation. MRDP already answers that question. Instead we ask for a reduction that respects a restrictive native interface: nonnegative stochastic coefficients, at most two actions per state, common positive transition support, a fixed discount, a trivial limiting value, and a complete, witness-controlled arithmetic description of bounded runs.

17.1 The central theorem

Let a state be owned by a minimizing player, a maximizing player, or have just one action. Write $p_{ia} \in \mathbb{Q}^N$ for the stochastic row of action a at state i . Put

$$S(v)_i = \text{opt}_{a \in A_i} p_{ia} v, \quad \Phi(v) = \frac{1}{4} \mathbf{1} + \frac{1}{2} S(v). \quad (45)$$

Here opt is the owner's minimum or maximum; for a one-action state there is no choice. The vector v is a terminal-payoff table, not a probability distribution over the states.

Theorem 17.1 (Fixed-game universality with a known fixed point). *There are a fixed positive integer N and a fixed finite game of the form (45) such that:*

- (i) *each state has one or two actions; every $p_{ia,j}$ is dyadic and $p_{ia,j} \geq 1/(2N)$;*
- (ii) *Φ maps $[0, 1]^N$ into itself, is a $1/2$ -contraction in the supremum norm, and has the unique fixed point $v_* = \frac{1}{2} \mathbf{1}$;*
- (iii) *the set of dyadic $v \in [0, 1]^N$ for which $\Phi^n(v) = v_*$ for some $n \geq 0$ is computably enumerable complete.*

The reduction is effective. More generally, the construction compiles every finite counter program into such a game, with an explicitly computed terminal-payoff encoding of each initial configuration.

The hard input is the initialization. The transition table, rewards, discount, and target in the fixed-game assertion do not depend on that input. A numerical transition table for a specifically chosen universal counter machine is not printed or shipped. Fixed-game existence follows by applying the fully specified compiler to a fixed universal counter machine. The shipped numerical tables are worked examples, not universal-machine records.

17.2 Why the source comparison matters

The repository contains an MRDP interface stating the existence of one ordinary integer polynomial and one finite witness dimension for an entire computably enumerable set [15]. Its computational-substrate reports already discuss contracting dynamics and quadratic neural certificates [7]. Consequently, neither the observation that contraction can coexist with computation nor the generic use of quadratic complementarity is claimed here as a discovery. The contribution is the *stochastic-game realization under the simultaneous restrictions in Theorem 17.1*, together with its complete arithmetic interface.

[write] *Comparison with the collection.* Within this collection a 2-adic $1/2$ -contraction already has an r.e.-complete exact extinction (Corollary `pqc:ad:cor:complete` in Part VII of [21], source 12 there), while a real four-dimensional piecewise-linear $1/2$ -contraction has decidable point-to-point reachability because its radial clock is divided by B at every step and never vanishes

(`pqc:sc:thm:main` (iii), source 14 there). The present construction is the real, order-preserving counterpart of the former: its live clock $s' = \sum_e e$ of (54) is erased at halting (Lemma 20.2), so the whole vector reaches the fixed point. It therefore shows that source 14’s decidable point reachability comes from that compiler’s non-erasable clock, not from real homogeneous contractions as such; nothing printed there is contradicted, and the picture agrees with the comparison section “Two routes to one phenomenon” of Part VII of [21]. Part V of [21] (source 17 there) has the same slogan for a different object: a uniform spectral gap with an undecidable exact stationary equality.

A particularly close external comparison is the January 26, 2026 revision of Varonka and Watanabe [10]. It proves point-reachability results for one-player min- or max-Bellman operators, including arbitrary dimension when source and target are componentwise comparable, and full decidability in dimension two. Our construction uses both owners, a larger fixed dimension, and initial errors of both signs. It therefore does not settle their remaining one-player problem or contradict their positive results. It does show why monotonicity and rapid fixed-point convergence alone cannot justify unrestricted extensions to two-player operators.

The chance-only scalar-equality boundary is related to established work, not a new undecidability assertion. Vahanwala proves Skolem and positivity completeness for ergodic Markov chains with a dimension-efficient reduction [11]. We supply a less dimension-efficient specialization with dyadic probabilities and a prescribed common reset. The general Skolem problem remains unresolved in the July 2026 work of Luca, Ouaknine, and Worrell, whose general decidability implication is conditional [12]. We do not resolve that problem.

17.3 What is and is not computed

There is no difficulty in obtaining the infinite-horizon game value in (45): every play receives the same running reward $1/4$ at every step, and

$$\sum_{t \geq 0} 2^{-t} \frac{1}{4} = \frac{1}{2}.$$

Every strategy profile therefore has infinite-horizon payoff $1/2$. The hard property is *finite exact attainment from an input terminal payoff table*. The usual initialization $v = \mathbf{0}$ is not the input family used in the reduction. In particular, this manuscript does not say that solving ordinary finite discounted games is undecidable.

Value iteration at horizon n is the finite-horizon game value with terminal payoff v and the indicated running reward. Its computation by backward induction is immediate: after choosing an action, average the remaining horizon’s values, then optimize according to the current state’s owner. Thus the construction uses actual game semantics, not an arbitrary piecewise-affine map merely renamed a Bellman operator. Stochastic games and discounted values originate in Shapley’s framework [8]; approximation stopping criteria, such as those studied in [13], address a different question from finite exact attainment.

18 Conventions and the three observation problems

We take $\mathbb{N} = \{0, 1, 2, \dots\}$. A dyadic rational is an element of $\mathbb{Z}[1/2]$. All finite rational objects have an effective exact encoding. “Computably enumerable complete” means complete under total computable many-one reductions; no polynomial-time reduction bound is asserted.

We distinguish three predicates for an exact trajectory v_n :

$$\text{scalar equality: } (v_n)_r = (v_n)_s, \quad (46)$$

$$\text{consensus: } v_n \in \mathbb{R}\mathbf{1}, \quad (47)$$

$$\text{fixed-point attainment: } v_n = v_*. \quad (48)$$

Each may be imposed at a specified $n = T$, or existentially over all n . These are different decision problems. In particular, a scalar zero in a linear recurrence is not the same as a whole vector becoming zero.

For a vector x , set

$$\|x\|_\infty = \max_i |x_i|, \quad \text{span}(x) = \max_i x_i - \min_i x_i.$$

The latter is a seminorm, since it vanishes on constant vectors. All stochastic rows in this article act on column vectors of payoffs from the left. The state of an exact computation is the entire payoff vector. It is not the state visited by a single random sample path.

A *bounded certificate* has an externally supplied natural horizon T . Its arity is allowed to depend on T . A *fixed-arity Diophantine representation* uses one fixed polynomial and a fixed number of existential variables for all inputs and all possible running times. Confusing these notions would invalidate the final MRDP discussion.

19 A homogeneous counter-machine compiler

19.1 Source instruction set

A program has m control states and $k \geq 1$ natural counters. Its instructions are

$$\text{INC}(i, q), \quad \text{DECJZ}(i, q_+, q_0), \quad \text{HALT}.$$

An increment increases counter i and jumps to q . A decrement instruction jumps to q_0 without changing counters when counter i is zero; otherwise it decreases that counter and jumps to q_+ . There is one halt state h . The ordinary operational convention makes it absorbing.

This instruction set supports universal computation with finitely many counters, indeed with two. The distinction between counter instruction sets matters: the present zero-test/decrement with independently specified targets contains the undecidable CM2 instruction set described and analyzed by Dudenhefner [9]. A fixed universal program can be obtained from a universal interpreter in this standard counter model. Universality of this source model is the sole external computability ingredient in the hardness proof; the stochastic and arithmetic translations are proved here.

19.2 Configuration coordinates and exact zero tests

Encode a configuration $(q, n_1, \dots, n_k)$ at scale $s > 0$ as

$$x = (s, p_1, \dots, p_m, c_1, \dots, c_k), \quad p_j = s \mathbf{1}_{j=q}, \quad c_i = s 2^{-n_i}. \quad (49)$$

The dimension is $d = 1 + m + k$. Every coordinate lies in $[0, s]$. The homogeneous gate

$$z_i = \max(0, 2c_i - s) \quad (50)$$

returns s when $n_i = 0$ and 0 otherwise. Indeed, a positive counter has $c_i \leq s/2$. No discontinuous test has been used.

For a nonbranching instruction at state j , its enable value is p_j . For a decrement/zero instruction testing counter i , its two enable values are

$$e_{j,0} = \min(p_j, z_i), \quad e_{j,+} = \min(p_j, s - z_i). \quad (51)$$

On encoded configurations each enable value is 0 or s , and precisely one is s when the active control state is included among the branches.

Associate to each branch e its target state $q(e)$ and multipliers $\alpha_{e,i} \in \{1/2, 1, 2\}$. An increment of counter i uses $1/2$, a positive decrement uses 2 , and an unchanged counter uses 1 . Put

$$p'_j = \sum_{e:q(e)=j} e, \quad (52)$$

$$c'_i = \sum_e \min(2e, \alpha_{e,i} c_i). \quad (53)$$

An inactive summand vanishes. An active summand equals its required counter value: in the decrement case $2c_i \leq s$, and otherwise the value is also at most s . The cap $2e = 2s$ is therefore harmless.

19.3 Two halt conventions

We use two related maps. The *absorbing map* F_{abs} includes a halt branch targeting h with all multipliers 1, and sets $s' = s$. It exactly implements the ordinary absorbing machine.

The *erasing map* F_{era} omits the halt branch and sets

$$s' = \sum_e e. \quad (54)$$

On a nonhalting configuration it performs the same machine step at the same scale. On a halting configuration it outputs the entire zero vector. Both maps take $\mathbf{0}$ to $\mathbf{0}$.

Lemma 19.1 (Counter simulation). *The two maps are continuous, positively homogeneous, dyadic circuits using only linear, minimum, and maximum gates. The absorbing map sends (49) to the encoding of its successor. If the source machine first reaches its halt state after H steps, then $F_{\text{era}}^{H+1}(x) = \mathbf{0}$, whereas $F_{\text{era}}^j(x)$ is a nonzero encoded configuration for $0 \leq j \leq H$. If the machine never halts, all these iterates remain nonzero and their clock coordinate remains s .*

Proof. Equation (50) gives the stated zero test. Exactly one branch is enabled at a nonhalting configuration, so (52) writes a one-hot successor state and (53) applies exactly the required multiplier to each counter coordinate. For the erasing map all branches have enable value zero at the halt configuration, including every counter summand, and (54) vanishes. Homogeneity and continuity hold gate by gate; the constant zero is a homogeneous linear gate. Induction proves all trajectory claims. $\square$

Using unrestricted fan-in for linear sums, the displayed circuit can be organized with depth at most seven: linear zero-test preparation, maximum, branch preparation, positive-branch selection, enable scaling, gated counter terms, and final sums. This is a bound for this particular source construction, not for arbitrary circuits or for binary fan-in arithmetic. The stochastic implementation permits a finite distribution with more than two successor states, so such sums do not violate the two-action restriction.

20 From signed circuits to genuine stochastic games

20.1 Layering and normalization

Consider any positively homogeneous dyadic circuit $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ whose gates are finite linear combinations and binary minima or maxima. Insert copy gates so every edge enters the immediately following layer. Pad *all outputs* to one common final layer L , chosen at least as deep as every gate retained in the circuit. The latter qualification handles unused gates and the initially halted edge case. Inputs are layer zero.

Choose powers of two $C_\ell \geq 1$ so that every linear gate $g = \sum_j a_j g_j$ at layer ℓ satisfies $\sum_j |a_j| \leq C_\ell$. Define

$$K_0 = 1, \quad K_\ell = C_1 \cdots C_\ell, \quad K = K_L, \quad M = L + 1. \quad (55)$$

For input x , every gate value at layer ℓ has absolute value at most $K_\ell \|x\|_\infty$. This follows by induction for linear gates and by $|\min(a, b)|, |\max(a, b)| \leq \max(|a|, |b|)$ for the other gates.

20.2 Paired channels and a baseline state

Create a baseline state o , with its only transition back to itself. For every circuit input and gate g , create two states g^+, g^- . They will hold a value and its negative. On a centered initialization the baseline value is zero. A retained gate at layer ℓ is intended to hold g/K_ℓ in its positive channel and $-g/K_\ell$ in its negative channel.

For a linear gate $g = \sum_j a_j g_j$ at layer ℓ , give g^+ a single action with probabilities

$$\frac{|a_j|}{C_\ell} \quad \text{to } g_j^+ \text{ if } a_j \geq 0, \quad \text{to } g_j^- \text{ if } a_j < 0, \quad (56)$$

and send the remaining probability to o . For g^- swap the signs of the selected channels. Repeated destinations are combined. Nonnegativity of the remaining probability is exactly the normalization inequality.

For $g = \min(g_1, g_2)$, make g^+ a MIN state. Its two actions assign probability $1/C_\ell$ to g_1^+ and g_2^+ , respectively, and the remaining probability to o . Make g^- a MAX state with the analogous negative-channel actions. For a maximum gate reverse the two owners. Thus the negative channel really is the negative output, including at ties. All rows are stochastic and dyadic; every state has at most two actions.

Finally, each input-register channel has a single action copying its corresponding output channel at layer L . This creates a cyclic pipeline. Denote its undiscounted, zero-reward operator by S_0 .

If there are G retained gates in addition to the d inputs, the number of states before padding is

$$N_0 = 1 + 2(d + G). \quad (57)$$

Pad to a power of two $N \geq N_0$, with every extra state copying o . Padding will keep a subsequent uniform reset dyadic. If the circuit has $g_{\min \max}$ min/max gates, then the game has exactly $b = 2g_{\min \max}$ binary-decision states. No optimality claim is attached to these counts.

20.3 The invariant subspace

Let $V \subseteq \mathbb{R}^N$ consist of vectors whose baseline and padding coordinates vanish and whose paired channels are negatives of each other. The construction gives

$$S_0(V) \subseteq V, \quad \frac{1}{N} \sum_i w_i = 0 \quad (w \in V). \quad (58)$$

The first assertion follows separately for linear gates, min/max pairs, copy registers, and the baseline. The second is cancellation of the pairs. In particular, signed arithmetic is represented inside a genuine order-preserving stochastic operator. The order on the full space does not prohibit this: distinct generic vectors of V are incomparable in the componentwise order.

20.4 Initialization is part of the compiler

Given $x \in [-1, 1]^d$, evaluate the finite circuit once on x . Initialize its input channels to $\pm x_j$, each layer- ℓ gate to its paired values $\pm g(x)/K_\ell$, and the baseline and padding coordinates to zero. Call the resulting vector $w(x) \in V \cap [-1, 1]^N$.

This is a finite, computable loader. It computes only the circuit's one-step gate values. It does not inspect whether the source program will halt. Zero-filling the internal gates instead would be unsound: the first feedback step would copy zero outputs into all registers and could falsely signal a halted or erased computation.

Lemma 20.1 (Every microstep is accounted for). *Let $r_n \in \mathbb{R}^d$ be the positive input-register coordinates of $S_0^n(w(x))$. Then*

$$r_n = K^{-q_n} F^{q_n}(x), \quad q_n = \left\lceil \frac{n}{M} \right\rceil, \quad (59)$$

where $q_0 = 0$. In particular, $r_0 = x$ and $r_1 = \dots = r_M = F(x)/K$.

Proof. Each layer computes its gates from the preceding layer with an additional factor $1/C_\ell$. Following an input-register vector through all L layers and then the feedback edge therefore gives, for every $n \geq 0$,

$$r_{n+M} = F(r_n)/K. \quad (60)$$

At time zero each layer was already evaluated consistently on x . For $0 \leq t \leq \ell$, the layer- ℓ values at time t are still the normalized gate values evaluated on x : prove this by induction on ℓ , using the preceding layer at time $t - 1$ for $t > 0$. Thus the final layer has its initial output for times $0, \dots, L$, and the feedback registers equal $F(x)/K$ for times $1, \dots, M$. Positive homogeneity now iterates (60). For each residue class modulo M , its initialized member has exactly the exponent specified in (59); every subsequent member increases that exponent by one. $\square$

Lemma 20.2 (Erasure drains the whole pipeline). *Use $F = F_{\text{era}}$. If the source first reaches its halt state after H steps, then*

$$S_0^n(w(x)) = \mathbf{0} \quad \text{for all } n \geq (H + 1)M. \quad (61)$$

If the source never halts, $S_0^n(w(x)) \neq \mathbf{0}$ for every n .

Proof. In the halting case $F^{H+1}(x) = \mathbf{0}$ and all its later iterates vanish. Lemma 20.1 makes all registers zero from time $HM + 1$ onward. After at most L more steps all internal gates have received only zero data. Every gate is homogeneous and hence maps zero inputs to zero. The time bound is $HM + 1 + L = (H + 1)M$. In the nonhalting case the clock register is $K^{-q_n}s > 0$ at every finite time, so the whole vector cannot vanish. $\square$

The extra L steps cannot in general be omitted. Registers can have become zero while stale, unused, or not-yet-overwritten internal gates remain nonzero. The full-vector theorem uses the drain argument, not just the register simulation.

21 Uniform mixing preserves exact computation

21.1 A common-reset lemma

For any zero-reward stochastic min/max operator S , any rational $0 < \eta < 1$, and $\bar{v} = N^{-1} \sum_j v_j$, define

$$\hat{S}(v) = (1 - \eta)S(v) + \eta\bar{v}\mathbf{1}. \quad (62)$$

It is itself a stochastic-game operator: replace every action row by

$$\hat{p}_{ia} = (1 - \eta)p_{ia} + \frac{\eta}{N}\mathbf{1}^\top. \quad (63)$$

The common term may be moved outside either a minimum or a maximum. Every new transition probability is at least η/N .

Lemma 21.1 (Exact preservation of all equality times). *Let $B = \delta\hat{S}$ with $0 < \delta < 1$. For every initial vector v there are scalars c_n such that*

$$B^n(v) = [\delta(1 - \eta)]^n S^n(v) + c_n\mathbf{1}. \quad (64)$$

Consequently, every pairwise equality or strict order between coordinates of $S^n(v)$ is preserved at exactly the same n by $B^n(v)$.

Proof. Stochastic min/max operators are positively homogeneous and additively homogeneous:

$$S(av + c\mathbf{1}) = aS(v) + c\mathbf{1} \quad (a \geq 0).$$

Put $a_n = [\delta(1 - \eta)]^n$. The assertion holds at $n = 0$ with $c_0 = 0$. Substitution into (62) gives the induction step with

$$c_{n+1} = \delta c_n + \delta\eta a_n \overline{S^n(v)}.$$

Subtracting two coordinates removes c_n , and a_n is strictly positive at every finite n . $\square$

This lemma does not assume that an input is a valid encoded computation. It is an identity for every real vector. In particular, the reset is not an approximation justified only on a finite test set.

21.2 Contraction estimates

Lemma 21.2 (Supremum and span contractions). *For any common reward r and $\Phi = r\mathbf{1} + \delta\hat{S}$,*

$$\|\Phi(v) - \Phi(u)\|_\infty \leq \delta\|v - u\|_\infty, \quad (65)$$

$$\text{span}(\Phi(v) - \Phi(u)) \leq \delta(1 - \eta) \text{span}(v - u). \quad (66)$$

Proof. If $m\mathbf{1} \leq v - u \leq M\mathbf{1}$, monotonicity and additive homogeneity give $m\mathbf{1} \leq S(v) - S(u) \leq M\mathbf{1}$. This proves its supremum nonexpansiveness and bounds the span of the difference by $M - m$. The same supremum argument applies to $\hat{S}$, which is a stochastic operator. For the span, the reset contribution $\eta(\bar{v} - \bar{u})\mathbf{1}$ is constant and therefore disappears. Multiplication by δ completes both inequalities. $\square$

On the invariant space V of (58), the mean vanishes at every iterate, so a stronger identity is available:

$$(\delta\hat{S})^n(w) = [\delta(1 - \eta)]^n S_0^n(w), \quad w \in V. \quad (67)$$

Now choose $\delta = \eta = 1/2$, define

$$v(x) = \frac{w(x) + \mathbf{1}}{2}, \quad \Phi(v) = \frac{1}{4}\mathbf{1} + \frac{1}{2}\hat{S}(v), \quad (68)$$

and use additive homogeneity. One obtains

$$\boxed{\Phi^n(v(x)) = \frac{1}{2}\mathbf{1} + \frac{4^{-n}}{2}S_0^n(w(x)).} \quad (69)$$

Every initial coordinate belongs to $[0, 1]$, all coefficients are dyadic, and all reset rows have entries at least $1/(2N)$. On these encoded inputs, the error is bounded by $4^{-n}/2$ in the supremum norm. For arbitrary $v \in [0, 1]^N$, the uniform bound is $2^{-n}\|v - \frac{1}{2}\mathbf{1}\|_\infty$.

22 Proof of fixed-game universality and its variants

Proof of Theorem 17.1. Fix a universal counter program and apply the erasing compiler. Choose the power-of-two state padding and the reset and reward in (68). All syntactic assertions about the resulting fixed game follow from the explicit row construction. Since a stochastic operator fixes constant vectors, $\Phi(\frac{1}{2}\mathbf{1}) = \frac{1}{2}\mathbf{1}$. The supremum contraction proves uniqueness. The cube is preserved because each convex average and its minimum or maximum lies in $[0, 1]$, whence every output of Φ lies in $[1/4, 3/4]$.

Encode an input of the fixed universal program by the dyadic vector $v(x)$. Equations (69) and (61) show

$$\exists n \Phi^n(v(x)) = \frac{1}{2}\mathbf{1} \iff \exists n S_0^n(w(x)) = \mathbf{0} \iff \text{the source computation halts.}$$

All factors discarded in these equivalences are strictly positive at finite times. The map from a source input to the full payoff vector is computable. Thus the fixed game's attainment set is c.e.-hard. It is c.e. because rational iteration and equality are decidable at every finite time, so successive times can be searched. The same reasoning applies uniformly when the source program is supplied to the compiler. $\square$

22.1 Uniformly stopping reachability games

The operator Φ also has a reachability interpretation. From every nonterminal state, every action goes to a success terminal with probability $1/4$, to a failure terminal with probability $1/4$, and otherwise continues according to its row $\hat{p}_{ia}$ with total probability $1/2$. There are now $N + 2$ states. Both terminals are absorbing.

The continuation probability for n steps is exactly 2^{-n} under every strategy profile, and the expected number of transitions before stopping is exactly two. Success probability is exactly $1/2$ from every nonterminal state. With a supplied boundary payoff table on nonterminal states, backward reachability iteration is exactly Φ .

Corollary 22.1 (Rapid stopping does not imply decidable exact attainment). *There is a fixed, uniformly stopping, finite turn-based reachability game with success-probability vector $\frac{1}{2}\mathbf{1}$ for which finite exact attainment of that vector from an input dyadic boundary table is c.e.-complete. The full-support lower bound applies to the conditional transition kernel on the nonterminal states; it does not apply to the absorbing terminal rows.*

Universality here belongs to the arithmetic of the iterates. A typical random path is short, and we do not assert that it physically executes an arbitrarily long counter computation.

22.2 Zero running rewards and scalar observations

For the absorbing source map, let $B = \frac{1}{2}\hat{S}$ have zero running rewards. The shifted encoding satisfies

$$B^n(v(x)) = \frac{2^{-n}}{2}\mathbf{1} + \frac{4^{-n}}{2}S_0^n(w(x)). \quad (70)$$

Let r be the positive halt-state register, and s the positive clock register. For initial scale one, put $q_n = \lceil n/M \rceil$. Before halting, their difference is

$$(B^n(v(x)))_r - (B^n(v(x)))_s = -\frac{4^{-n}}{2K^{q_n}}; \quad (71)$$

after the simulated state is halted, the difference is zero. If the source's first halting time is $H \geq 1$, the first equality time is

$$n_{\text{eq}} = (H - 1)M + 1. \quad (72)$$

For an initially halted source it is zero. The separate observation “halt-register value greater than baseline value” is true exactly at simulated halted configurations, giving c.e.-complete strict-inequality reachability as well.

For the erasing source map, consensus in (70) is equivalent to $S_0^n(w(x)) = \mathbf{0}$: the latter vector belongs to V , and the only constant vector in V is zero. Thus finite consensus is c.e.-complete in a suitable fixed zero-reward game, although its unique fixed point is zero and all initial payoffs in the reduction are nonnegative.

22.3 One-sided errors are easy

Proposition 22.2. *For the full-support game in Theorem 17.1, if $v \leq v_*$ or $v \geq v_*$ componentwise, finite attainment occurs if and only if $v = v_*$ already.*

Proof. Suppose $v - v_* \leq 0$ and some coordinate is strictly negative. Every full-support action average of $v - v_*$ is then strictly negative, so every coordinate of $\Phi(v) - v_*$ is strictly negative. This remains true at every later iterate. The nonnegative case is identical with signs reversed. $\square$

The encoded hard initializations straddle the fixed point through their paired positive and negative errors. This is essential, not a dispensable technicality.

23 Two decidable boundaries and one Skolem boundary

The combination of averaging and mixed min/max control is important. Each of two natural restrictions has a direct decision procedure for full fixed-point attainment.

23.1 No averaging: finite selector dynamics

Suppose every action row is a point mass. Then S chooses one of the current coordinates at each state, possibly after a minimum or maximum. If v has $m_0 \leq N$ distinct coordinate values, every vector $S^n(v)$ has all coordinates in this fixed finite set. There are at most m_0^N such vectors.

Proposition 23.1 (Deterministic-selector decidability). *Scalar equality, consensus, and attainment of the uniform fixed point are decidable for zero-reward or common-reward discounted games with point-mass action rows. At most m_0^N distinct undiscounted vectors need be visited before a repetition proves that no new observation will occur.*

Proof. For scalar equality and consensus, positive discount factors and common additive rewards do not affect the observations. Iterate the finite selector map and test each vector until it repeats. For the uniform fixed point $v_* = r/(1 - \delta)\mathbf{1}$, translate to the error $v - v_*$. The error at time n is $\delta^n S^n(v - v_*)$, so fixed-point attainment is zero-vector reachability in the same finite abstraction. $\square$

23.2 No choices: whole-vector zeros are easy

Suppose each state has only one action, so $S(v) = Pv$ for one rational stochastic matrix. For a common reward and its uniform fixed point,

$$\Phi^n(v) - v_* = \delta^n P^n(v - v_*).$$

Proposition 23.2 (Kernel-stabilization test). *Finite fixed-point attainment for a chance-only N -state game is equivalent to $P^N(v - v_*) = \mathbf{0}$ and is therefore decidable by exact linear algebra.*

Proof. The kernels $\ker P^j$ form an increasing chain. Their dimensions can strictly increase at most N times. If two consecutive kernels agree, all subsequent kernels agree, by taking inverse images under P . Hence $\bigcup_{j \geq 0} \ker P^j = \ker P^N$. $\square$

This conclusion does *not* solve the scalar zero problem. A single coordinate may vanish while the whole orbit vector stays nonzero.

23.3 Scalar equality: a dyadic, full-support Skolem specialization

Consider the integer-matrix form of the Skolem problem:

$$\text{given } A \in \mathbb{Z}^{d \times d}, x \in \mathbb{Z}^d, \quad \exists n \geq 0 \quad e_1^\top A^n x = 0 ? \quad (73)$$

Every integer linear recurrence is put into this form by its companion matrix. Conversely, Cayley–Hamilton gives a recurrence for every such matrix sequence.

Choose powers of two

$$C \geq \max \left(1, \max_i \sum_j |A_{ij}| \right), \quad R \geq \max(1, \|x\|_\infty).$$

Use a baseline and d paired channels. For the positive channel i^+ , put mass $|A_{ij}|/C$ on j^+ or j^- according to the sign of A_{ij} ; put the remaining mass on the baseline. Swap channels for i^- . Pad to a power of two $N \geq 2d + 1$ and add a half-uniform reset. Initialize centered channels to $\pm x_i/R$, then shift to $[0, 1]$. There are no min/max decisions.

For either zero rewards or a common reward, let $r = 1^+$ and let o be the baseline. With discount $1/2$, the difference is

$$v_{n,r} - v_{n,o} = \frac{e_1^\top A^n x}{2R(4C)^n}. \quad (74)$$

Every coefficient is dyadic and every transition has probability at least $1/(2N)$. The formula follows by induction on the paired channels before reset and by Lemma 21.1 after reset.

Proposition 23.3 (Scalar-equality equivalence). *For the class of chance-only dyadic games with discount $1/2$ and a half-uniform reset, scalar-equality reachability is computably many-one equivalent to the Skolem problem. The game is part of the input in this statement; no fixed chance-only game is asserted c.e.-complete.*

Proof. Equation (74) gives one reduction. In the reverse direction, cancel common rewards and the positive discount. Equality of two coordinates is the zero of $(e_r - e_s)^\top P^n v$. Clear the denominators of P and v : with DP integral and av integral, the integer sequence $(e_r - e_s)^\top (DP)^n (av)$ has exactly the same zeros. Cayley–Hamilton supplies its integer recurrence effectively. $\square$

This is an explicit specialization of an established hardness phenomenon. It is not a claim that Skolem-hardness of ergodic chains was previously unknown, nor an improvement over the smaller dimension achieved in [11]. The specialization is useful because its probability and reset constraints match the nonlinear compiler exactly.

Substrate	Scalar equality	Uniform fixed-point attainment
Point-mass rows; MIN and MAX allowed	Decidable by finite selector states	Decidable by the same finite abstraction
Chance only; rational stochastic rows	Skolem-equivalent, even with the stated dyadic reset	Decidable by kernel stabilization
MIN, MAX, and stochastic rows	c.e.-complete for a suitable fixed game	c.e.-complete for a suitable fixed game

The table is not a classification of every one-player subclass. In particular, allowing stochastic rows and only MAX states, or only MIN states, is a separate frontier.

[write] *Bears on a question of [21]*. Part VII of [21] asks (Question “Structural subclasses”) which restrictions force half-space reachability to be decidable: coordinatewise monotonicity, nonnegative weights, boundedly many activation patterns, or a genuine Bellman-operator structure. Theorem 17.1 and Section 22 answer this negatively in part: the operators here are coordinatewise monotone, have nonnegative (stochastic) coefficients and a genuine two-player Bellman structure, and still the strict-inequality observation “halt-register value greater than baseline value”, scalar equality and exact fixed-point attainment are c.e.-complete for suitable fixed games. The one-player case (only MIN or only MAX states) stays open (Section 29.2).

24 Canonical quadratic Diophantine certificates

We now compile the native game recursion directly, rather than importing an MRDP polynomial. Every existential variable below ranges over $\mathbb{N}$. The nonnegative domain is essential.

24.1 Integer scaling of zero-reward runs

Let every transition probability be $P_{ia,j}/D$, with $P_{ia,j} \in \mathbb{N}$ and $\sum_j P_{ia,j} = D$. Let the discount be $1/2$ and $v_0 = A/d_0$, where $A \in \mathbb{N}^N$ and $d_0 \geq 1$. Define

$$X_t = d_0(2D)^t v_t. \quad (75)$$

For zero rewards the recurrence becomes

$$X_{t+1,i} = \text{opt}_{a \in A_i} L_{ia}(X_t), \quad L_{ia}(X) = \sum_j P_{ia,j} X_j, \quad X_0 = A. \quad (76)$$

In particular, every X_t is integral and nonnegative. The exponentially large denominator is not an uncharged existential exponentiation gadget: T is external, and (76) itself has no exponentiation in its transition constraints.

24.2 A min/max gate with unique gaps

For two nonnegative integers L_0, L_1 , a claimed output Z , and natural gaps u, v , define

$$q_{\min}(L_0, L_1, Z; u, v) = (L_0 - Z - u)^2 + (L_1 - Z - v)^2 + uv, \quad (77)$$

$$q_{\max}(L_0, L_1, Z; u, v) = (Z - L_0 - u)^2 + (Z - L_1 - v)^2 + uv. \quad (78)$$

Lemma 24.1 (Canonical complementarity). *On natural inputs, (77) vanishes exactly when $Z = \min(L_0, L_1)$ and $(u, v) = (L_0 - Z, L_1 - Z)$. Formula (78) has the analogous statement for the maximum. These polynomials are nonnegative on the entire real nonnegative orthant.*

Proof. Every displayed summand is nonnegative on that orthant. A zero therefore forces both linear residuals and uv to vanish. In the minimum case, $L_0 = Z + u$ and $L_1 = Z + v$, so Z is a lower bound; since $uv = 0$, it equals at least one of the two values. The gaps are then forced. The maximum case is identical with inequalities reversed. Conversely, the asserted output and gaps make every term zero. $\square$

When the two actions tie, both gaps are zero. There is no selector bit whose value can be chosen in two ways. Also, uv is *not squared*. Squaring it would unnecessarily raise the degree from two to four. Its sign is already controlled by the domain. This use of complementarity is consistent with the repository's prior quadratic neural-certificate work [7]; the contribution here is its full game-level specialization and resource ledger. ([write] The repository work meant is source 15 of Part VII of [21]: Lemma `pqc:pm:lem:clamp`, an unsquared complementarity certificate on $\mathbb{N}$ of degree two, and Theorem `pqc:pm:thm:trace`, one degree-two polynomial with canonical scales and a unique natural zero; compare also `pqc:sc:lem:relucert`. Lemma 24.1 specializes that device to minimum and maximum gates; the game-level gate types and counts are this source's.)

For a one-action state use the single square $(Z - L_0)^2$.

24.3 The complete polynomial

For each step $t = 0, \dots, T - 1$ introduce the next state vector X_{t+1} , and two gaps for each binary-decision state. Sum the state terms (77), (78), or the one-action square. Add an endpoint sum E_T . The resulting *one polynomial* is

$$Q_T(A; W) = \sum_{t=0}^{T-1} \sum_{i=1}^N q_{t,i} + E_T. \quad (79)$$

For scalar equality use $E_T = (X_{T,r} - X_{T,s})^2$. For consensus use $E_T = \sum_{i=2}^N (X_{T,i} - X_{T,1})^2$. To test a specified integer endpoint vector Y_T , use $E_T = \sum_i (X_{T,i} - Y_{T,i})^2$.

Theorem 24.2 (Unique natural witnesses at every fixed horizon). *For fixed game data and an externally specified T , the polynomial (79) has integer coefficients and degree at most two, and is nonnegative on the real nonnegative orthant. For each supplied initial vector, its natural zero set is empty when the actual endpoint does not meet the observation, and consists of exactly one complete witness tuple when the observation holds.*

With b binary-decision states and e endpoint squares, it has exactly

$$\boxed{(N + 2b)T \text{ natural witnesses, } (N + b)T + e \text{ squared linear forms, } bT \text{ products.}} \quad (80)$$

Here $e = 1$ for a scalar pair, $e = N - 1$ for consensus, and $e = N$ for a specified full endpoint. These are counts of the stated construction, not minimality results.

Proof. There are N next-state coordinates and $2b$ gaps per step. A one-action state contributes one squared linear form; a binary state contributes two squares and one complementarity product. This gives the ledger.

All residuals are linear in the variables and all products have degree two. Nonnegativity prevents cancellation between any two terms. At a zero, starting from the supplied X_0 , Lemma 24.1 and the one-action constraints force every X_1 coordinate and every first-step gap. Induction forces the whole trajectory and all gaps. The endpoint squares then hold exactly when the actual endpoint satisfies the observation. Conversely that trajectory and those forced gaps make all terms vanish whenever the observation holds. The case $T = 0$ has no existential variables and reduces to the endpoint test on X_0 ; the unique tuple in the true case is the empty tuple. $\square$

This proves uniqueness of the *entire* witness tuple, not just of a projected trajectory. It does not assert that a first-hit witness is unique when time is also allowed to vary.

24.4 The fixed-point theorem's nonzero reward

For $\Phi(v) = \frac{1}{4}\mathbf{1} + \frac{1}{2}S(v)$ use the slightly enlarged scaling

$$X_t = 4d_0(2D)^t v_t, \quad X_0 = 4A. \quad (81)$$

Then

$$X_{t+1,i} = \text{opt}_a \left(\sum_j P_{ia,j} X_{t,j} + d_0(2D)^{t+1} \right). \quad (82)$$

For fixed T , every power appearing here is a known integer coefficient. The expression is linear in X_t and d_0 . To test the actual fixed point $\frac{1}{2}\mathbf{1}$, use

$$E_T = \sum_{i=1}^N \left(X_{T,i} - 2d_0(2D)^T \right)^2. \quad (83)$$

Thus the full fixed-point-attainment certificate is still quadratic, including in its displayed input parameters A, d_0 . Its count is $(N + 2b)T$ witnesses, $(N + b)T + N$ squares, and bT products. The external input conditions are $d_0 \geq 1$ and $0 \leq A_i \leq d_0$ when the payoff vector is required to lie in the unit cube.

For completeness, the implementation permits any rational discount $\delta = a/b_0 \in (0, 1)$ and any nonnegative common rational reward r . Let D clear the transition probabilities, let q be a rational constant point target when one is prescribed, and choose

$$m = \text{lcm}(\text{den}(r), \text{den}(q)), \quad \lambda = md_0,$$

with target denominator one when there is no point target. Then

$$X_t = \lambda(b_0 D)^t v_t, \quad X_0 = mA,$$

uses the integer recurrence

$$X_{t+1,i} = \text{opt}_a \left(a \sum_j P_{ia,j} X_{t,j} + \lambda(b_0 D)^{t+1} r \right).$$

Every forcing term is an emitted integer; no unnamed relation remains in the exported polynomial. The endpoint $X_{T,i} = \lambda(b_0 D)^T q$ is also integral. For the main theorem this rule gives exactly (81)–(83).

24.5 Size and precision

For a zero-reward run, $\max_i X_{t,i} \leq D^t \max_i A_i$ and each gap is at most $D^{t+1} \max_i A_i$. The bit length of every witness is therefore $O(\text{bits}(A) + T \log(D + 1))$. For the common-reward construction and cube inputs, $X_t \leq 4d_0(2D)^t \mathbf{1}$, giving the analogous linear-in- T bit-length estimate. The number of witnesses is also linear in T .

These bounds concern a horizon supplied in unary or otherwise charged by its numerical value. They are not polynomial bounds in $\log T$. The large dense reset rows can make the unexpanded polynomial sizeable: the count of variables alone does not count the coefficient storage or the work needed to evaluate all dense linear forms.

25 Worked examples and complete emitted objects

25.1 A two-state polynomial that can be checked by hand

Consider the zero-reward game with discount $1/2$ and effective action rows

$$\begin{aligned} \text{MIN state:} & \quad (1/2, 1/2), (1/4, 3/4), \\ \text{MAX state:} & \quad (3/4, 1/4), (1/2, 1/2). \end{aligned}$$

It is obtained from a half-uniform reset on an averaging/copy game. Start at $v_0 = (1, 0)$. The discounted trajectory begins

$$v_1 = (1/8, 3/8), \quad v_2 = (1/8, 1/8).$$

Here $D = 4$, and $X_t = 8^t v_t$ gives $X_1 = (1, 3)$ and $X_2 = (8, 8)$. Use trajectory variables $(x, y) = X_1$, $(z, w) = X_2$, and gaps a, b, c, d, e, f, g, h . The complete scalar-equality polynomial is

$$\begin{aligned} Q = & (2 - x - a)^2 + (1 - x - b)^2 + ab \\ & + (y - 3 - c)^2 + (y - 2 - d)^2 + cd \\ & + (2x + 2y - z - e)^2 + (x + 3y - z - f)^2 + ef \\ & + (w - 3x - y - g)^2 + (w - 2x - 2y - h)^2 + gh \\ & + (z - w)^2. \end{aligned} \tag{84}$$

Its unique natural zero is

$$(x, y, z, w) = (1, 3, 8, 8), \quad (a, b, c, d, e, f, g, h) = (1, 0, 0, 1, 0, 2, 2, 0).$$

The expanded polynomial has 45 nonzero monomials. Both the complete factored polynomial and its expansion are delivered. The machine-readable variable ordering interleaves each time slice's coordinates and gaps; its label array specifies that ordering explicitly.

This small game is not universal. Its purpose is to expose the actual Diophantine construction, including its degree, natural domain, tie behavior, and full witness count.

25.2 A compiled countdown program

The source has one counter and two control states:

State	Instruction
0	If the counter is zero, go to 1; otherwise decrement it and stay at 0.
1	Halt.

From state zero with counter two, halting is reached after three source steps. The complete compiler exports two versions.

Quantity	Absorbing/scalar version	Erasing/fixed-point version
Padded game states N	128	128
Binary-decision states b	12	10
Layered nodes, including inputs	52	41
Pipeline period M	8	8
Normalization K	128	64
Common reward	0	1/4
Initial payoff denominator	512	256
First observed event	equality at 17	full fixed point at 30
Natural witnesses at that horizon	2,584	4,440
Squared linear forms	2,381	4,268
Complementarity products	204	300

Both use discount $1/2$, a half-uniform reset, and dyadic full-support rows. The first equality time is $2M + 1 = 17$, as (72) predicts. In the erasing version, Lemma 20.2 gives the safe full-drain bound $(H + 1)M = 32$; the actual full fixed point is attained earlier, at time 30. The test checks all coordinates at every preceding time, not only the two observed registers.

The numerical tables above describe this illustrative one-counter program. They are not a claim of universality with 128 states. The game exports include every base action row and the explicit common-reset fraction. The certificate exports have the reset already expanded into all integer linear forms; they do not leave transition semantics as an oracle.

26 Reproducibility, checks, and verification limits

The archive has one standard-library implementation, a test/export driver, and a separate checker that does not import the generator. Python 3.13.5 was used; the code uses language features available in Python 3.10 and later. No floating-point approximations, numerical optimization, or random search for integer roots are used. The reproducible pseudorandom seed is 20261002; it controls only finite test cases, not theorem selection or proof validity.

The code exports all linear forms and all complementary products. Thus `countdown_certificate.json` and `fixedpoint_certificate.json` each specify one complete polynomial in factored form, a full variable list, a complete natural assignment, and a resource ledger. Factored representation avoids a gratuitous expansion of the dense reset terms. The 12-witness example additionally includes its fully expanded sparse polynomial.

[write] The two large certificates are *not stored* in this report: `countdown_certificate.json` (11,921,969 bytes) and `fixedpoint_certificate.json` (21,813,796 bytes) are regenerated exactly by the shipped programs, and the README of this report says how and gives their SHA-256 hashes. They also survive in the original archive of the arrival commit. The other delivered artifacts are shipped as `data/10-exact-convergence-*.json`; the delivered `artifacts/test_run.txt`, a byte copy of `verification.json`, is not.

To regenerate the examples and tests:

```
python code/run_checks.py
python code/check_certificate.py artifacts/fixedpoint_certificate.json \
    --game artifacts/fixedpoint_game.json
```

[write] These commands use the delivered layout: `code/run_checks.py` imports `bellman_diophantine` and `check_certificate` by their delivered names and rewrites every file of `artifacts/`. In this report the programs are `code/10-exact-convergence-*.py`; the README gives a recipe that runs on a copy.

The recorded run contains 26,077 assertions. Selected categories are:

Check category	Assertions
Exhaustive bounded local min/max certificate equivalences	17,150
Raw counter-map successor comparisons	640
Layered counter-map successor comparisons	640
Positive homogeneity comparisons	640
Common-reset coordinate-difference identities	1,920
Original-pipeline checks at every microstep	248
Erasing-pipeline checks at every microstep	100
Chance-only Skolem-lift exact identities	225
Expanded versus factored small-polynomial evaluations	100
Each countdown witness coordinate incremented individually	2,584
Independent complete-export checks	3

The full category list, including ties, input rejection, time-zero cases, nonzero rewards, rational targets, and initially halted pipelines, is in `artifacts/verification.json`. ([write] Shipped as `data/10-exact-convergence-verification.json`. The suite was rerun at the write on a copy with the delivered layout: 26,077 assertions, and all eight regenerated exports equal the delivered files apart from line endings.)

For the 2,584 coordinate-mutation checks, the exact change in the polynomial under a unit increment of one coordinate is computed from its squared-linear coefficients and the paired gap values; the test does not rerun a dense polynomial thousands of times. Separate small-instance tests directly evaluate mutated assignments. The distinction is recorded to avoid inflating what the test suite independently establishes.

The independent checker validates natural domains, variable indices, counts, and the integer value of the complete exported polynomial. When a game export is supplied, it reconstructs the reset rows independently, checks the input payoff vector, and recomputes every scaled trajectory coordinate and every gap. This catches serialization or exporter mistakes that a generator-only self-check could miss.

[write] *A defect in the delivered checker.* The research programme’s review of this manuscript [24] found that the checker compares the declared input with the game’s initialization but replays the trajectory from a separate scaled initial vector that it does not bind to that input. Changing only the declared input numerators of a genuine one-step certificate, and supplying a game with the same changed initial payoffs, still yields polynomial value zero and a successful independent game check although the actual endpoint fails the observation. The integer polynomial itself remains an honest certificate for its own input; the false assertion is the checker’s binding. The review’s patch `exact_convergence_input_binding.patch` binds the input vector, denominator, witness scale, reward, discount, endpoint mode and game, and leaves every valid delivered example accepted; the review also reconstructs every coefficient of the three delivered polynomials from their games and inputs. The checker shipped here, `code/10-exact-convergence-check_certificate.py`, is the unpatched original.

None of this proves an infinite theorem by testing. The inductive proofs in Sections 19–24 carry that obligation. No Lean, Rocq, or Isabelle formalization of the new game compiler was produced or run. The repository’s MRDP interface is an inspected source, not a formal verification of these additional constructions.

27 From bounded quadratics to a fixed Diophantine relation

For the fixed game of Theorem 17.1, encode each valid rational payoff vector by a natural number. Its finite-attainment set E is c.e. and c.e.-complete. MRDP, in precisely the fixed-arity form exposed by the repository [15], yields

$$\exists k \exists P \in \mathbb{Z}[n, y_1, \dots, y_k] \forall n \in \mathbb{N} : \quad n \in E \iff \exists y \in \mathbb{N}^k P(n, y) = 0. \quad (85)$$

The ordinary polynomial P and its witness dimension are fixed before the input. We do not print its coefficients or claim an operation count for it. A constructive formal route to MRDP through arithmetic computation is also described in the published Coq development [14].

The bounded quadratics do not themselves give a fixed quadratic polynomial for unbounded time: Q_T has $(N + 2b)T$ witnesses, and its time-dependent coefficients in the reward version also depend on T . Writing “ $\exists T, W \ Q_T = 0$ ” does not turn a varying-arity family into one first-order polynomial equation.

At each fixed horizon the transition graph is described by finitely many linear equalities and inequalities, one finite collection for each action comparison pattern. It is consequently Presburger-definable on its integer parameters. The quadratic complementarity package compresses this finite union into one equation over the orthant. The computational difficulty arises from unbounded iteration, not from a hidden claim that one fixed quadratic polynomial represents every enumerable set.

27.1 A quartic existence corollary, with the correct caveat

One can convert any fixed integer polynomial in (85) into a fixed sum of squares of quadratic residuals. Separate its positive and negative parts, evaluate each by additions and multiplications on natural registers, and impose equality of their outputs. A multiplication gate has residual $z - xy$ and an addition gate residual $z - x - y$. Squaring and summing produces a fixed nonnegative polynomial of degree at most four with the same projected zero set.

The added circuit registers are uniquely determined by the original witness y . The original MRDP witnesses are not thereby made unique. Thus neither the single-fold nor the finite-fold Diophantine representation problem is settled. The horizon-indexed uniqueness in Theorem 24.2 is deliberately a different assertion.

[write] The circuit argument of this subsection is the unique-extension quadratization `cdc:wf:lem:quadratization` of [22] and `pqc:lem:quarticlower` of [21]; Part I proves the same corollary for exact erasure (Proposition 11.1). No novelty is claimed for it.

27.2 No computable global search cutoff

Corollary 27.1. *For the fixed universal game there is no total computable bound, as a function of the length of the input payoff encoding, on the first exact attainment time of all inputs that eventually*

attain the fixed point. Likewise, for any fixed polynomial representation of its attainment set, there is no total computable input-length bound that guarantees a solution with all witness coordinates below that bound whenever a solution exists.

Proof. Either bound would turn semidecision into decision: compute the bound and search the finite time interval, or the finite witness box. That would decide a c.e.-complete set. $\square$

[write] This is the bounded-search argument of `cdc:bd:prop:nobound` in [22]; for contractions in the collection it appears as `pqc:sc:cor:gap` and `pqc:ad:cor:nobound` in Part VII of [21], and in Part I as Theorem 11.2.

The repository README read during this audit reports 75 arithmetic operations for its complete certificate and 87 for one universal polynomial [5]. Nothing here changes either figure. Our counted objects are native game tables and horizon-dependent quadratic certificates, not a competing numerically instantiated universal Diophantine polynomial.

28 Precision, robustness, and why approximation is not a decision

The main operator’s fixed point is explicit and its approximation error is uniformly bounded. On compiled initializations the stronger estimate

$$\|\Phi^n(v(x)) - v_*\|_\infty \leq 4^{-n}/2$$

holds whether or not the source computation halts. Therefore any prescribed absolute accuracy is obtained after a computably bounded number of steps, while exact finite attainment remains undecidable.

For the absorbing scalar construction, a nonzero prehalting register gap has the exact value in (71). If $K = 2^\kappa$, its magnitude is

$$2^{-(1+2n+\kappa\lceil n/M \rceil)}.$$

For a known finite horizon, its bit requirement is explicit. Two separately approximated coordinates with absolute error less than one quarter of this gap can distinguish it from zero, provided a corresponding promise about the two alternatives is available. There is no contradiction with the lack of a computable bound on how far one must look for a future exact zero. Precision requirements and time requirements must not be conflated.

Proposition 28.1 (A fixed positive threshold restores a finite cutoff). *For a zero-reward discounted game on $[0, 1]^N$ and a supplied rational $\varepsilon > 0$, the property $\exists n (v_{n,r} - v_{n,s} \geq \varepsilon)$ is decidable.*

Proof. Every coordinate of v_n lies in $[0, \delta^n]$, so the difference is at most δ^n . Once $\delta^n < \varepsilon$, no later time can qualify. Compute such a cutoff and check the finite prefix exactly. $\square$

[write] The same elementary argument decides threshold reachability for source 14’s map in Part VII of [21] (Propositions `pqc:sc:prop:away` and `pqc:sc:prop:threshold`).

A positive absolute tolerance does not solve the zero-threshold predicate: it changes the predicate. Strict inequalities with threshold zero can still encode halting, as Section 22 shows. Similarly, full support is not arbitrary robustness against perturbing the transition table. The theorem proves preservation under one very specific, exactly known common reset. It does not say that all nearby

games have the same halting language, or that rounded floating-point implementations retain universality.

Finite-precision implementations have finitely many representable states under a specified rounding and overflow policy. Their exact reachability can in principle be decided by finite exploration. That fact is compatible with the exact-rational result, because the required precision is not bounded uniformly over all running times.

29 Research agenda and concrete next targets

29.1 Instantiate and minimize a fixed universal game

The most direct next deliverable is to choose an explicit published universal counter program, enter its entire instruction table, and export the resulting game and proof certificate. The output should include the natural-input loader, not merely a claim about a universal circuit. Minimize the number of states, binary decisions, stochastic successors, normalization factors, and the maximum common denominator separately. These objectives need not agree. The present depth-seven source compiler makes the microstep period bounded by eight, but it is not a size-optimal construction.

A particularly concrete optimization is to share delay copies, normalized linear forms, and zero tests while preserving the two-channel invariant. Any change must be checked against the actual feedback schedule and full pipeline drain, not only against one-step circuit equivalence.

29.2 Remove one player’s genuine choices

Does an analogous fixed-game finite-attainment theorem hold when every nontrivial decision state belongs to MAX, or all belong to MIN? The present proof does not remove the opposite owner: the negative channel of a minimum is a maximum. Any attempted one-player translation must replace this exact dependency, rather than silently switching objectives. This question is close to the remaining Bellman-operator reachability frontier in [10], but their exact hypotheses and observation must be matched before claiming a resolution.

Even a restricted subclass with either a proof of undecidability or a uniform decision procedure would clarify which portion of the two-player compiler carries the essential nonlinearity.

29.3 Prescribed randomness and sparse supports

Can all chance computations be routed through one shared randomizing state, or through a constant-size family of fair-coin gadgets? Can one retain a constant number of stochastic successors per action before the common reset? A global reset and sparse support cannot both describe the same expanded row when N grows, so the cost model must distinguish a native “reset to uniform” instruction from its explicitly expanded probability table.

The current theorem uses an exact common reset. It would be valuable to characterize all reset kernels that preserve the simulated quotient modulo $\mathbb{R}\mathbf{1}$, and all nonuniform resets that still admit an invariant paired-channel subspace. This is a precise algebraic question about the reset’s action on V , not a claim of unrestricted noise robustness.

29.4 One-dimensional observations versus whole-vector attainment

The scalar/whole-vector distinction is already decisive for chance-only systems: Skolem-equivalence on one side, kernel stabilization on the other. Investigate the minimum number of observed coordinates required for hardness in one-player stochastic systems. Related targets include a specified affine subspace, a consensus line, a nonzero point, and a fixed point. They should be studied as distinct problems rather than collapsed under the word “reachability.”

For the two-player construction, can full fixed-point attainment be reduced to equality of one designated pair with no additional input promises? The present reductions give completeness of each problem separately; they do not furnish a global equivalence on every arbitrary initialization.

29.5 Witness-economical native certificates

The quadratic compiler spends two natural gaps per binary comparison. Determine the minimum arity of a single-fold orthant-nonnegative quadratic representation of the binary min and max graphs under a fixed interface. One gap can encode distance from one action but loses the sign of the other difference; naive elimination risks increasing degree or introducing multiple witnesses. A useful result would be a proved lower bound under clearly stated interface restrictions, or an explicit smaller compiler.

Time-dependent reward terms are emitted as integer coefficients. A shared canonical clock could trade coefficient size for additional variables. Compare the two representations using bit complexity and literal evaluator costs, not only total polynomial degree.

29.6 Integrate with the repository’s fixed-arity MRDP route

A formal interface should expose exact game-step evaluation, its computability, and the semidecision procedure for the chosen observation. The repository’s MRDP theorem can then supply the abstract fixed polynomial. Producing a fully expanded native-to-MRDP compiler with audited operation counts is a separate engineering and mathematical project.

Track three properties independently: preservation of the accepted input set, preservation of projected trajectories, and preservation of complete witness tuples. The present bounded compiler proves all three at fixed T . An unbounded history compression must not inherit the strongest property without a separate proof.

29.7 Formalize the proof-critical interfaces

The highest-value formalization targets are the counter encoding, layer-normalization invariant, every-microstep pipeline lemma, erasing pipeline drain, common-reset identity, and the zero-set uniqueness theorem. These are short enough to be individually inspectable and strong enough to expose most plausible compiler mistakes.

A proof-carrying exporter could attach a machine-checkable derivation of each stochastic row and each quadratic residual. The independent checker in this archive is a useful testing boundary, but it does not replace a kernel-checked theorem connecting exported coefficients to the original counter semantics.

30 Conclusion

The main result isolates a computationally universal exact iteration inside a tightly constrained stochastic-game interface. Its transition rows are positive and dyadic; its discount is fixed; its limiting value is obvious; and its error decays geometrically. None of these facts supplies a decision procedure for finite exact attainment from arbitrary rational initial values.

The construction’s nontrivial obligations are explicit: signed values are realized by paired channels, nested computation is realized by a cyclic pipeline, initialization accounts for every phase, and halting erases the entire pipeline rather than only its visible registers. Uniform resetting preserves the computation in a precise invariant subspace.

At fixed time, the same substrate has a particularly economical arithmetic semantics: one orthant-nonnegative quadratic polynomial with a unique complete natural witness whenever its endpoint condition holds. At unbounded time, MRDP supplies a fixed Diophantine relation, but not the bounded family’s uniqueness. The chance-only and selector-only boundaries explain why both the observation and the computational primitives must be specified carefully. These distinctions are the foundation for further native compilers and formal developments, rather than shortcuts around them.

Part III

Thermal arithmetic (source 03)

[write] Part III prints batch-78 manuscript 03, *Thermal Arithmetic: Diophantine Witnesses, Effective Excited Spectra, and Undecidable Confinement Boundaries*, under the kicker “ProveIt research continuation”, “Research report prepared for Vladimir Reshetnikov”, October 2, 2026 (the manuscript names no assistant). Its PDF metadata gave the shorter subtitle *Diophantine Witnesses and Undecidable Confinement Boundaries*; the printed subtitle is used here. Its labels carry the prefix `ste:ta:`. Its abstract and status statement follow; its appendix is [Appendix E](#).

Abstract (source 03). An existential Diophantine representation controls whether witnesses exist, but generally not how many there are or how far away they lie. We construct a witness-faithful spectral normal form that separates these difficulties. A natural arithmetic circuit with k witness inputs and g materialized gates gives a diagonal, nonnegative, integer-valued Hamiltonian on $m = k + g + 1$ occupation modes, of joint polynomial degree at most five, with an explicit decomposition into positive commuting terms supported on at most four modes. Its zero-energy basis states are in bijection with the original witnesses, while every positive-energy state satisfies $E \geq 1 + \sum_i n_i$.

This radial isolation makes the positive-energy resolvent effectively compact and the excited thermal trace uniformly computable, even when the zero-energy space is infinite. Compact resolvent and finite unconfined thermal trace are equivalent to finite witness multiplicity. A free-witness padding construction gives unconditional undecidability of both properties on a fixed universal family; requiring compactness on every input instead is exactly a finite-fold representation condition.

A second construction produces a uniformly computable critical fugacity $t_*(c) \in (0, 1]$ for each encoded computation. The exact assertion $t_*(c) < 1$ is equivalent to halting, although every regulated partition function and every finite-precision approximation to $t_*(c)$ is computable. We prove explicit tail estimates, quantitative witness-height bounds, and the absence of uniform

computational cutoffs. An exact Python compiler and 7,936 passing regression assertions accompany the proofs.

Status (source 03). The proofs below establish the stated normal-form and boundary results from standard Diophantine representability. Their priority in the literature is not established. This report does not resolve the finite-fold conjecture, claim physical hypercomputation, provide a numerical universal polynomial, or claim proof-assistant verification.

31 Research question, scope, and relationship to ProveIt

31.1 Why another computational substrate?

The central question of this report is not simply whether a Turing-complete model has a Diophantine halting predicate. Once a model has an effective finite observation semantics, standard representability already supplies such a predicate. The more informative question is whether an arithmetic representation can separate three independent obstacles:

existence of witnesses, multiplicity of witnesses, height of witnesses.

We use a finite number of occupation-number modes, each with infinitely many basis states. The substrate is a static polynomial energy landscape, not a step-by-step simulator. Its zero-energy questions represent computation; its positive-energy states have effective, uniform confinement. In particular, infinitely many witnesses can coexist with an entirely computable excited-state thermal trace.

The repository’s October 2 review of computational substrates distinguishes horizon-dependent certificates from one fixed universal polynomial, and existence-preserving compilation from witness-preserving compilation. It also identifies uncontrolled multiplicity in a fixed-quartic MRDP route [6]. ([write] The fixed-quartic route meant is Corollary `pqc:tu:cor:fixedquartic` of [21], Part VIII, and the remark after it, which the cited review discusses.) Here multiplicity is no longer an optional implementation detail: it is exactly the obstruction to compact resolvent and to a finite unconfined thermal trace. The present construction builds on that distinction rather than proposing another uncharged bounded-history encoding.

31.2 Source snapshot and limits of the audit

The repository was inspected at commit

6914ccca6685baf53b7a35f25efc89366c76ba74.

The relevant audit file has blob identifier

8bd07142c8b4298fb636e47b40d2d84e993a40ce.

The repository’s reported 87-operation fixed universal polynomial remains a separate arithmetic-circuit result [5]. This report neither improves that bound nor counts an existential MRDP polynomial as a free primitive in a numerical operation ledger. The source inspection was not a rebuild of the Lean or Rocq developments, and the proofs below do not assume that every research manuscript in the repository has been formally verified.

31.3 Established foundations and the proposed contribution

We use the standard fact that every computably enumerable relation over natural numbers has an existential polynomial representation, and that one fixed polynomial can represent a universal such relation [16]. A natural arithmetic-circuit expansion then replaces a high-degree polynomial equation by quadratic gate equations with uniquely determined intermediate values. We give that expansion explicitly because its witness multiplicity matters here. ([write] This expansion is the unique-extension quadratization of the collection’s report [22], Lemma `cdc:wf:lem:quadratization`, with the definition of witness-faithfulness given there; see also `cdc:cs:lem:quartic` in [22] and, without the uniqueness statement, `pqc:lem:quarticlower` in [21]. In Theorem 32.1 the zero-set bijection and the quartic sum of squares are that lemma; new here are the marker $a - 1$, the radial factor $1 + \sum_i n_i$, the bound $H > 0 \Rightarrow H \geq 1 + \sum_i n_i$, the degree five and the $r(m + 1)$ four-mode ledger.)

The basic substitution of number operators into a squared Diophantine polynomial is not new. Kieu already writes a Hamiltonian of the form $D(N_1, \dots, N_k)^2$ and identifies zero energy with an integer solution [17]. We use only that mathematical observation, not any proposed physical decision procedure. Our additional factor $1 + \sum_i N_i$ removes all nonzero-energy flat directions while leaving the zero set unchanged. The effective spectral decomposition, multiplicity classification, and critical-fugacity construction are the results developed and proved here. ([write] Part V of [21] (source 13 there, Theorem `pqc:ql:thm:compiler`) builds a witness-faithful quartic energy landscape for Boolean circuits over $\mathbb{R}$ and states that it treats no Gibbs distribution; this Part is the unbounded natural counterpart, with a thermal trace.)

Finite-fold and single-fold representation questions are discussed in the primary literature [18, 19]. The equivalences below are reformulations of those representation conditions, not answers to them. In particular, we do not silently replace an arbitrary MRDP representation by one with finitely many witnesses.

31.4 Main conclusions at a glance

Let $Q = \sum_j f_j^2$ be the compiled quartic penalty and $s = \sum_i n_i$. The construction is

$$H = (1 + s)Q. \tag{86}$$

Its essential features are:

- (i) $\ker H$ has one occupation basis vector for each original witness, with no extra auxiliary multiplicity.
- (ii) The positive-energy sector is effectively confined: $H > 0$ implies $H \geq 1 + s$.
- (iii) Compact resolvent and finite thermal trace hold exactly when there are finitely many witnesses.
- (iv) A regulated partition function has a computable threshold location whose exact contact with the boundary $t = 1$ encodes nonhalting.

The novelty claim is deliberately limited: this is a proved package of normal-form, effective-estimation, and boundary results proposed for further research. The literature search and repository audit do not establish that every individual consequence first appears here.

32 Arithmetic conventions and the witness-faithful compiler

32.1 Natural domains and representations

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A parameter tuple $p \in \mathbb{N}^d$ is fixed when studying an individual operator. Witness coordinates range independently over $\mathbb{N}$. A polynomial $P \in \mathbb{Z}[p, x_1, \dots, x_k]$ represents a relation by

$$A(p) \iff \exists x \in \mathbb{N}^k P(p; x) = 0. \quad (87)$$

A representation is *finite-fold* if each parameter fiber has finitely many witnesses, including zero witnesses outside the relation. It is *single-fold* if each fiber has at most one witness. A representation originally written over positive integers can be moved to our convention by the bijective substitution $x_i = u_i + 1$.

All polynomial degrees stated below are total degrees jointly in parameters and occupation variables unless explicitly qualified. Parameter values can be arbitrarily large, but the polynomial expression and the number of coordinates do not change with their values.

32.2 Natural arithmetic circuits

Write

$$P = P^+ - P^-, \quad P^+, P^- \in \mathbb{N}[p, x].$$

Choose finite directed acyclic circuits for these two nonnegative polynomials using constants in $\mathbb{N}$, parameter and witness inputs, and binary addition and multiplication. Common subexpressions may be shared. Each actual gate introduces one natural coordinate u_j , in a topological order. The corresponding residual is one of

$$u_j - v - w, \quad u_j - vw, \quad (88)$$

where v, w are constants, parameters, original witness coordinates, or outputs of earlier gates. The final residual is the difference between the two output atoms. A final, separate coordinate a has residual $a - 1$.

The purpose of a is not arithmetical expressiveness. It makes the occupation vacuum a positive-energy state, which will ensure that the thermal threshold starts strictly below its target level. It adds exactly one coordinate and one equation, and is uniquely determined at every zero.

Theorem 32.1 (Witness-faithful degree-five compiler). *Suppose the chosen natural circuit has k witness inputs and g materialized gates. There is an effectively constructed integer polynomial $H(p; n)$ with*

$$m = k + g + 1, \quad r = g + 2, \quad (89)$$

occupation coordinates and residual equations, respectively, satisfying the following properties.

For every fixed p , projection onto the first k coordinates is a bijection

$$\{n \in \mathbb{N}^m : H(p; n) = 0\} \longrightarrow \{x \in \mathbb{N}^k : P(p; x) = 0\}.$$

Every occupation value of H is a nonnegative integer, $H(p; 0) \geq 1$, and

$$H(p; n) > 0 \implies H(p; n) \geq 1 + \sum_{i=1}^m n_i. \quad (90)$$

The joint degree of H is at most five. It has a displayed decomposition into $r(m + 1)$ nonnegative commuting terms, each supported on at most four occupation modes.

Proof. Let $f_1, \dots, f_r$ be all gate residuals, the final output difference, and $a - 1$. Put

$$Q(p; n) = \sum_{j=1}^r f_j(p; n)^2, \quad H(p; n) = \left(1 + \sum_{i=1}^m n_i\right) Q(p; n).$$

Since every residual is an integer, $Q = 0$ exactly when all residuals vanish. Fix p and the original witness tuple x . Topological induction through the circuit forces each u_j to be the natural value of its gate. Thus there is exactly one assignment of intermediate coordinates satisfying the gate equations. The marker is then uniquely $a = 1$. At this assignment the remaining output residual is $P(p; x)$. Consequently a source root has exactly one zero-energy lift, and every zero-energy occupation tuple is such a lift.

Each residual has joint degree at most two. Therefore $\deg Q \leq 4$ and $\deg H \leq 5$. Every factor $1 + \sum_i n_i$ is positive on the natural domain, and every nonzero Q is an integer at least one. This proves (90). At the vacuum, the marker residual equals -1 , so $Q(p; 0) \geq 1$.

Finally, expand only the radial factor, not the residual squares:

$$H = \sum_{j=1}^r f_j^2 + \sum_{i=1}^m \sum_{j=1}^r n_i f_j^2. \quad (91)$$

A gate residual involves at most three occupation modes; the output residual involves at most two, and the marker one. Multiplication by n_i adds at most one mode to the support. Each displayed term is nonnegative on the natural domain. As diagonal multiplication operators all these terms commute. There are $r + mr$ terms in this unsimplified display. $\square$

Remark 32.2 (What this ledger does not count). The number $r(m + 1)$ counts displayed positive summands, not additions and multiplications in an optimized evaluation circuit. Formula (86) is usually cheaper to evaluate than fully expanding (91). Constant bit lengths are also not charged by this ledger. Four-mode support does not mean nearest-neighbor geometry, bounded local Hilbert-space dimension, or bounded operator norm. If the degree is measured in creation and annihilation operators rather than number operators, substituting $N_i = a_i^\dagger a_i$ gives a bound of ten instead of five.

Corollary 32.3 (Fixed universal family). *There is a fixed m and one fixed integer polynomial $H(c; n)$ of joint degree at most five such that*

$$c \in \mathbf{K} \iff \ker H_c \neq \{0\},$$

where $\mathbf{K}$ is a fixed computably enumerable complete halting set, and every fiber has the properties of Theorem 32.1.

Proof. Choose once and for all a polynomial $U(c; x)$ representing $\mathbf{K}$, as supplied by Diophantine representability. Apply the fixed finite compiler to U . Only c changes from instance to instance. $\square$

No numerical value of m is claimed here: the universal polynomial is not instantiated in the accompanying code. This distinction prevents a low-degree existence theorem from being mistaken for a competitive numerical universal equation.

32.3 A small example with a complete resource count

For $P(p; x, y) = xy - p$, use one gate $u = xy$. The four occupation coordinates are (x, y, u, a) and

$$Q_p = (u - xy)^2 + (u - p)^2 + (a - 1)^2, \quad (92)$$

$$H_p = (1 + x + y + u + a)Q_p. \quad (93)$$

Thus $k = 2$, $g = 1$, $m = 4$, $r = 3$, and there are 15 displayed positive terms. For $p = 6$, the four zero-energy tuples are

$$(1, 6, 6, 1), (2, 3, 6, 1), (3, 2, 6, 1), (6, 1, 6, 1).$$

For $p = 0$, infinitely many tuples occur because $xy = 0$ has infinitely many natural solutions. The circuit has preserved that distinction exactly; it has not attempted to regularize the source multiplicity.

33 The operator and its effectively confined positive spectrum

33.1 Domains are part of the construction

Fix the parameters and suppress them from the notation. Let

$$\mathcal{H}_m = \ell^2(\mathbb{N}^m), \quad N_i |n\rangle = n_i |n\rangle.$$

For the polynomial $H(n)$, define the multiplication operator with maximal domain

$$\mathcal{D}(H) = \left\{ \psi = (\psi_n) : \sum_n H(n)^2 |\psi_n|^2 < \infty \right\}, \quad (H\psi)_n = H(n)\psi_n. \quad (94)$$

Although the formal expression is a polynomial in commuting number operators, (94) specifies the actual closed operator used in all statements.

Lemma 33.1 (Self-adjoint realization). *The operator (94) is nonnegative and self-adjoint. Its restriction to finitely supported vectors is essentially self-adjoint, with closure equal to (94).*

Proof. Symmetry follows from the real diagonal coefficients. If ψ belongs to the adjoint domain of the finite-support restriction, testing against each basis vector forces the adjoint coordinates to be $H(n)\psi_n$. Membership of that vector in ℓ^2 is precisely (94). Thus the adjoint of the restriction is the maximal multiplication operator. The maximal operator is symmetric, and the same coordinate argument identifies its adjoint with itself.

For the core assertion, truncate any $\psi \in \mathcal{D}(H)$ to a finite box. Both the ℓ^2 tail of ψ and that of $H\psi$ tend to zero. Hence finite-support truncations converge in graph norm. Nonnegativity follows from $H(n) \geq 0$ coordinatewise. $\square$

33.2 Energy layers and an exact finite search

For $n \in \mathbb{N}^m$, put $s(n) = \sum_i n_i$. Let

$$\mathcal{Z} = \{n : Q(n) = 0\}, \quad M = \#\mathcal{Z} \in \mathbb{N} \cup \{\infty\}.$$

The radial lower bound is assumed in the next results; consequently they apply both to the compiler and to any directly supplied nonnegative integer polynomial Q used in (86).

Theorem 33.2 (Effective positive-energy layers). *For every integer $E \geq 1$, all basis states with $0 < H(n) \leq E$ lie in the finite simplex $s(n) \leq E - 1$. Their number, with multiplicity, is at most*

$$\binom{E + m - 1}{m}. \quad (95)$$

Every positive eigenvalue is an integer of finite multiplicity, and its exact multiplicity is computable uniformly from the polynomial and the eigenvalue. There is no finite accumulation point of distinct eigenvalues.

Proof. The inequality $H(n) \geq 1 + s(n)$ on nonzero states gives the simplex restriction. The number of natural m -tuples of sum at most $E - 1$ is (95): add a slack coordinate to get a tuple of sum $E - 1$, and count placements of separators. Polynomial evaluation on this finite set decides each positive-energy membership exactly.

The spectrum of a diagonal multiplication operator is the closure of its diagonal values. To see the only potentially nontrivial direction directly, a complex number at positive distance from all those values has a bounded diagonal inverse for $H - \lambda I$; the other direction follows from basis approximate eigenvectors. Here the values are nonnegative integers, so the set of distinct values has no finite accumulation point and is already closed. $\square$

There is also a useful divisor refinement. If $H(n) = E > 0$, then

$$1 + s(n) \mid E, \quad Q(n) = \frac{E}{1 + s(n)}. \quad (96)$$

Thus one may enumerate only shells $s = d - 1$ for divisors $d \mid E$, rather than the entire simplex. This changes the enumeration strategy, not the decidability result.

33.3 Separating the zero projection from a computable compact part

Let P_0 be the diagonal projection onto $\ell^2(\mathcal{Z})$, and let C_B denote projection onto

$$C_B = \{0, 1, \dots, B\}^m.$$

Define K and T_q , for $0 < q < 1$, by their diagonal entries:

$$K(n) = \begin{cases} (1 + H(n))^{-1}, & H(n) > 0, \\ 0, & H(n) = 0, \end{cases} \quad (97)$$

$$T_q(n) = \begin{cases} q^{H(n)}, & H(n) > 0, \\ 0, & H(n) = 0. \end{cases} \quad (98)$$

Theorem 33.3 (Effective compact and trace-class remainders). *The following decompositions hold:*

$$(H + I)^{-1} = P_0 + K, \quad q^H = P_0 + T_q. \quad (99)$$

The operator K is compact, and T_q is trace class. For the finite-rank truncations $K_B = C_B K C_B$ and $T_{q,B} = C_B T_q C_B$,

$$\|K - K_B\| \leq \frac{1}{B + 3}, \quad (100)$$

$$\|T_q - T_{q,B}\|_1 \leq \frac{mq^{B+2}}{(1 - q)^m}. \quad (101)$$

For rational q , these are effective approximation procedures using exact rational finite matrices, uniformly in all polynomial parameters.

Proof. The identities hold on each basis vector. Outside $\mathcal{C}_B$, at least one coordinate is at least $B+1$, so $s(n) \geq B+1$. On positive-energy states this gives $H(n) \geq B+2$. The supremum of the omitted diagonal entries of K is at most $1/(B+3)$, proving (100) and compactness.

For the heat remainder, nonnegative diagonal entries give a trace norm equal to their sum. Since $q^{H(n)} \leq q^{1+s(n)}$ on positive states, a union bound over the coordinate that leaves the box yields

$$\begin{aligned} \sum_{n \notin \mathcal{C}_B, H(n) > 0} q^{H(n)} &\leq q \sum_{i=1}^m \left(\sum_{n_i=B+1}^{\infty} q^{n_i} \right) \prod_{j \neq i} \left(\sum_{n_j=0}^{\infty} q^{n_j} \right) \\ &= \frac{mq^{B+2}}{(1-q)^m}. \end{aligned}$$

The same geometric calculation over the whole space bounds the trace by $q/(1-q)^m$. Exact evaluation of $Q(n)$ tells the finite algorithm which entries are zero and which belong to the remainder. $\square$

Remark 33.4 (Effective strong information is not effective norm information). For each explicitly supplied basis index n , membership in $\mathcal{Z}$ is decidable by evaluating $Q(n)$. The action of P_0 is consequently computable on vectors presented with effective ℓ^2 approximations: project a finite approximation and use $\|P_0\| \leq 1$ to control the tail. This does not supply a uniform finite-rank approximation to P_0 in operator norm. If any zero state lies outside a proposed cutoff, the omitted projection has norm one. The distinction is exactly where an attempted finite spectral decision procedure can fail.

34 Multiplicity, compactness, and thermal finiteness

Theorem 34.1 (Exact multiplicity classification). *For the radially isolated Hamiltonian $H = (1+s)Q$,*

$$\dim \ker H = M.$$

The following are equivalent:

- (i) M is finite;
- (ii) H has compact resolvent;
- (iii) q^H is trace class for some $q \in (0, 1)$;
- (iv) q^H is trace class for every $q \in (0, 1)$.

When $M < \infty$,

$$\mathrm{Tr}(q^H) = M + \mathrm{Tr}(T_q). \tag{102}$$

When $M = \infty$, the essential spectrum is exactly $\{0\}$. When $M < \infty$, the essential spectrum is empty.

Proof. A vector is killed by H exactly when its support lies in $\mathcal{Z}$, proving the dimension statement. By Theorem 33.3, $(H + I)^{-1}$ differs from P_0 by a compact operator. An orthogonal projection is compact exactly when its range is finite dimensional: an infinite orthonormal sequence in its range

has no norm-convergent image subsequence, while a finite-rank projection is compact. This proves (i) $\Leftrightarrow$ (ii).

Similarly, $q^H = P_0 + T_q$ with T_q positive trace class. Its diagonal sum is $M + \text{Tr}(T_q)$, with the usual extended interpretation when $M = \infty$. Thus (i), (iii), and (iv) are equivalent, and (102) follows. The positive eigenvalues are isolated and of finite multiplicity by Theorem 33.2; only the eigenvalue zero can have infinite multiplicity. This proves the essential-spectrum assertions, using the characterization by failure to be an isolated finite-multiplicity eigenvalue. $\square$

Remark 34.2 (Ground states versus zero-energy states). If $M = 0$, the operator still has a ground-state eigenvalue: a nonempty subset of positive integers has a minimum. What fails is specifically the existence of a *zero-energy* state. If $M = 1$, there is a unique zero-energy state up to scalar multiples. These are different statements from uniqueness of a positive-energy ground state in a root-free instance.

34.1 A spectral equivalent of finite-fold representability

Definition 34.3. A polynomial diagonal family is a single integer polynomial $F(p; n)$ in the parameter and finitely many occupation coordinates, nonnegative on all natural tuples, interpreted by the maximal multiplication operator. The number of occupation coordinates is fixed, independent of p .

Theorem 34.4 (Finite-fold/compact-family equivalence). *For a relation $A \subseteq \mathbb{N}^d$, the following conditions are equivalent:*

- (i) *A has a finite-fold natural Diophantine representation.*
- (ii) *There is a polynomial diagonal family F_p with compact resolvent for every p , such that $p \in A$ exactly when $\ker F_p \neq \{0\}$.*

In (ii), whenever (i) holds, one may choose joint degree at most five and the four-mode positive decomposition of Theorem 32.1. The analogous single-fold condition is equivalent to a polynomial diagonal representation with $\dim \ker F_p \in \{0, 1\}$ for every p ; the forward construction can also have compact resolvent on every fiber.

Proof. For (i) $\Rightarrow$ (ii), compile a finite-fold representation. The zero-set bijection preserves finite fiber sizes, and Theorem 34.1 supplies compact resolvent on every fiber.

Conversely, $F(p; n) = 0$ is already a natural polynomial representation of A . Compact resolvent forces every eigenspace, in particular the zero eigenspace, to be finite dimensional. Distinct zero tuples supply orthogonal basis vectors, so there are finitely many such tuples. This is a finite-fold representation. Replacing “finitely many” by “at most one” proves the single-fold assertion in both directions. $\square$

The joint polynomial requirement matters. An arbitrary algorithm that selects a polynomial from p does not itself give one polynomial relation in (p, n) , and invoking representability for that selection may introduce additional witnesses whose multiplicity is uncontrolled. No such substitution occurs in Theorem 34.4.

This equivalence explains why the apparently stronger assertion “there is a universal halting family with compact resolvent on every input” must not be obtained by simply announcing that

circuit auxiliaries are unique. Circuit auxiliaries are unique *over each source witness*; the source fiber may still be infinite.

[write] *Credit and a consequence.* Theorem 34.4 follows the template of the finite-fold substrate equivalence `cdc:cs:thm:finitfold` of [22]: a unique-extension quartic in the forward direction, and the polynomial itself as a representation in the backward direction. What it adds is the replacement of “finite fibre” by “compact resolvent” through Theorem 34.1. Combined with the equivalence `cdc:bd:thm:universal` of [22] it gives the following statement, recorded here as an editorial consequence of the two printed theorems, not as a result of source 03.

Remark 34.5 ([write] All-fibre compactness for universal halting). The following are equivalent: (i) every computably enumerable subset of $\mathbb{N}$ has a finite-fold Diophantine representation over $\mathbb{N}$; (ii) there is one polynomial diagonal family $F(e, x; n)$, in the sense of the definition above, with compact resolvent for every (e, x) and $\ker F_{(e, x)} \neq \{0\}$ exactly when program e halts on input x . When they hold, F can be chosen of joint degree at most five with the four-mode positive decomposition of Theorem 32.1. Indeed, (ii) gives by Theorem 34.4 a finite-fold representation of universal halting, which is statement (b) of `cdc:bd:thm:universal` and hence equivalent to (i), its statement (a); conversely (i) gives (b), and Theorem 34.4 compiles that representation into a family as in (ii). The equivalence decides neither statement.

34.2 An unconditional undecidable compactness family

Theorem 34.6 (Padding transfers halting to loss of compactness). *Let $Q_c(n)$ be the fixed universal compiled penalty in Corollary 32.3. Add one free coordinate $y \in \mathbb{N}$ and define*

$$\tilde{H}_c(n, y) = \left(1 + y + \sum_i n_i\right) Q_c(n). \quad (103)$$

Then

$$\begin{aligned} c \notin \mathbf{K} &\iff \tilde{H}_c \text{ has compact resolvent} \\ &\iff \mathrm{Tr}(q^{\tilde{H}_c}) < \infty \text{ for some/every } 0 < q < 1, \end{aligned}$$

whereas $c \in \mathbf{K}$ gives an infinite-dimensional zero eigenspace and infinite thermal trace for every such q . Thus compactness and thermal finiteness are co-computably-enumerable complete on this fixed padded family; their negations are computably enumerable complete.

Proof. If Q_c has no root, the padded penalty has no root. If it has one root n , every (n, y) is a root, so its root set is infinite. The radial factor includes the new coordinate and hence satisfies the positive-energy lower bound in $m + 1$ modes. Theorem 34.1 gives the asserted analytic alternatives. The exact equivalences with membership and nonmembership in $\mathbf{K}$ give both reductions and the stated enumerability classifications on this family. $\square$

No assumption about finite-fold Diophantine representations occurs in this proof. Also, no claim is made that the unrestricted finite-root problem has exactly this complexity. For a general polynomial, finite root cardinality has the elementary upper-bound description

$$\exists B \forall n (s(n) > B \implies Q(n) \neq 0),$$

but identifying a sharper unrestricted classification would require a separate argument.

35 Regulated partition functions and their analytic boundary

35.1 Two thermal parameters

Fix $0 < q < 1$. It is convenient to regard $q = e^{-\beta}$ as an inverse-temperature parameter in exponential coordinates. The second parameter $0 \leq t < 1$ damps large total occupation independently of the energy:

$$Z(q, t) = \sum_{n \in \mathbb{N}^m} q^{H(n)} t^{s(n)}. \quad (104)$$

Equivalently, this is the trace of the positive diagonal operator $q^H t^{\sum_i N_i}$. We use $0^0 = 1$ in evaluating $t = 0$. The terminology *fugacity* is a mathematical parametrization; no physical particle reservoir or laboratory implementation is assumed.

Separate the zero-state and excited-state contributions:

$$G(t) = \sum_{n \in \mathcal{Z}} t^{s(n)}, \quad (105)$$

$$R(q, t) = \sum_{n \notin \mathcal{Z}} q^{H(n)} t^{s(n)}, \quad (106)$$

$$Z(q, t) = G(t) + R(q, t). \quad (107)$$

For finite M , G is a polynomial recording witness heights. For infinite M , it is an infinite power series with nonnegative integer coefficients. Unique circuit completion makes these coefficients count the source witnesses at their *compiled* total heights; the heights generally differ from those of the original witness coordinates alone.

Theorem 35.1 (Effective partition functions). *For rational q, t in the domains below, the following sums are uniformly computable with exact rational enclosures:*

- (i) $Z(q, t)$ for $0 < q < 1$ and $0 \leq t < 1$;
- (ii) $R(q, t)$ for $0 < q < 1$ and $0 \leq t \leq 1$.

If Z_B and R_B denote sums over the box $\mathcal{C}_B$, then

$$0 \leq Z(q, t) - Z_B(q, t) \leq \frac{mt^{B+1}}{(1-t)^m}, \quad (108)$$

$$0 \leq R(q, t) - R_B(q, t) \leq \frac{mq(qt)^{B+1}}{(1-qt)^m}. \quad (109)$$

In particular,

$$0 \leq R(q, t) \leq \frac{q}{(1-qt)^m} \leq \frac{q}{(1-q)^m}. \quad (110)$$

At $t = 0$, the sums are determined by the single vacuum tuple.

Proof. Since $q^{H(n)} \leq 1$, the full sum is dominated by

$$\sum_{n \in \mathbb{N}^m} t^{s(n)} = (1-t)^{-m}.$$

A union bound over coordinates exceeding B gives (108). For the excited sum, use $q^{H(n)} \leq q^{1+s(n)}$ and apply the same argument with qt in place of t , proving (109) and (110). For rational parameters

every finite term is a rational number computed by integer polynomial evaluation and exponentiation. Increasing B until the displayed rational tail is below a prescribed tolerance gives an effective enclosure. The argument is uniform in the instance parameters, since neither tail uses them. $\square$

This theorem does not assert a fast algorithm close to $t = 1$. In the full trace, the sufficient box size from (108) can be very large. It asserts termination for each rational $t < 1$ and each requested precision. The excited trace, unlike the full trace, remains effectively computable at $t = 1$.

35.2 A sharp analytic dichotomy

Theorem 35.2 (All boundary obstructions lie in the witness series). *For fixed $q \in (0, 1)$, the function $R(q, t)$ is holomorphic in the complex disk $|t| < q^{-1}$. On every smaller closed disk $|t| \leq \rho < q^{-1}$, its box tails satisfy*

$$|R(q, t) - R_B(q, t)| \leq \frac{mq(q\rho)^{B+1}}{(1 - q\rho)^m}. \quad (111)$$

Consequently:

- (i) If $M < \infty$, then $Z(q, t)$ is holomorphic on $|t| < q^{-1}$ and $Z(q, 1) = M + R(q, 1)$.
- (ii) If $M = \infty$, the Taylor series of $Z(q, t)$ at zero has radius of convergence exactly one and $Z(q, t) \rightarrow +\infty$ as $t \uparrow 1$.

Thus a finite-valued holomorphic continuation of $Z(q, t)$ through $t = 1$ exists exactly when $M < \infty$.

Proof. The majorant $q(q\rho)^{s(n)}$ proves locally uniform absolute convergence of the series for R in the larger disk. Each term is a polynomial in t , so the limit is holomorphic; the same majorant gives (111).

If M is finite, G is a polynomial and the first conclusion follows. If M is infinite, monotone convergence along real $t \uparrow 1$ gives $G(t) \uparrow \infty$, since every zero-state term tends to one. The full series converges absolutely for $|t| < 1$ by the geometric bound. It cannot have radius greater than one, because that would give a finite value at $t = 1$, contradicting the divergence just proved. A finite-valued holomorphic continuation in a neighborhood of 1 would similarly be locally bounded on its real approach and is impossible. $\square$

Corollary 35.3 (A promised analytic-continuation problem is undecidable). *For the padded fixed universal family of Theorem 34.6, choose any fixed rational $q \in (0, 1/2)$. The functions*

$$Z_c(q, t) = \sum_{s=0}^{\infty} b_s(c) t^s, \quad b_s(c) = \sum_{s(n,y)=s} q^{\tilde{H}_c(n,y)},$$

have uniformly computable positive rational coefficients and are uniformly computable at rational $0 \leq t < 1$. Nevertheless, deciding whether their radius of convergence is one or at least $q^{-1} > 2$ is undecidable. Equivalently, deciding whether they extend holomorphically through $t = 1$ is co-computably-enumerable complete on this family.

Proof. Each coefficient is a finite exact sum over a simplex shell. A halting input has infinitely many padded zero states and hence radius one; a nonhalting input has no zero states and hence radius at least q^{-1} . Theorem 34.6 gives the completeness statement. $\square$

This corollary has a genuine promise gap in the possible radii. It is not caused by uncertainty in the Taylor coefficients or by an inability to evaluate the function at individual interior points. The missing information is a global boundary property of the entire computable series.

36 A computable critical fugacity with an undecidable endpoint

36.1 An instance-independent thermal separation

The marker coordinate in Theorem 32.1 guarantees $H(0) \geq 1$. For a system of $m \geq 1$ modes, fix

$$q_m = \frac{1}{8(m+1)}, \quad \varepsilon_m = \frac{q_m}{(1-q_m)^m}. \quad (112)$$

Bernoulli's inequality gives the explicit estimate

$$\varepsilon_m \leq \frac{q_m}{1-mq_m} = \frac{1}{7m+8} < \frac{1}{8}. \quad (113)$$

Thus the entire excited contribution, at any $0 \leq t \leq 1$, is less than $1/8$. Let

$$F(t) = Z(q_m, t), \quad c_0 = F(0) = q_m^{H(0)} \leq q_m \leq \frac{1}{16}.$$

All Taylor coefficients of F are positive, since each occupation tuple has a positive weight and each total-occupation shell is nonempty. Therefore F is continuous and strictly increasing on $[0, 1)$.

Theorem 36.1 (The critical fugacity). *For every compiled instance define*

$$t_* = \begin{cases} \text{the unique } t \in (0, 1) \text{ such that } F(t) = 1/2, & \mathcal{Z} \neq \emptyset, \\ 1, & \mathcal{Z} = \emptyset. \end{cases} \quad (114)$$

This definition is well posed. The real number t_ is uniformly computable from the finite polynomial description. For the fixed universal family,*

$$c \in \mathbf{K} \iff t_*(c) < 1, \quad c \notin \mathbf{K} \iff t_*(c) = 1. \quad (115)$$

Hence its strict interior condition is computably enumerable complete, and its exact endpoint condition is co-computably-enumerable complete.

Proof. If there is no zero state, $F = R \leq \varepsilon_m < 1/8$, so there is no crossing. If there is a zero state of total occupation L , then $F(t) \geq t^L$ and consequently $\lim_{t \uparrow 1} F(t) \geq 1$. Since $F(0) \leq 1/16$, continuity gives a crossing of $1/2$; strict monotonicity makes it unique.

It remains to prove computability without assuming that equality of computable reals is decidable. Write $F(t) = \sum_{s \geq 0} b_s t^s$, with $b_s \geq 0$. Termwise differentiation inside the unit disk gives, for $0 < t < 1$,

$$tF'(t) = \sum_{s \geq 1} s b_s t^s \geq F(t) - c_0. \quad (116)$$

In particular, on any interval where $F(t) \geq 3/8$, one has $F'(t) \geq 1/4$.

Given a requested rational width $0 < \eta \leq 1$, maintain an interval $[a, b] \subseteq [0, 1]$ containing t_* , initially $[0, 1]$. At its rational midpoint $u < 1$, compute a rational enclosure $[L, U]$ of $F(u)$ with

width at most $\delta = \eta/16$, using Theorem 35.1. If $U < 1/2$, replace a by u . If $L > 1/2$, replace b by u . These updates preserve the enclosure, including the case $t_* = 1$.

In the remaining case, $1/2 \in [L, U]$ and hence $|F(u) - 1/2| \leq \delta$. Such a value is impossible in a no-root instance, whose entire range lies below $1/8$. Thus a crossing exists. Throughout the interval joining u to the crossing, the function is at least $1/2 - \delta \geq 7/16 > 3/8$. Integrating the derivative lower bound gives

$$|u - t_*| \leq 4 |F(u) - 1/2| \leq 4\delta.$$

Intersecting $[a, b]$ with $[u - 4\delta, u + 4\delta]$ produces a valid interval of width at most $8\delta = \eta/2$. Otherwise ordinary bisection halves the width at every iteration and terminates when it is at most η . Every function evaluation is at a rational point strictly below one and terminates. This proves uniform computability.

Finally, the fixed universal polynomial has a zero state exactly on K . The definition of t_* then gives (115), and with it the completeness assertions. $\square$

Remark 36.2 (The exact boundary is not a noncomputable real). It would be incorrect to describe $t_*(c)$ as a noncomputable number. There is one uniform algorithm that encloses it to any requested precision, even for nonhalting inputs. What is undecidable is the exact assertion that its value equals one. A finite-precision interval very close to the boundary does not distinguish the boundary itself from a sufficiently nearby interior crossing.

36.2 No computable uniform distance from the boundary

Corollary 36.3 (No instance-computable positive separation). *There is no total computable function $d : \mathbb{N} \rightarrow \mathbb{Q}_{>0}$ such that every halting input in the universal family satisfies*

$$1 - t_*(c) \geq d(c).$$

There is likewise no total computable choice of rational $0 \leq t(c) < 1$ guaranteed to satisfy $F_c(t(c)) \geq 3/4$ on every halting input.

Proof. For the first assertion, replace $d(c)$ by $\min(d(c), 1)$ if necessary and compute $t_*(c)$ to error less than $d(c)/4$. A nonhalting input has value one; a halting input is at most $1 - d(c)$. These disjoint alternatives would decide K .

For the second assertion, evaluate $F_c(t(c))$ to error less than $1/8$. On nonhalting inputs its value is below $1/8$, while the promised halting value is at least $3/4$. Again the alternatives are effectively separated and would decide halting. $\square$

36.3 Witness heights determine the crossover scale

Theorem 36.4 (Finite-multiplicity height bounds). *Suppose $1 \leq M < \infty$, and let*

$$L = \min\{s(n) : n \in \mathcal{Z}\} \geq 1.$$

Then the critical fugacity satisfies

$$\frac{\log 2}{L} \leq -\log t_* \leq \frac{\log(M/(1/2 - \varepsilon_m))}{L}. \quad (117)$$

For $M = 1$, more precisely,

$$\frac{1}{2} - \varepsilon_m \leq t_*^L \leq \frac{1}{2}. \quad (118)$$

If the zero set is infinite but all its heights are at least L , one still has the unconditional estimate

$$G(t) \leq \frac{t^{L/2}}{(1 - \sqrt{t})^m}, \quad 0 < t < 1. \quad (119)$$

Proof. At the crossing, $G(t_*) = 1/2 - R(q_m, t_*)$, so

$$1/2 - \varepsilon_m \leq G(t_*) \leq 1/2.$$

A minimum-height zero contributes t_*^L , while every zero contributes at most t_*^L . Hence

$$t_*^L \leq G(t_*) \leq M t_*^L.$$

Combining these inequalities and taking negative logarithms proves (117); with one zero, $G(t) = t^L$ and (118) follows. For the final estimate, use $t^{s(n)} \leq t^{L/2}(\sqrt{t})^{s(n)}$ whenever $s(n) \geq L$, and sum over all natural tuples. $\square$

The theorem identifies a separate height obstruction even in a single-witness example. A witness can be unique and still require a fugacity arbitrarily close to one before its contribution becomes visible. Moreover, the relevant height is that of the compiled tuple. Large multiplication-gate values can make it much larger than the source witness height.

37 What remains undecidable despite the positive-energy gap

37.1 Bottom-of-spectrum separation

Theorem 37.1 (A gap does not provide a search cutoff). *For the fixed universal family, the bottom spectral value satisfies*

$$\inf \text{spec } H_c = 0 \quad \text{or} \quad \inf \text{spec } H_c \geq 1,$$

and distinguishing the alternatives is undecidable. There is no total computable bound $B(c)$ such that a zero state, whenever one exists, always has a representative in $\{0, \dots, B(c)\}^m$. In particular, no total finite-truncation procedure can uniformly approximate the bottom spectral value to error less than $1/4$.

Proof. All energies are nonnegative integers, and energy zero occurs exactly on halting inputs. A computable occupation cutoff with the stated property would permit exact finite search for a zero and thereby decide halting. Any uniform approximation to the bottom spectral value with error less than $1/4$, whether obtained by truncation or otherwise, would distinguish zero from values at least one by comparison with $1/2$. $\square$

This does not conflict with Theorem 33.2. To find all *positive-energy* states below E , one needs only a known finite simplex. A zero state can lie outside every proposed computable instance-dependent cutoff, because multiplication by the radial factor leaves every true zero of Q at energy zero. The two searches have different logical content.

[write] The second sentence of Theorem 37.1 is the proposition `cdc:bd:prop:nobound` of [22] applied to the compiled penalty, whose zero set is that of H_c ; the spectral form “ $\inf \text{spec } H_c \in \{0\} \cup [1, \infty)$, undecidable which” is source 03’s restatement.

37.2 Bounded operator perturbations versus coefficient errors

Proposition 37.2 (A precise perturbative stability statement). *Let V be any bounded self-adjoint operator with $\|V\| \leq \delta < 1/4$. Then*

$$|\inf \text{spec}(H + V) - \inf \text{spec } H| \leq \delta.$$

Thus zero-root and root-free instances retain disjoint bottom-energy ranges, respectively $[-\delta, \delta]$ and $[1 - \delta, \infty)$. Moreover, $H + V$ has compact resolvent exactly when H does.

Proof. For every unit vector in the operator domain, the quadratic form of V lies between $-\delta$ and δ . Taking infima proves the spectral-bottom inequality. Bounded self-adjoint perturbation preserves self-adjointness on $\mathcal{D}(H)$; it can also be seen by a resolvent Neumann series at a sufficiently large imaginary argument. At any common nonreal resolvent point, the resolvent identity expresses the difference of the two resolvents as a product containing either resolvent. If one resolvent is compact, the identity makes the other compact; interchanging the operators proves the converse. $\square$

The proposition is deliberately not a statement of coefficient-noise robustness. For every $\delta > 0$, the coefficient perturbation δN_i has infinite operator norm on the full occupation space. A very small perturbation of polynomial coefficients can become arbitrarily large at high occupation. Exact coefficient specification, unbounded occupation, and exact boundary predicates are substantive features of this representation, not implementation details that a spectral-gap statement removes.

Nor is diagonal evolution a Turing-complete time evolution in the usual operational sense. For an occupation basis vector,

$$e^{-itH}|n\rangle = e^{-itH(n)}|n\rangle :$$

it changes a phase and does not execute a sequence of machine configurations. The universality proved here belongs to static spectral predicates and to the partition-function boundary, not to autonomous simulation or to a physical algorithm for an undecidable problem.

38 Worked examples and exact computational checks

38.1 A single witness that moves toward the boundary

For $N \in \mathbb{N}$, consider the direct two-mode system

$$H_N(x, a) = (1 + x + a)((x - N)^2 + (a - 1)^2). \quad (120)$$

It has exactly one zero, $(N, 1)$, of total height $L = N + 1$. Here $m = 2$, $q_m = 1/24$, and

$$G_N(t) = t^{N+1}.$$

The Hamiltonian in this special example has degree three; the degree-five normal form is a general upper bound, not a requirement that every instance have exact degree five.

Table 5 reports certified crossings obtained by combining the exact ground polynomial with a rational enclosure of the excited trace. Each enclosure has width at most 2^{-30} ; the displayed decimals have absolute error below 2×10^{-9} . The rational endpoints, not just the rounded decimals, are saved in `verification.json`. For example, the enclosure for $N = 1000$ is exactly

N	Zero-state height L	Critical fugacity t_*
0	1	0.458301564
1	2	0.704142627
3	4	0.840894904
10	11	0.938930911
100	101	0.993160652
1000	1001	0.999307785

Table 5: Single-witness crossover values for (120), at $q = 1/24$.

$$\frac{1072998563}{1073741824} \leq t_* \leq \frac{268249641}{268435456}.$$

Every member of this elementary family is decidable directly. Its role is to display the height effect cleanly, not to demonstrate an undecidable individual instance.

38.2 Infinite zero multiplicity with a computable excited trace

Take

$$H(x, a) = (1 + x + a)(a - 1)^2. \quad (121)$$

All $(x, 1)$ are zero states, so

$$G(t) = \frac{t}{1 - t}.$$

For $a \neq 1$, summing the geometric series in x gives the independent formula

$$R(q, t) = \sum_{\substack{a \geq 0 \\ a \neq 1}} \frac{t^a q^{(1+a)(a-1)^2}}{1 - tq^{(a-1)^2}}. \quad (122)$$

At $t = 1$ this excited series is finite by Theorem 35.1, even though the full thermal trace is infinite. The critical fugacity at $q = 1/24$ is enclosed by

$$\frac{337179061}{1073741824} \leq t_* \leq \frac{168589531}{536870912},$$

with midpoint approximately 0.314022472. It is below $1/3$, the crossing of the ground series alone, because the excited contribution is strictly positive.

For an integer $E > 0$, the multiplicity of energy E equals the number of natural $a \neq 1$ satisfying

$$(a - 1)^2 \mid E, \quad \frac{E}{(a - 1)^2} - a - 1 \geq 0.$$

The corresponding x is uniquely the expression on the left of the inequality. This divisor formula was checked independently against finite occupation enumeration for every $1 \leq E \leq 60$.

38.3 An arithmetic infinite fiber: Pell witnesses

A less degenerate infinite zero set comes from

$$H(x, y, a) = (1 + x + y + a)((x^2 - 2y^2 - 1)^2 + (a - 1)^2). \quad (123)$$

Starting with $(x_0, y_0) = (1, 0)$, define

$$x_{j+1} = 3x_j + 4y_j, \quad y_{j+1} = 2x_j + 3y_j.$$

Direct expansion gives

$$(3x + 4y)^2 - 2(2x + 3y)^2 = x^2 - 2y^2.$$

The positive recurrence strictly increases both coordinates after the initial step and therefore produces infinitely many distinct natural solutions. Thus (123) has infinitely many zero-energy states and no compact resolvent, while its excited heat operator has the explicit effective tail bound of Theorem 33.3. The proof needs only this infinite subfamily, not a classification of every Pell solution. The first 32 recurrence steps were checked with exact integers.

38.4 Why the radial factor must contain every mode

The root-free polynomial $Q(x, y) = x^2 + 1$ has an infinitely degenerate eigenvalue one at $(0, y)$ when used as an unweighted energy. Multiplication by $1 + x$ does not help: the omitted y coordinate is still a flat positive-energy direction. Multiplication by $1 + x + y$ does help, because every positive state then has energy at least $1 + x + y$.

The same issue applies to the padding coordinate in Theorem 34.6 and to intermediate circuit variables. Omitting a mode from the radial factor invalidates the uniform positive-energy bounds unless an additional independent argument controls that mode.

38.5 Compiler and verification package

The accompanying `compiler.py` implements natural addition/multiplication circuit construction, commutative gate sharing, exact residual evaluation, canonical witness lifting, rational thermal enclosures, and critical-fugacity enclosure. Sparse polynomial inputs use exponent tuples with parameters first and source witnesses second. Floats and Boolean values are rejected where exact integer inputs are required.

[write] *Editorial note: this holds for `compile_polynomial`; the low-level `Compiler` and `Atom` interface assumes well-formed natural DAG descriptors and does not check them.* The research programme’s review [23] found that the low-level builder accepts a self-referential gate and non-natural constants: `c = Compiler(1); z = c.add(c.variable(0), Atom('mode', 1)); f = c.finish(z, z)` defines the gate $u = x + u$ and gives `f.energy((0,7,1)) == f.energy((0,8,1)) == 0`, two completions over one input; `Atom('const', 0.5)` is accepted and yields energy 0.5; Boolean atoms are accepted. Both behaviours were reproduced on a copy of the shipped compiler at the write. No theorem is affected: the theorems assume a natural DAG whose gate inputs are constants in $\mathbb{N}$, parameters, witnesses or outputs of *earlier* gates (Section 31 and the conventions before Theorem 32.1). The review’s tested patch `thermal_exact_dag_guards.patch` validates atoms and gate descriptors; with it the 7,936-assertion suite still passes and both JSON outputs are unchanged. The shipped `code/03-thermal-arithmetic-compiler.py` is the unpatched original. In this report the package files are shipped as `code/03-thermal-arithmetic-compiler.py`, `code/03-thermal-arithmetic-verify.py`, `code/03-thermal-arithmetic-build.sh`, `data/03-thermal-arithmetic-examples.json`, `data/03-thermal-arithmetic-verification.json`, `data/03-thermal-arithmetic-verification_console.txt`, `data/03-thermal-arithmetic-requirements.txt` and `03-thermal-arithmetic-PROVENANCE.md`; the example below imports the compiler by its delivered name `compiler`.

For example, the product compiler can be invoked as follows:

```
from compiler import compile_polynomial

#  $P(p; x, y) = x*y - p$ ; tuple order is  $(p, x, y)$ .
model = compile_polynomial(
    {(0, 1, 1): 1, (1, 0, 0): -1},
    inputs=2, parameters=1,
)
n = model.canonical((2, 3), (6,))
assert n == (2, 3, 6, 1)
assert model.energy(n, (6,)) == 0
assert (model.modes, model.equations) == (4, 3)
```

The verification suite performs 7,936 assertions. These are reproducible finite and symbolic regression checks, not a substitute for the general proofs and not a proof-assistant formalization. Table 6 groups the actual assertions without claiming statistical independence.

Check group	Assertions
Canonical gate/marker values, signed outputs, penalty identities, source-zero equivalence	1,240
Mutation of a single auxiliary coordinate is rejected	1,120
Exhaustive small supplied-tuple zero-fiber equivalence	2,384
Radial positive-energy lower bounds on supplied tuples	2,384
Symbolic degree, exact resource counts, and mode supports	170
Exact input-type rejection	5
Dimension-uniform thermal bound for $1 \leq m \leq 200$	200
Finite energy shells and independent multiplicities	192
Pell recurrence identity and strict growth	64
Certified partition widths, independent overlap, global bounds	60
Critical enclosures, height bounds, endpoint and algorithm checks	17
Omitted-mode confinement counterexample	100
Total	7,936

Table 6: Executed exact and symbolic checks. Detailed categories and rational outputs appear in `verification.json`.

Six source polynomials are exported in `examples.json`: $x-p$, x^3-p , $xy-p$, x^2+1 , $x^2+xy-py-3$, and the zero polynomial. Their complete compiled polynomial expressions are included. Symbolic expansion independently confirms the degree and support bounds for these examples. Small supplied tuples include noncanonical gate assignments; the suite is not limited to evaluating the intended lifts.

38.6 Certified truncation without enormous denominators

The rational partition routine uses two explicit error budgets. First choose a box so that (108) or (109) is at most half the requested tolerance. Let $D = (B+1)^m$ be the number of tuples in that box. Next choose an integer energy cutoff C such that

$$Dq^C \leq \frac{\text{tolerance}}{2}.$$

Terms with $H(n) \geq C$ may be omitted from the finite rational sum, since their total contribution is at most Dq^C when $0 \leq t \leq 1$. This avoids constructing denominators such as $24^{H(n)}$ for unnecessarily large energies. The output interval is

$$[\text{retained exact sum}, \text{retained exact sum} + \text{box tail} + Dq^C].$$

No omitted term is silently discarded. A configurable resource limit raises an exception rather than claiming a certificate from unfinished work. The optional exact ground-series interface speeds up the examples near $t = 1$; supplying such a formula is an extra, independently justified input, not an assumed ability to solve the general zero-set problem.

39 Further research questions

The following questions are selected to extend the proved results rather than restate the general halting problem.

39.1 Can radial isolation be achieved at degree four?

The direct multiplier converts a quartic sum of gate squares into a quintic. Is there a witness-faithful finite-variable polynomial normal form of degree at most four with both the same zero fibers and an effective bound excluding positive low-energy states at large occupation? The target need not be the exact inequality $H \geq 1 + s$; any computable proper lower bound on the positive sector would preserve much of the analysis. A construction must account for the possibility that fresh auxiliaries create their own low-energy flat directions. A lower bound for a clearly specified class of constructions would also be useful. ([write] Degree at most three is impossible for a universal family: H is nonnegative on the real nonnegative orthant, and an orthant-nonnegative polynomial of degree at most three has a semilinear natural zero set (`cdc:of:thm:classification` and `cdc:of:cor:no-cubic-universal` in [22]; for globally nonnegative coercive polynomials compare `pqc:ql:prop:degree` in [21]), whereas the zero sets of Corollary 32.3 project onto a c.e.-complete set. For universal families the open gap is therefore four versus five.)

39.2 Can four-mode positive terms be reduced to three-mode terms?

The extra mode enters when n_i multiplies a residual square already involving three modes. A useful refinement would lower the maximum support without losing positivity, introducing uncontrolled witness multiplicity, or increasing degree beyond a stated budget. The appropriate initial task is to classify local identities for one multiplication residual and one external radial coordinate. Geometric locality is a separate, harder requirement.

39.3 Instantiate a numerical universal arithmetic family

Take one audited universal polynomial from ProveIt and apply the compiler literally. Record source witness count, shared gate count, coefficient bit lengths, expanded degree, positive-term supports, and the cost of exact energy evaluation. This would turn Corollary 32.3 from a fixed-family existence result into a reproducible numerical normal form. It should not be reported as an improved universal-operation bound unless that operation count is actually lower under the same convention.

39.4 Optimize compiled witness height, not only gate count

Two zero-set-bijective arithmetic circuits can have very different sums of intermediate values. Theorem 36.4 makes this a measurable analytic cost. Define and compare the height overhead of alternative circuits, including product trees, shared powers, redundant checks, and parameter loaders. A circuit with fewer gates might create a much larger critical-fugacity boundary layer. A useful objective is a joint frontier in gate count, maximum intermediate value, and total zero-state occupation.

39.5 Find unconditional all-fiber compact subclasses

Theorem 34.4 identifies the exact representation requirement for all-fiber compactness. Rather than assume it for universal halting, identify interesting restricted computational relations with explicit finite-fold representations and carry out the compact normal form. Preserving known fiber cardinalities under composition is especially important. Any genuinely universal solution would need to confront the finite-fold issue, not merely make the gate completion unique.

39.6 Characterize effective compactness on restricted families

Even where compact resolvent is mathematically guaranteed, a uniform computable modulus for its finite-rank approximation may fail. Which syntactic or arithmetic hypotheses provide such a modulus? An explicit computable witness-height bound is sufficient. Determine whether weaker information, such as an effective bound on zero-state count combined with a controlled arithmetic geometry, also suffices for useful subclasses. The answer should distinguish compactness, computable compactness, and computability of individual matrix entries.

39.7 Classify witness-series singularities

For infinite fibers, $G(t)$ controls all singular behavior at $t = 1$, while R extends to a larger disk. For families with structured witnesses, determine asymptotic expansions of $G(t)$ as $t \uparrow 1$: polynomial poles for free coordinates, logarithmic growth for exponentially sparse families, and possible logarithmic oscillations. The Pell example is a concrete starting point. Any all-orders expansion would directly become an expansion of the full partition function after adding the analytic excited contribution. This offers a connection with the repository's generating-function and transseries work without importing an unproved asymptotic claim into this article.

39.8 What survives non-diagonal bounded couplings?

Proposition 37.2 preserves compactness and a separated bottom-energy test under bounded self-adjoint perturbations. It does not preserve the exact ground projection or the simple partition decomposition. Seek quantitative heat-trace comparison bounds and effective truncations for a precisely specified class of bounded couplings. An important first distinction is whether the perturbation is effectively approximable in norm or merely has computable matrix entries.

39.9 Can a physically motivated cutoff retain a useful conditional theorem?

A fixed maximum occupation makes the space finite dimensional and every zero test decidable. The meaningful question is therefore not whether this reproduces unrestricted halting, but which bounded computational questions it represents with certified precision and resource costs. Compare known runtime or witness-height bounds with occupation limits, coefficient errors, and energy resolution. A valid theorem should state explicitly which promise supplies the missing cutoff.

39.10 Formalize the arithmetic and effective-analysis interface

A modular formalization can begin with the exact zero-fiber bijection, the radial inequality, and finite-box geometric tails. These are independent of a full formalization of unbounded operators. The next layer can treat diagonal operators directly on ℓ^2 , prove the maximal-domain realization, and connect trace and compactness to diagonal sequences. The universal result should depend on a separately audited MRDP interface, and the finite-fold equivalence should remain visibly conditional. The present executable code is a test oracle for this project, not a formal proof artifact.

40 Dependency and claim ledger

Statement	Status and dependencies
Universal Diophantine representation	Established foundation, used from the MRDP/universal-equation literature; no new proof or numerical instance supplied.
Squared number-operator embedding	Established prior construction; explicitly attributed to Kieu. No physical decision claim used.
Witness-faithful compiler	Proved here by topological induction; classical circuit method with an explicit marker and quantitative ledger. [write] The circuit core is <code>cdc:wf:lem:quadratisation</code> of [22] (note in Section 31).
Degree-five radial isolation and four-mode positive display	Proved here by integer positivity and support counting.
Positive spectral cutoffs and effective remainders	Proved here from the radial inequality, geometric sums, and diagonal operator arguments.
Compactness and thermal multiplicity classification	Proved here; exactly preserves the source root count.
All-fiber compact universal representation	Not asserted unconditionally. The exact finite-fold equivalence is proved.
Padded compactness and analytic-radius undecidability	Unconditional consequence of fixed universal representability and free-witness padding.
Uniformly computable critical fugacity	Proved here using certified interior evaluations and a positive derivative bound.
Undecidable exact endpoint and absent cutoffs	Proved by reductions to halting. The critical real itself is computable.
Numerical and symbolic checks	7,936 executed assertions; evidence about the implementation and selected examples, not proof-assistant verification.

Statement	Status and dependencies
Priority or breakthrough status	Not established by this targeted search. The article supplies explicit proved results and a research program, not a certified historical first.

41 Conclusion

A small structural modification of an arithmetic penalty yields a useful separation: difficult zero states are left exactly where the Diophantine witnesses put them, while every positive state is forced into an effective radial energy bound. This separates unknown witness multiplicity from an otherwise computable excited spectrum and gives exact correspondences between finite-fold arithmetic, compact resolvent, and finite thermal trace.

The boundary construction adds a second distinction. A family of partition functions can be uniformly computable throughout its regulated domain, and its threshold location can itself be a uniformly computable real, while the exact endpoint predicate remains complete for nonhalting. No inaccessible approximation oracle is hidden in the proof; the obstruction is the impossibility of a uniform positive separation from the boundary.

The resulting substrate is unconventional but mathematically precise: a fixed finite collection of unbounded occupation modes with polynomial diagonal energies. It is not a physical solution of an undecidable problem. Its value is to expose which parts of a computational representation are existential, which are counting-theoretic, which are controlled by witness height, and which become visible only at an exact analytic boundary.

A A fully instantiated horizon-two polynomial schema

[write] Appendix A of source 02.

For the three-state alphabet (44), let $e_{t,0}, e_{t,R}, e_{t,L}$ be the natural selectors at time $t = 1, 2$. Its compressed information matrix is

$$A(e_t) = e_{t,0}H_0 + e_{t,R}H_R + e_{t,L}H_L = 3 \begin{pmatrix} e_{t,R} + e_{t,L} & e_{t,R} \\ e_{t,L} & e_{t,R} + e_{t,L} \end{pmatrix}.$$

Let $U = \mathbf{1}_2\mathbf{1}_2^\top$, $Y_0 = I_2 + U$, $p_0 = 1$, $p_1 = 39$, and $p_2 = 1521$. The following is one explicit integer polynomial in fourteen natural variables (six selectors and eight entries of Y_1, Y_2):

$$\begin{aligned} \mathcal{I}_2 = & \sum_{t=1}^2 (e_{t,0} + e_{t,R} + e_{t,L} - 1)^2 \\ & + \sum_{t=1}^2 \sum_{a,b=1}^2 \left((Y_t)_{ab} - p_t - [A(e_t)(Y_{t-1} - p_{t-1}U)]_{ab} \right)^2 \\ & + \sum_{a,b=1}^2 ((Y_2)_{ab} - 1521)^2. \end{aligned}$$

All finite sums, coefficients, dimensions, and constant matrices are specified; expanding the squares is optional and does not change the polynomial. The word $(0, L)$ gives selectors $(1, 0, 0)$ and $(0, 0, 1)$,

with $Y_1 = 39U$ and $Y_2 = 1521U$. Every displayed residual is then visibly zero. A word containing only R, L produces an invertible transverse product and cannot satisfy the terminal conditions.

This example illustrates the natural-domain uniqueness directly. A selector assignment such as $(2, -1, 0)$ is inadmissible, even though its entries sum to one; using arbitrary integer selectors without additional conditions would invalidate the one-hot proof. The checker enforces the natural domain exactly.

B Reproduction and source ledger

[write] Appendix B of source 02.

The archive contains the standalone `article.tex`, its compiled `article.pdf`, the Python implementation and independent evaluator, numerical example JSON files, a test receipt, and separate source/proof-status notes. From the package directory, run

```
python3 code/run_checks.py
python3 code/check_certificate.py \
    examples/seven_state_information_certificate.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command rewrites the delivered examples and verification receipt. The checker returns exit status zero for an accepted certificate and nonzero for a rejected or ill-typed certificate. The deliberately incorrect uniform-target example is expected to fail, not to be treated as a regression.

[write] These commands use the delivered layout and names; the first one overwrites the recorded examples and receipt, and on Windows writes them with CRLF line endings. In this report the programs are `code/02-exact-erasure-*.py`, the examples and the three verification records are `data/02-exact-erasure-*.json`, and the delivered `article.tex` is printed as Part I; the duplicate `verification/run_output.txt` (byte-identical to the check results) and the source's PDF are not shipped. The README of this report gives a recipe that runs on a copy and leaves the shipped files untouched.

The numerical implementation uses only the Python standard library. The PDF build uses pdfLaTeX and standard TeX packages named in the preamble. The version used for the recorded Python checks was 3.13.5. The final PDF is rendered and checked separately; the PDF preflight record is included in the verification folder.

The external dependencies are narrow. The mortality bounds come from [1]; the halting/PCP interface is supported by [2]; the computably-enumerable/Diophantine correspondence is DPRM [3]. The MathOverflow discussion [4] supplies the precise research question, not a theorem used as an unproved premise. The repository sources [5, 6, 7] supply project context and claim boundaries, not automatic mathematical authority for the present results.

C Source audit and provenance

[write] Appendix A of source 10.

Repository inspection used the connected GitHub interface on October 2, 2026. The relevant reads were:

1. Computability/HilbertTenthProblem/README.md, first 210 lines, returned blob 880d2a13393ea55110b49e573180b51b011d9bc9. ([write] The link was the unpinned /blob/main/. Blob 880d2a13 is the README introduced by commit db3b377f0 and replaced by b7a9404d4, both after the pin 297eb58e4 used for items 2 and 3, whose README blob is 138f234b; both versions state the 75- and 87-operation figures quoted in Section 27.)
2. Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean, pinned at commit 297eb58e43157b7723cc28982083fecf8d2f29bf.
3. The probabilistic/quantum/continuous substrate collection’s README, pinned at the same commit; it identifies the strict-contraction and neural-precision reports as existing research contributions.
4. Targeted repository searches for Diophantine work, contraction, self-assembly, and Shapley operators. These are a scope check, not an exhaustive audit of every file in the repository.

The probabilistic collection is itself an unrefereed research collection. This article uses it for provenance and nonduplication, not as a substitute for proving the new simulation. The code in this archive was written for this manuscript and does not copy the repository’s imported compilers. No repository files were modified.

External sources were checked through primary publisher or author-preprint pages. In particular, the Bellman comparison uses arXiv revision 3 dated January 26, 2026, rather than only the February 2025 abstract. The Skolem status comparison includes the July 2026 preprint. No claim of exhaustive priority clearance follows from this targeted audit.

D A compact implementation contract

[write] Appendix B of source 10.

Program validates a finite source table and natural initial counters. **counter_circuit** emits the absorbing or erasing map. **layer_circuit** inserts shared delay copies and pads all outputs. **compile_program** emits paired-channel actions, normalization constants, and optional common reset and reward. **initialize** emits the complete dyadic payoff vector. ([write] The review [24] found that an explicitly supplied *integer* scale in **Program.encode** is divided in floating point, so that a counter value 1075 becomes 0.0 instead of 2^{-1075} ; the default rational scale, used by every shipped loader and export, is exact. Its patch **exact_convergence_integer_scale.patch** normalizes integer or rational scales and rejects floating-point and Boolean ones. The shipped **code/10-exact-convergence-bellman_diophantine.py** is the unpatched original.)

certificate accepts the game, initial numerator vector and positive denominator, a natural horizon, and either a scalar pair, consensus test, or rational constant full-vector target. It emits every squared linear form and complementary product, including all reward forcing coefficients. Its metadata reports the exact witness count, scaling convention, and endpoint-square count.

A JSON linear form is a list of integer pairs (j, a_j) ; index -1 denotes the constant monomial, and all other indices refer to the supplied variable label array. A complementarity term is an index pair (u, v) and means $W_u W_v$, not a residual that still needs squaring. The complete polynomial is the sum of the squares of all listed linear forms plus all listed complementarity terms. This specifies

an ordinary integer polynomial without auxiliary syntax such as “run,” “minimum,” or “there exists a schedule.”

The independent checker evaluates that polynomial and, optionally, reconstructs the game trajectory from the serialized stochastic rows. Unsupported domains and malformed instructions are rejected. The implementation is intended as a transparent research reference, not as a hardened service for untrusted, resource-exhausting inputs.

E Reproduction and file inventory

[write] Appendix A of source 03.

The archive contains the following principal files:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Source and compiled research article
<code>compiler.py</code>	Exact natural circuit and thermal routines
<code>verify.py</code>	Finite and symbolic verification suite
<code>verification.json</code>	Exact results and rational threshold enclosures
<code>examples.json</code>	Gate-level and expanded compiled examples
<code>README.md</code>	Usage, limitations, and reproduction instructions
<code>PROVENANCE.md</code>	Repository snapshot and primary-source links
<code>requirements.txt</code>	Symbolic verification dependency
<code>build.sh</code>	PDF build commands

Run the tests with Python 3.10 or later and SymPy installed:

```
python -m pip install -r requirements.txt
python verify.py
```

The arithmetic compiler and interval routines themselves use only the Python standard library. The manuscript can be rebuilt with a conventional LaTeX installation containing the packages listed in its preamble:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

[write] The delivered `build.sh` (shipped as `code/03-thermal-arithmetic-build.sh`) runs *three* `pdflatex` passes, not the two shown. File names in the table above are the delivered ones; their shipped names are listed in Section 38.5. `verify.py` imports `compiler` by its delivered name and rewrites `examples.json` and `verification.json` in its own directory, so it must be run on a copy (README of this report); `article.tex` is printed as Part III, and the source’s `README.md`, `article.pdf` and `SHA256SUMS.txt` are not shipped.

The verification receipt is generated by the delivered code. In particular, the optional accelerated ground-series path used for the near-boundary examples is identified in the implementation; it is not substituted for the general computability proof.

F Provenance of this report

[write] This appendix was written when the three manuscripts were merged.

F.1 Sources

The report was built from three manuscripts of batch 78 of ProveIt’s incoming reports, cluster H2. Two arrived in the archives of commit 1977e6ea6 and one in those of commit 808b53ed8; all three were placed by commit 798b0c5d4, which staged their programs, data and audit notes byte-identical to the delivery.

Source	Archive and title	Arrival	Pin	Printed as
02 (base)	ProveIt_Exact_Erasure_Rese arch.zip; <i>Exact Erasure without a Common Invariant Line</i> ; 29-page PDF	1977e6ea6	928ea9701	Part I (Sections 2–15 and conclusion), Appendices A–B
10	ProveIt_Exact_Convergence_ Research.zip; <i>Exact Convergence as Computation</i> ; 24-page PDF	808b53ed8	297eb58e4 (README read unpinned; Appendix C)	Part II (Sections 17–29 and conclusion), Appendices C–D
03	Thermal_Arithmetic_Diophan tine_Hamiltonians.zip; <i>Thermal Arithmetic</i> ; 25-page PDF	1977e6ea6	6914ccca6; blob 8bd07142 of the review it cites	Part III (Sections 31–39, ledger and conclusion), Appendix E

Contributions: source 02 the positive stochastic erasure compiler, its PCP front end, higher guards, decidable boundaries, zero–one law and natural quartics; source 10 the stochastic-game compiler, its boundaries and its quadratic certificates; source 03 the radially isolated Hamiltonian, its spectral and thermal classification and the critical fugacity. Every result, proof, example, remark, limitation, research question and appendix of the three manuscripts is printed. They share no theorem with one another, so no statement had to be printed once for two sources.

F.2 Where the merge had to choose

- *One report, three Parts.* The three manuscripts’ nearest neighbour is the report [21], which already had nine Parts; they were placed together as a sibling report because they share the spine of Section 1.2, not a construction. Placing sources 02 and 10 as a tenth Part of [21] and source 03 as an eleventh was the alternative considered at placement.
- *Base and order.* Source 02 is the base (the longest text, the earliest arrival, the widest set of results); its `article.tex` was staged unprefixd and has been replaced in place by this merged text. Source 10 follows because stochastic games extend stochastic matrices by minima and maxima; source 03 shares no lemma with either and comes last.
- *Re-proofs of collection results.* The core of Theorem 32.1, the fixed-arity quartic corollaries of Parts I and II, the no-cutoff clauses of all three Parts and the template of Theorem 34.4 are results of [22] (and partly of [21]). They are printed in place, with the sources’ own proofs, and marked by [write] pointers naming the host labels; no novelty is claimed for them. Remark 34.5 is an editorial consequence of two printed theorems.
- *Reviews and defects.* The research programme’s reviews of all three manuscripts [23, 24] are cited where their findings apply: the thermal compiler’s unchecked low-level descriptors (Section 38.5), the Bellman checker’s unbound initial vector (Section 26) and the Bellman

loader’s floating-point integer scale (Appendix D). The delivered programs are shipped unpatched; the patches stay in the research programme.

- *Labels.* Every label carries the prefix `ste:.`. The labels of source 02 are `ste:ee:` plus their delivered names, those of source 10 `ste:ec:` plus theirs, and those of source 03 `ste:ta:` plus theirs; no source label was dropped or renamed apart from the prefix. Added labels: 50 in all; 206 delivered labels give 256. Front section, Parts and provenance (12): `ste:sec:front`, `ste:sec:parts`, `ste:sec:spine`, `ste:tab:spine`, `ste:sec:notation`, `ste:tab:notation`, `ste:sec:relation`, `ste:sec:status`, `ste:part:ee`, `ste:part:ec`, `ste:part:ta`, `ste:app:provenance`. Part I (9), the nine research questions: `ste:ee:q:overhead`, `ste:ee:q:irreducible`, `ste:ee:q:lowdim`, `ste:ee:q:schedules`, `ste:ee:q:certificates`, `ste:ee:q:fixedalphabet`, `ste:ee:q:nonerasure`, `ste:ee:q:robust`, `ste:ee:q:formal`. Part II (13), its four unlabelled statements, two appendices and seven research directions: `ste:ec:cor:stopping`, `ste:ec:prop:onesided`, `ste:ec:prop:selectors`, `ste:ec:prop:threshold`, `ste:ec:q:instantiate`, `ste:ec:q:oneplayer`, `ste:ec:q:sparse`, `ste:ec:q:observations`, `ste:ec:q:economical`, `ste:ec:q:mrp`, `ste:ec:q:formalize`, `ste:ec:app:audit`, `ste:ec:app:contract`. Part III (16), four sections, its appendix, the ten research questions and the editorial remark: `ste:ta:sec:intro`, `ste:ta:rem:universalcompact`, `ste:ta:sec:gap`, `ste:ta:sec:package`, `ste:ta:sec:questions`, `ste:ta:q:degree4`, `ste:ta:q:threemode`, `ste:ta:q:instantiate`, `ste:ta:q:height`, `ste:ta:q:compact`, `ste:ta:q:effective`, `ste:ta:q:singularities`, `ste:ta:q:couplings`, `ste:ta:q:cutoff`, `ste:ta:q:formalize`, `ste:ta:app:reproduction`.
- *Numbering.* Source 02’s Section n is Section $n + 1$ here (Sections 2–16) and its Appendices A–B keep their letters; source 10’s Section n is Section $n + 16$ (Sections 17–30) and its Appendices A–B are Appendices C–D; source 03’s Section n is Section $n + 30$ (Sections 31–41) and its Appendix A is Appendix E. Theorem numbers follow the sections.
- *Bibliography.* The three bibliographies are merged. Source 03’s `matiyasevich` (the 2010 finite-fold paper) is `matiyasevichFF`, because source 02 uses that key for the 1993 book. The repository items read by several sources at different pins are merged into one entry each, recording every pin: `repoH10` (02’s `repoH10`, 03’s `reporeadme`, 10’s `repo-h10`), `repoReview` (02’s `repoReview`, 03’s `repoaudit`) and `repoProb` (02’s `repoProb`, 10’s `repo-substrates`). Entries marked [write] were added.
- *Front matter and typography.* The three delivered title pages are replaced by one; each source’s title, subtitle, author line, abstract and status statement open its Part verbatim. Source 02’s remarks used the definition style; all remarks now use the remark style. No symbol was renamed (Section 1.3).
- *Files.* Delivered programs, data and audit notes are shipped byte-identical under prefixed names (the README maps them). Not shipped: the three PDFs, the delivered READMEs of all three and the manuscripts of sources 10 and 03 (printed as Parts II and III), the checksum ledgers of sources 02 and 03 (verified at placement), the two in-archive byte copies `verification/run_output.txt` (source 02) and `artifacts/test_run.txt` (source 10), and source 10’s two regenerable certificates `countdown_certificate.json` and `fixedpoint_certificate.json`, which the README shows how to rebuild. All survive in the archives of their arrival commits.
- *Added text.* Every sentence written in the merge is marked [write]: Section 1; the openings of the three Parts; the notes in Part I on the MathOverflow claim, the repository relationship and

review, the (3, 6) attribution, the nonnegative-mortality boundary, the radix-three encoding, geometric clocks, the circuit quadratization, the seven-state wording, the no-bound argument and shipped file names; the notes in Part II on the collection’s contractions, the structural-subclasses question, the complementarity device, the quartic corollary, the no-cutoff and threshold arguments, the excluded certificates, the checker defect, the unpinned README read, the loader-scale defect and shipped file names; the notes in Part III on the fixed-quartic route, the circuit quadratization, the Boolean energy landscape, the finite-fold template with Remark 34.5, the no-cutoff clause, the compiler defect, the degree floor, the build passes and shipped file names; the notes opening each appendix; and this appendix.

References

- [1] J. Cassaigne, V. Halava, T. Harju, and F. Nicolas, *Tighter Undecidability Bounds for Matrix Mortality, Zero-in-the-Corner Problems, and More*, arXiv:1404.0644v3, September 5, 2014. <https://arxiv.org/html/1404.0644v3>. The adopted bounds are $(d, m) = (3, 6), (5, 4), (9, 3), (15, 2)$; the paper’s general reducibility convention is oracle Turing reducibility.
- [2] Y. Forster, E. Heiter, and G. Smolka, *Verification of PCP-Related Computational Reductions in Coq*, arXiv:1711.07023v2, July 18, 2018. <https://arxiv.org/abs/1711.07023v2>. Related publication DOI: https://doi.org/10.1007/978-3-319-94821-8_15.
- [3] Y. V. Matiyasevich, *Hilbert’s Tenth Problem*, MIT Press, 1993. Author-hosted preface, including the statement of the computably-enumerable/Diophantine correspondence: <https://logic.pdmi.ras.ru/~yumat/H10Pbook/preface.htm>.
- [4] V. Miller, *Decidability of a matrix product being rank 1*, MathOverflow question 511960, June 2, 2026; subsequent discussion with B. Steinberg, June 2–4, 2026. The common-invariant-line restriction appears in the June 3 discussion. Accessed October 2, 2026. <https://mathoverflow.net/questions/511960/decidability-of-a-matrix-product-being-rank-1>.
- [5] V. Reshetnikov and contributors, *ProveIt: Hilbert’s tenth problem* (titled *Hilbert’s tenth problem: Jones’s articles on Diophantine representation* by sources 10 and 03), repository README Computability/HilbertTenthProblem/README.md, accessed October 2, 2026. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/Computability/HilbertTenthProblem/README.md>. Source 03 read it at the snapshot of [6], commit 6914ccca6685baf53b7a35f25efc89366c76ba74: <https://github.com/VladimirReshetnikov/ProveIt/blob/6914ccca6685baf53b7a35f25efc89366c76ba74/Computability/HilbertTenthProblem/README.md>. Source 10 read it through the unpinned link <https://github.com/VladimirReshetnikov/ProveIt/blob/main/Computability/HilbertTenthProblem/README.md>; the returned blob is recorded in Appendix C. [write] The three sources’ keys repoH10 (02), reporeadme (03) and repo-h10 (10) are merged into this entry.
- [6] *Review of the imported computational-substrate reports, 2 October 2026*, ProveIt research review, Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/imported_substrate_review_20261002.md, blob 8bd07142c8b4298fb636e47b40d2d84e993a40ce. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/Computability/HilbertTenthProblem/README.md>.

- em/Papers/research-wip/native-stream-queue/imported_substrate_review_20261002.md. Source 03 read it at snapshot commit 6914ccca6685baf53b7a35f25efc89366c76ba74: https://github.com/VladimirReshetnikov/ProveIt/blob/6914ccca6685baf53b7a35f25efc89366c76ba74/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/imported_substrate_review_20261002.md. [write] The keys `repoReview` (02) and `repoaudit` (03) are merged; both pins carry the same blob.
- [7] *Diophantine Laws of Probabilistic, Quantum and Continuous Computation*, ProveIt combined report README, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md`, an AI-assisted, unrefereed research collection; its source inventory includes *Strict Contractions with Diophantine-Complete Transients* and *Precision Is Memory*. Source 02 read it at commit 928ea97017a25ebe56d240c84f27d2275d818c75: <https://github.com/VladimirReshetnikov/ProveIt/blob/928ea97017a25ebe56d240c84f27d2275d818c75/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md>. Source 10 read it at commit 297eb58e43157b7723cc28982083fecf8d2f29bf, October 2, 2026: <https://github.com/VladimirReshetnikov/ProveIt/blob/297eb58e43157b7723cc28982083fecf8d2f29bf/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/README.md>. [write] The keys `repoProb` (02) and `repo-substrates` (10) are merged. The report itself is [21].
- [8] L. S. Shapley. Stochastic games. *Proceedings of the National Academy of Sciences* 39(10) (1953), 1095–1100. doi:10.1073/pnas.39.10.1095.
- [9] A. Dudenhefner. Certified decision procedures for two-counter machines. *FSCD 2022*, LIPIcs 228, 16:1–16:18. doi:10.4230/LIPIcs.FSCD.2022.16.
- [10] A. Varonka and K. Watanabe. On piecewise affine reachability with Bellman operators. [arXiv:2502.19923v3](https://arxiv.org/abs/2502.19923), January 26, 2026. Extended version of the MFCS 2025 paper, LIPIcs 345, 92:1–92:18.
- [11] M. Vahanwala. Skolem and positivity completeness of ergodic Markov chains. *Information Processing Letters* 186 (2024), 106481. doi:10.1016/j.ipl.2024.106481. Author preprint: [arXiv:2305.04881](https://arxiv.org/abs/2305.04881).
- [12] F. Luca, J. Ouaknine, and J. Worrell. Conjectural decidability of the Skolem problem. [arXiv:2607.15510](https://arxiv.org/abs/2607.15510), July 2026.
- [13] M. Grobelna, J. Křetínský, and M. Weininger. Stopping criteria for value iteration on concurrent stochastic reachability and safety games. [arXiv:2505.21087v2](https://arxiv.org/abs/2505.21087), September 10, 2025. Full version of the LICS 2025 paper.
- [14] D. Larchey-Wendling and Y. Forster. Hilbert’s tenth problem in Coq. *FSCD 2019*, LIPIcs 131, 27:1–27:20. doi:10.4230/LIPIcs.FSCD.2019.27.
- [15] V. Reshetnikov, ProveIt repository. *Diophantine/MRDP.lean*. Pinned repository source, commit 297eb58e43157b7723cc28982083fecf8d2f29bf, read October 2, 2026.
- [16] James P. Jones, *Universal Diophantine equation*, *Journal of Symbolic Logic* 47 (1982), no. 3, 549–571. DOI: 10.2307/2273588. Primary publisher page consulted October 2, 2026.

- [17] Tien D. Kieu, *A reformulation of Hilbert's tenth problem through quantum mechanics*, arXiv:quant-ph/0111063, version 2 (2003; initial submission 2001). <https://arxiv.org/abs/quant-ph/0111063>. The squared number-operator construction is the only external operator representation used here; proposed physical consequences are not assumed.
- [18] Yuri Matiyasevich, *Towards finite-fold Diophantine representations*, Journal of Mathematical Sciences **171** (2010), no. 6, 745–752. DOI: [10.1007/s10958-010-0179-4](https://doi.org/10.1007/s10958-010-0179-4). [write] Source 03's key `matiyasevich`, renamed because source 02 uses that key for a different item.
- [19] D. Cantone, A. Casagrande, F. Fabris, and E. G. Omodeo, *The quest for Diophantine finite-fold-ness*, Le Matematiche **76** (2021), no. 1. <https://lematematiche.dmi.unict.it/index.php/lematematiche/article/view/2044>.
- [20] [write] T. Neary, *Undecidability in Binary Tag Systems and the Post Correspondence Problem for Five Pairs of Words*, in *32nd International Symposium on Theoretical Aspects of Computer Science (STACS 2015)*, LIPIcs **30** (2015), 649–661. A copy is kept at `Computability/HilbertTenthProblem/Lean/LIPIcs.STACS.2015.649.pdf`.
- [21] [write] ProveIt research report *Diophantine Laws of Probabilistic, Quantum and Continuous Computation*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/`; Parts II (quantum substrates), IV (questions), V (exact probabilities), VII (exact observation in continuous-state dynamics), VIII (continuous time) and IX (exact stopping laws). Cited by label names, as at the write of this report.
- [22] [write] ProveIt research report *Canonical Diophantine Certificates*, collection `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/canonical-diophantine-certificates/`; its witness-faithful foundations, Part III (accelerators), Part IX (the boundary of fixed-arity compression) and Part XI (positivity-restricted degree threshold). Cited by label names, as at the write of this report.
- [23] [write] ProveIt research programme, *Six new substrate reports: review, constructor repairs and local reductions*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_review_1977e6ea6.md` (commit 44b28c395), sections *Exact erasure* and *Thermal arithmetic and its repair*, with its receipt `incoming_substrate_review_1977e6ea6.json` (keys `erasure`, `thermal`, `thermal_repair`) and the patch `thermal_exact_dag_guards.patch` in the same directory.
- [24] [write] ProveIt research programme, *Seven incoming computational-substrate reports and Exact Convergence and Total Quadratic Semantics: review of 808b53ed8*, `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/incoming_substrate_review_808b53ed8.md` and `incoming_convergence_review_808b.md` (commit 126028588), with the receipt `incoming_substrate_review_808b53ed8.json` (keys `independent.convergence`, `independent.convergence_repairs`, `independent.convergence_scale`) and the patches `exact_convergence_input_binding.patch` and `exact_convergence_integer_scale.patch` in the same directory.
- [25] [write] ProveIt research programme, research note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/matrix_pcp_trace.md`, Section 1, as at the write of this report.